RecoTrackerEntranceCompare: entrance seeding diagnostic

A: original saved trajectory. B: truth position only. C: full truth boundary.
D: truth momentum only, when enabled. No truth constraint upstream of entrance.
Control: fresh backward propagation from all-reconstructed boundary quantities.
All distances: mm. Momenta: MeV/c. Times: ns. Phase: degrees. Pitch: mrad.
Signed width s68 = (q84.1345-q15.8655)/2. Standard deviation includes tails.
Nonnegative errors use 68% and 95% containment, not signed Gaussian widths.
All statistics use unbinned finite values, including histogram under/overflow.
NaN values and unavailable states are counted, not converted to zero.
dt is corrected for the common proton-bunch clock; dt_beam includes that clock.
Fixed-z comparisons do not measure an independent longitudinal resolution.
Target branch is unique, has initial production direction, and precedes entrance.
Principal target/foil results require completed transport; partial roots are counted.
Residual plots use available valid states; always compare their sample sizes.
Event efficiencies keep the identical generated-primary event denominator.
Full-truth residuals can include stochastic material effects and transport error.
Linear angular moments depend on wrapping; consult the circular statistics.
Input and propagated covariance-invalid counts are separate; legacy status bits overlap.
ROOT histograms remain in the input file. JSON/CSV retain full-precision tables.

A: tracker entrance / baseline_direct

```
ent_dx         N=    853 NaN=    0
  mean=0.83837 med=1.4645 std=17.824 s68=6.3519
  |err| q68=6.3244 q95=23.021 q99=50.438 tail |err|>40: 12 (1.407%)
ent_dy         N=    853 NaN=    0
  mean=0.7008 med=0.62587 std=35.517 s68=6.0488
  |err| q68=5.9049 q95=21.016 q99=75.903 tail |err|>40: 15 (1.758%)
ent_dr         N=    853 NaN=    0
  mean=12.621 med=6.5656 std=37.697 s68=6.4438
  |err| q68=9.9646 q95=29.307 q99=101.4 tail |err|>40: 30 (3.517%)
ent_dp         N=    853 NaN=    0
  mean=-0.20161 med=-0.05753 std=2.1263 s68=0.21166
  |err| q68=0.20423 q95=0.92193 q99=6.7125 tail |err|>2: 22 (2.579%)
ent_dpz        N=    853 NaN=    0
  mean=-0.12794 med=-0.045624 std=1.4798 s68=0.35599
  |err| q68=0.34252 q95=1.0004 q99=4.4869 tail |err|>2: 20 (2.345%)
ent_pitch      N=    853 NaN=    0
  mean=-0.040134 med=0.02716 std=16.527 s68=8.8837
  |err| q68=8.8406 q95=26.224 q99=75.175 tail |err|>20: 65 (7.620%)
ent_dt         N=    853 NaN=    0
  mean=0.30476 med=0.38205 std=0.7213 s68=0.35207
  |err| q68=0.57516 q95=1.3383 q99=2.8918 tail |err|>3: 5 (0.586%)
ent_dt_beam    N=    853 NaN=    0
  mean=0.29805 med=0.35639 std=0.90394 s68=0.65895
  |err| q68=0.83444 q95=1.6894 q99=3.0468 tail |err|>3: 10 (1.172%)
```

A: tracker entrance / baseline_reflected

```
ent_dx         N=    679 NaN=    0
  mean=0.53968 med=1.533 std=29.237 s68=6.3403
  |err| q68=6.7364 q95=26.818 q99=75.776 tail |err|>40: 20 (2.946%)
ent_dy         N=    679 NaN=    0
  mean=0.13725 med=0.66618 std=21.935 s68=6.2186
  |err| q68=6.3895 q95=24.306 q99=79.409 tail |err|>40: 18 (2.651%)
ent_dr         N=    679 NaN=    0
  mean=13.758 med=6.8717 std=33.867 s68=7.2868
  |err| q68=10.538 q95=37.949 q99=117.76 tail |err|>40: 32 (4.713%)
ent_dp         N=    679 NaN=    0
  mean=-0.20446 med=-0.059252 std=1.4113 s68=0.21031
  |err| q68=0.21286 q95=1.0533 q99=4.3327 tail |err|>2: 16 (2.356%)
ent_dpz        N=    679 NaN=    0
  mean=-0.10984 med=-0.01036 std=1.056 s68=0.38196
  |err| q68=0.3638 q95=1.1717 q99=3.2161 tail |err|>2: 11 (1.620%)
ent_pitch      N=    679 NaN=    0
  mean=-0.50211 med=-0.59873 std=18.085 s68=8.8095
  |err| q68=8.8406 q95=29.908 q99=54.576 tail |err|>20: 62 (9.131%)
ent_dt         N=    679 NaN=    0
  mean=0.33526 med=0.37933 std=0.72736 s68=0.35516
  |err| q68=0.56373 q95=1.263 q99=3.403 tail |err|>3: 9 (1.325%)
ent_dt_beam    N=    679 NaN=    0
  mean=0.32804 med=0.3191 std=0.89501 s68=0.6688
  |err| q68=0.76821 q95=1.7074 q99=3.7938 tail |err|>3: 12 (1.767%)
```

A: tracker entrance / control_direct

```
ent_dx         N=    853 NaN=    0
  mean=0.83837 med=1.4645 std=17.824 s68=6.3519
  |err| q68=6.3244 q95=23.021 q99=50.438 tail |err|>40: 12 (1.407%)
ent_dy         N=    853 NaN=    0
  mean=0.7008 med=0.62587 std=35.517 s68=6.0488
  |err| q68=5.9049 q95=21.016 q99=75.903 tail |err|>40: 15 (1.758%)
ent_dr         N=    853 NaN=    0
  mean=12.621 med=6.5656 std=37.697 s68=6.4438
  |err| q68=9.9646 q95=29.307 q99=101.4 tail |err|>40: 30 (3.517%)
ent_dp         N=    853 NaN=    0
  mean=-0.20161 med=-0.05753 std=2.1263 s68=0.21166
  |err| q68=0.20423 q95=0.92193 q99=6.7125 tail |err|>2: 22 (2.579%)
ent_dpz        N=    853 NaN=    0
  mean=-0.12794 med=-0.045624 std=1.4798 s68=0.35599
  |err| q68=0.34252 q95=1.0004 q99=4.4869 tail |err|>2: 20 (2.345%)
ent_pitch      N=    853 NaN=    0
  mean=-0.040134 med=0.02716 std=16.527 s68=8.8837
  |err| q68=8.8406 q95=26.224 q99=75.175 tail |err|>20: 65 (7.620%)
ent_dt         N=    853 NaN=    0
  mean=0.30476 med=0.38205 std=0.7213 s68=0.35207
  |err| q68=0.57516 q95=1.3383 q99=2.8918 tail |err|>3: 5 (0.586%)
ent_dt_beam    N=    853 NaN=    0
  mean=0.29805 med=0.35639 std=0.90394 s68=0.65895
  |err| q68=0.83444 q95=1.6894 q99=3.0468 tail |err|>3: 10 (1.172%)
```

A: tracker entrance / control_reflected

```
ent_dx         N=    679 NaN=    0
  mean=0.53968 med=1.533 std=29.237 s68=6.3403
  |err| q68=6.7364 q95=26.818 q99=75.776 tail |err|>40: 20 (2.946%)
ent_dy         N=    679 NaN=    0
  mean=0.13725 med=0.66618 std=21.935 s68=6.2186
  |err| q68=6.3895 q95=24.306 q99=79.409 tail |err|>40: 18 (2.651%)
ent_dr         N=    679 NaN=    0
  mean=13.758 med=6.8717 std=33.867 s68=7.2868
  |err| q68=10.538 q95=37.949 q99=117.76 tail |err|>40: 32 (4.713%)
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  mean=-0.20446 med=-0.059252 std=1.4113 s68=0.21031
  |err| q68=0.21286 q95=1.0533 q99=4.3327 tail |err|>2: 16 (2.356%)
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  |err| q68=0.3638 q95=1.1717 q99=3.2161 tail |err|>2: 11 (1.620%)
ent_pitch      N=    679 NaN=    0
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ent_dt_beam    N=    679 NaN=    0
  mean=0.32804 med=0.3191 std=0.89501 s68=0.6688
  |err| q68=0.76821 q95=1.7074 q99=3.7938 tail |err|>3: 12 (1.767%)
```

A: tracker entrance / momentum_direct

```
ent_dx         N=    853 NaN=    0
  mean=0.83837 med=1.4645 std=17.824 s68=6.3519
  |err| q68=6.3244 q95=23.021 q99=50.438 tail |err|>40: 12 (1.407%)
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  |err| q68=0.34252 q95=1.0004 q99=4.4869 tail |err|>2: 20 (2.345%)
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```

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  |err| q68=6.7364 q95=26.818 q99=75.776 tail |err|>40: 20 (2.946%)
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  mean=0.13725 med=0.66618 std=21.935 s68=6.2186
  |err| q68=6.3895 q95=24.306 q99=79.409 tail |err|>40: 18 (2.651%)
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  mean=13.758 med=6.8717 std=33.867 s68=7.2868
  |err| q68=10.538 q95=37.949 q99=117.76 tail |err|>40: 32 (4.713%)
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```

A: tracker entrance / position_direct

```
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ent_dr         N=    679 NaN=    0
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```

A: tracker entrance / truth_direct

```
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  |err| q68=0.57516 q95=1.3383 q99=2.8918 tail |err|>3: 5 (0.586%)
ent_dt_beam    N=    853 NaN=    0
  mean=0.29805 med=0.35639 std=0.90394 s68=0.65895
  |err| q68=0.83444 q95=1.6894 q99=3.0468 tail |err|>3: 10 (1.172%)
```

A: tracker entrance / truth_reflected

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  mean=0.13725 med=0.66618 std=21.935 s68=6.2186
  |err| q68=6.3895 q95=24.306 q99=79.409 tail |err|>40: 18 (2.651%)
ent_dr         N=    679 NaN=    0
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  |err| q68=10.538 q95=37.949 q99=117.76 tail |err|>40: 32 (4.713%)
ent_dp         N=    679 NaN=    0
  mean=-0.20446 med=-0.059252 std=1.4113 s68=0.21031
  |err| q68=0.21286 q95=1.0533 q99=4.3327 tail |err|>2: 16 (2.356%)
ent_dpz        N=    679 NaN=    0
  mean=-0.10984 med=-0.01036 std=1.056 s68=0.38196
  |err| q68=0.3638 q95=1.1717 q99=3.2161 tail |err|>2: 11 (1.620%)
ent_pitch      N=    679 NaN=    0
  mean=-0.50211 med=-0.59873 std=18.085 s68=8.8095
  |err| q68=8.8406 q95=29.908 q99=54.576 tail |err|>20: 62 (9.131%)
ent_dt         N=    679 NaN=    0
  mean=0.33526 med=0.37933 std=0.72736 s68=0.35516
  |err| q68=0.56373 q95=1.263 q99=3.403 tail |err|>3: 9 (1.325%)
ent_dt_beam    N=    679 NaN=    0
  mean=0.32804 med=0.3191 std=0.89501 s68=0.6688
  |err| q68=0.76821 q95=1.7074 q99=3.7938 tail |err|>3: 12 (1.767%)
```

C: target production plane / baseline_direct

```
tar_dx         N=    741 NaN=    0
  mean=0.74614 med=-0.16321 std=141.86 s68=109.47
  |err| q68=109.2 q95=343.35 q99=416.73 tail |err|>400: 14 (1.889%)
tar_dy         N=    741 NaN=    0
  mean=-6.9655 med=-1.5433 std=138 s68=94.948
  |err| q68=90.919 q95=334.18 q99=403.07 tail |err|>400: 9 (1.215%)
tar_dr         N=    741 NaN=    0
  mean=144.84 med=93.527 std=135.05 s68=149.41
  |err| q68=173.68 q95=423.35 q99=455.6 tail |err|>400: 60 (8.097%)
tar_phase      N=    741 NaN=    0
  mean=-1.6176 med=-1.5228 std=62.879 s68=48.494
  |err| q68=47.729 q95=150.31 q99=173.22 tail |err|>90: 120 (16.194%)
tar_center     N=    741 NaN=    0
  mean=8.2107 med=5.9091 std=10.685 s68=4.4655
  |err| q68=8.3998 q95=20.698 q99=46.882 tail |err|>40: 11 (1.484%)
tar_radius     N=    741 NaN=    0
  mean=-2.5976 med=-1.1575 std=11.001 s68=3.7408
  |err| q68=3.8899 q95=15.256 q99=44.313 tail |err|>20: 31 (4.184%)
tar_dp         N=    741 NaN=    0
  mean=-0.50057 med=-0.17071 std=2.3582 s68=0.52009
  |err| q68=0.43449 q95=2.7593 q99=7.9471 tail |err|>2: 53 (7.152%)
tar_dpz        N=    741 NaN=    0
  mean=0.47301 med=0.11132 std=2.8377 s68=1.529
  |err| q68=1.4769 q95=6.5219 q99=10.839 tail |err|>2: 173 (23.347%)
tar_pitch      N=    741 NaN=    0
  mean=-12.424 med=-4.989 std=57.751 s68=32.682
  |err| q68=32.986 q95=132.04 q99=223.88 tail |err|>20: 356 (48.043%)
tar_dt         N=    741 NaN=    0
  mean=0.61385 med=0.47426 std=1.6172 s68=1.0581
```

C: target production plane / baseline_direct

```
  |err| q68=1.1745 q95=3.8278 q99=6.4883 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=    741 NaN=    0
  mean=0.60572 med=0.44193 std=1.7022 s68=1.2524
  |err| q68=1.3084 q95=3.7773 q99=6.7335 tail |err|>10: 0 (0.000%)
```

C: target production plane / baseline_reflected

```
tar_dx         N=    589 NaN=    0
  mean=-19.998 med=-13.403 std=219.29 s68=239.4
  |err| q68=250.16 q95=409.7 q99=450.19 tail |err|>400: 34 (5.772%)
tar_dy         N=    589 NaN=    0
  mean=3.9613 med=8.149 std=221.51 s68=247.98
  |err| q68=247.04 q95=406.17 q99=447.28 tail |err|>400: 35 (5.942%)
tar_dr         N=    589 NaN=    0
  mean=280.79 med=306.82 std=136.85 s68=152.4
  |err| q68=383.39 q95=451.15 q99=479.33 tail |err|>400: 147 (24.958%)
tar_phase      N=    589 NaN=    0
  mean=3.7564 med=8.0696 std=105.06 s68=123.8
  |err| q68=122.88 q95=172.74 q99=177.55 tail |err|>90: 298 (50.594%)
tar_center     N=    589 NaN=    0
  mean=11.052 med=8.2445 std=9.8237 s68=6.572
  |err| q68=11.436 q95=29.187 q99=46.048 tail |err|>40: 12 (2.037%)
tar_radius     N=    589 NaN=    0
  mean=-3.7688 med=-2.2034 std=9.8867 s68=5.4255
  |err| q68=5.8422 q95=19.236 q99=34.111 tail |err|>20: 28 (4.754%)
tar_dp         N=    589 NaN=    0
  mean=-0.72318 med=-0.26133 std=2.0452 s68=0.85775
  |err| q68=0.76974 q95=4.0638 q99=8.75 tail |err|>2: 74 (12.564%)
tar_dpz        N=    589 NaN=    0
  mean=-0.67219 med=-0.37701 std=3.2712 s68=2.8012
  |err| q68=2.8136 q95=6.7077 q99=10.452 tail |err|>2: 270 (45.840%)
tar_pitch      N=    589 NaN=    0
  mean=16.906 med=8.6688 std=65.955 s68=54.176
  |err| q68=58.478 q95=132.28 q99=212.53 tail |err|>20: 387 (65.705%)
tar_dt         N=    589 NaN=    0
  mean=-0.31289 med=-0.14307 std=3.6838 s68=2.8482
```

C: target production plane / baseline_reflected

```
  |err| q68=2.8077 q95=7.8533 q99=12.115 tail |err|>10: 13 (2.207%)
tar_dt_beam    N=    589 NaN=    0
  mean=-0.33201 med=-0.16854 std=3.7031 s68=2.8057
  |err| q68=2.7878 q95=7.7924 q99=12.164 tail |err|>10: 12 (2.037%)
```

C: target production plane / control_direct

```
tar_dx         N=    692 NaN=    0
  mean=2.1729 med=0.28247 std=146.28 s68=117.78
  |err| q68=116.95 q95=344.95 q99=417.63 tail |err|>400: 14 (2.023%)
tar_dy         N=    692 NaN=    0
  mean=-6.6482 med=-0.98603 std=142.4 s68=101.39
  |err| q68=97.658 q95=340.02 q99=404.44 tail |err|>400: 9 (1.301%)
tar_dr         N=    692 NaN=    0
  mean=152.14 med=100.76 std=136.3 s68=152.96
  |err| q68=196.34 q95=424.97 q99=457.95 tail |err|>400: 60 (8.671%)
tar_phase      N=    692 NaN=    0
  mean=-0.79048 med=-1.5309 std=64.848 s68=53.261
  |err| q68=52.474 q95=153.9 q99=173.38 tail |err|>90: 120 (17.341%)
tar_center     N=    692 NaN=    0
  mean=8.2077 med=5.9081 std=10.905 s68=4.5768
  |err| q68=8.3424 q95=20.259 q99=47.935 tail |err|>40: 11 (1.590%)
tar_radius     N=    692 NaN=    0
  mean=-2.741 med=-1.2209 std=10.993 s68=3.6794
  |err| q68=3.7217 q95=14.512 q99=44.556 tail |err|>20: 26 (3.757%)
tar_dp         N=    692 NaN=    0
  mean=-0.52749 med=-0.18265 std=2.4109 s68=0.5249
  |err| q68=0.42831 q95=2.7634 q99=8.1815 tail |err|>2: 48 (6.936%)
tar_dpz        N=    692 NaN=    0
  mean=0.50568 med=0.1207 std=2.9142 s68=1.5831
  |err| q68=1.5079 q95=6.8725 q99=11.018 tail |err|>2: 167 (24.133%)
tar_pitch      N=    692 NaN=    0
  mean=-13.138 med=-5.2576 std=58.281 s68=33.364
  |err| q68=32.812 q95=135.01 q99=228.49 tail |err|>20: 337 (48.699%)
tar_dt         N=    692 NaN=    0
  mean=0.6647 med=0.48559 std=1.5927 s68=1.0788
```

C: target production plane / control_direct

```
  |err| q68=1.2007 q95=3.8429 q99=6.3898 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=    692 NaN=    0
  mean=0.64446 med=0.45915 std=1.6858 s68=1.2743
  |err| q68=1.3858 q95=3.8331 q99=6.7058 tail |err|>10: 0 (0.000%)
```

C: target production plane / control_reflected

```
tar_dx         N=    595 NaN=    0
  mean=-20.253 med=-10.706 std=217.75 s68=235.18
  |err| q68=246.02 q95=408.64 q99=450.74 tail |err|>400: 34 (5.714%)
tar_dy         N=    595 NaN=    0
  mean=3.8609 med=7.865 std=220.8 s68=246.8
  |err| q68=246.74 q95=406.23 q99=447.42 tail |err|>400: 36 (6.050%)
tar_dr         N=    595 NaN=    0
  mean=278.35 med=304.65 std=138.25 s68=158.85
  |err| q68=383.52 q95=451.47 q99=480.05 tail |err|>400: 149 (25.042%)
tar_phase      N=    595 NaN=    0
  mean=3.7428 med=7.5869 std=104.65 s68=123.46
  |err| q68=122.61 q95=172.54 q99=177.77 tail |err|>90: 295 (49.580%)
tar_center     N=    595 NaN=    0
  mean=11.09 med=8.1976 std=9.8166 s68=6.5864
  |err| q68=11.507 q95=29.045 q99=45.909 tail |err|>40: 13 (2.185%)
tar_radius     N=    595 NaN=    0
  mean=-3.6645 med=-2.0897 std=9.8617 s68=5.4021
  |err| q68=5.7493 q95=19.156 q99=33.908 tail |err|>20: 28 (4.706%)
tar_dp         N=    595 NaN=    0
  mean=-0.69723 med=-0.25346 std=2.0443 s68=0.85887
  |err| q68=0.76765 q95=4.0275 q99=8.7427 tail |err|>2: 73 (12.269%)
tar_dpz        N=    595 NaN=    0
  mean=-0.67987 med=-0.384 std=3.257 s68=2.7544
  |err| q68=2.7878 q95=6.7003 q99=10.446 tail |err|>2: 270 (45.378%)
tar_pitch      N=    595 NaN=    0
  mean=16.932 med=8.6558 std=65.628 s68=53.246
  |err| q68=58.063 q95=131.5 q99=208.65 tail |err|>20: 389 (65.378%)
tar_dt         N=    595 NaN=    0
  mean=-0.30934 med=-0.1131 std=3.6672 s68=2.8028
```

C: target production plane / control_reflected

```
  |err| q68=2.7902 q95=7.8454 q99=12.07 tail |err|>10: 13 (2.185%)
tar_dt_beam    N=    595 NaN=    0
  mean=-0.32571 med=-0.15858 std=3.687 s68=2.754
  |err| q68=2.7734 q95=7.746 q99=12.117 tail |err|>10: 12 (2.017%)
```

C: target production plane / momentum_direct

```
tar_dx         N=    704 NaN=    0
  mean=8.983 med=2.999 std=127.11 s68=79.801
  |err| q68=77.375 q95=299.41 q99=427.46 tail |err|>400: 13 (1.847%)
tar_dy         N=    704 NaN=    0
  mean=-3.3572 med=1.592 std=134.97 s68=84.4
  |err| q68=83.847 q95=332 q99=423.78 tail |err|>400: 12 (1.705%)
tar_dr         N=    704 NaN=    0
  mean=130.02 med=71.274 std=132.51 s68=137.82
  |err| q68=151.41 q95=422.66 q99=461.55 tail |err|>400: 50 (7.102%)
tar_phase      N=    704 NaN=    0
  mean=0.44909 med=-0.1688 std=58.322 s68=40.543
  |err| q68=40.723 q95=146.07 q99=173.1 tail |err|>90: 95 (13.494%)
tar_center     N=    704 NaN=    0
  mean=11.587 med=8.3141 std=16.237 s68=6.7993
  |err| q68=11.301 q95=27.139 q99=63.314 tail |err|>40: 17 (2.415%)
tar_radius     N=    704 NaN=    0
  mean=-1.8178 med=-0.73805 std=5.3796 s68=2.947
  |err| q68=2.6025 q95=9.5346 q99=27.274 tail |err|>20: 11 (1.562%)
tar_dp         N=    704 NaN=    0
  mean=-0.34807 med=-0.063568 std=1.0108 s68=0.37703
  |err| q68=0.2486 q95=1.5147 q99=5.0849 tail |err|>2: 30 (4.261%)
tar_dpz        N=    704 NaN=    0
  mean=0.41268 med=0.076725 std=2.377 s68=1.2645
  |err| q68=1.2081 q95=5.7537 q99=9.6819 tail |err|>2: 125 (17.756%)
tar_pitch      N=    704 NaN=    0
  mean=-9.7514 med=-2.6697 std=47.38 s68=26.422
  |err| q68=25.59 q95=110.8 q99=189.98 tail |err|>20: 276 (39.205%)
tar_dt         N=    704 NaN=    0
  mean=0.57127 med=0.43682 std=1.3551 s68=0.83378
```

C: target production plane / momentum_direct

```
  |err| q68=0.93258 q95=3.4275 q99=5.4062 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=    704 NaN=    0
  mean=0.54903 med=0.38257 std=1.4635 s68=1.121
  |err| q68=1.1534 q95=3.52 q99=5.1672 tail |err|>10: 0 (0.000%)
```

C: target production plane / momentum_reflected

```
tar_dx         N=    609 NaN=    0
  mean=0.5433 med=5.0376 std=210.54 s68=236.17
  |err| q68=234.86 q95=398.23 q99=452.41 tail |err|>400: 29 (4.762%)
tar_dy         N=    609 NaN=    0
  mean=-1.241 med=-1.6627 std=216.54 s68=241.99
  |err| q68=241.39 q95=398.87 q99=427.67 tail |err|>400: 29 (4.762%)
tar_dr         N=    609 NaN=    0
  mean=268.33 med=292.89 std=138.63 s68=160.22
  |err| q68=367.86 q95=448.14 q99=484.03 tail |err|>400: 138 (22.660%)
tar_phase      N=    609 NaN=    0
  mean=10.986 med=14.369 std=99.473 s68=113.93
  |err| q68=116.28 q95=169.55 q99=177.4 tail |err|>90: 285 (46.798%)
tar_center     N=    609 NaN=    0
  mean=15.348 med=10.673 std=19.235 s68=9.6419
  |err| q68=15.876 q95=39.028 q99=74.221 tail |err|>40: 30 (4.926%)
tar_radius     N=    609 NaN=    0
  mean=-2.4733 med=-1.135 std=7.4422 s68=4.4412
  |err| q68=4.9216 q95=15.66 q99=30.901 tail |err|>20: 18 (2.956%)
tar_dp         N=    609 NaN=    0
  mean=-0.45794 med=-0.12432 std=1.5241 s68=0.62962
  |err| q68=0.57013 q95=3.147 q99=6.9913 tail |err|>2: 66 (10.837%)
tar_dpz        N=    609 NaN=    0
  mean=-0.59403 med=-0.29443 std=2.7621 s68=2.3424
  |err| q68=2.438 q95=5.8902 q99=8.3418 tail |err|>2: 237 (38.916%)
tar_pitch      N=    609 NaN=    0
  mean=14.091 med=7.1605 std=54.624 s68=46.498
  |err| q68=49.803 q95=116.6 q99=165.75 tail |err|>20: 343 (56.322%)
tar_dt         N=    609 NaN=    0
  mean=-0.16687 med=0.076285 std=3.1285 s68=2.5302
```

C: target production plane / momentum_reflected

```
  |err| q68=2.4423 q95=6.3673 q99=9.1729 tail |err|>10: 4 (0.657%)
tar_dt_beam    N=    609 NaN=    0
  mean=-0.17462 med=0.040362 std=3.14 s68=2.5506
  |err| q68=2.5474 q95=6.5524 q99=9.5908 tail |err|>10: 4 (0.657%)
```

C: target production plane / position_direct

```
tar_dx         N=    685 NaN=    0
  mean=0.70616 med=1.343 std=148.59 s68=125.82
  |err| q68=125.71 q95=340.36 q99=417.12 tail |err|>400: 14 (2.044%)
tar_dy         N=    685 NaN=    0
  mean=-7.766 med=-1.6251 std=142.68 s68=100.79
  |err| q68=98.164 q95=339.31 q99=409.91 tail |err|>400: 12 (1.752%)
tar_dr         N=    685 NaN=    0
  mean=154.38 med=108.33 std=136.61 s68=154.08
  |err| q68=204.67 q95=423.82 q99=461.89 tail |err|>400: 63 (9.197%)
tar_phase      N=    685 NaN=    0
  mean=-1.9243 med=-1.8348 std=65.033 s68=54.306
  |err| q68=54.617 q95=148.11 q99=176.86 tail |err|>90: 120 (17.518%)
tar_center     N=    685 NaN=    0
  mean=11.348 med=7.8919 std=17.98 s68=6.1783
  |err| q68=11.328 q95=25.359 q99=62.247 tail |err|>40: 12 (1.752%)
tar_radius     N=    685 NaN=    0
  mean=-2.8059 med=-1.2257 std=8.7087 s68=3.7066
  |err| q68=3.6994 q95=14.352 q99=35.828 tail |err|>20: 23 (3.358%)
tar_dp         N=    685 NaN=    0
  mean=-0.57777 med=-0.16767 std=1.8661 s68=0.52506
  |err| q68=0.42825 q95=2.7704 q99=7.7829 tail |err|>2: 48 (7.007%)
tar_dpz        N=    685 NaN=    0
  mean=0.40588 med=0.095161 std=2.9939 s68=1.6237
  |err| q68=1.5753 q95=7.0797 q99=11.671 tail |err|>2: 166 (24.234%)
tar_pitch      N=    685 NaN=    0
  mean=-11.152 med=-4.1344 std=60.116 s68=33.952
  |err| q68=31.865 q95=142.45 q99=225.34 tail |err|>20: 344 (50.219%)
tar_dt         N=    685 NaN=    0
  mean=0.62664 med=0.46125 std=1.6416 s68=1.1026
```

C: target production plane / position_direct

```
  |err| q68=1.1952 q95=3.831 q99=6.3903 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=    685 NaN=    0
  mean=0.61382 med=0.42211 std=1.7319 s68=1.3341
  |err| q68=1.3318 q95=3.8306 q99=6.6968 tail |err|>10: 0 (0.000%)
```

C: target production plane / position_reflected

```
tar_dx         N=    591 NaN=    0
  mean=-21.778 med=-10.683 std=221.28 s68=240.52
  |err| q68=253.46 q95=411.22 q99=454.39 tail |err|>400: 39 (6.599%)
tar_dy         N=    591 NaN=    0
  mean=-5.6701 med=0.65817 std=219.81 s68=241.04
  |err| q68=238.04 q95=408.55 q99=449.46 tail |err|>400: 39 (6.599%)
tar_dr         N=    591 NaN=    0
  mean=278.48 med=317.86 std=142.27 s68=164.41
  |err| q68=384.55 q95=457.04 q99=491.89 tail |err|>400: 154 (26.058%)
tar_phase      N=    591 NaN=    0
  mean=3.469 med=1.6439 std=105.21 s68=123.09
  |err| q68=122.97 q95=172.57 q99=178.86 tail |err|>90: 305 (51.607%)
tar_center     N=    591 NaN=    0
  mean=15.046 med=11.003 std=20.983 s68=8.7706
  |err| q68=15.104 q95=38.755 q99=69.944 tail |err|>40: 26 (4.399%)
tar_radius     N=    591 NaN=    0
  mean=-3.5701 med=-2.0833 std=9.8918 s68=5.757
  |err| q68=6.0848 q95=19.219 q99=34.079 tail |err|>20: 29 (4.907%)
tar_dp         N=    591 NaN=    0
  mean=-0.69332 med=-0.25852 std=2.0168 s68=0.84222
  |err| q68=0.76592 q95=3.9525 q99=8.6214 tail |err|>2: 73 (12.352%)
tar_dpz        N=    591 NaN=    0
  mean=-0.58055 med=-0.34716 std=3.3399 s68=2.8467
  |err| q68=2.9243 q95=6.7943 q99=10.573 tail |err|>2: 276 (46.701%)
tar_pitch      N=    591 NaN=    0
  mean=15.095 med=8.8802 std=67.421 s68=57.197
  |err| q68=57.87 q95=133.94 q99=223.7 tail |err|>20: 393 (66.497%)
tar_dt         N=    591 NaN=    0
  mean=-0.24985 med=-0.13802 std=3.7579 s68=2.9118
```

C: target production plane / position_reflected

```
  |err| q68=2.9449 q95=7.9746 q99=13.121 tail |err|>10: 13 (2.200%)
tar_dt_beam    N=    591 NaN=    0
  mean=-0.27018 med=-0.11261 std=3.7765 s68=2.9313
  |err| q68=2.9304 q95=7.8956 q99=13.07 tail |err|>10: 14 (2.369%)
```

C: target production plane / truth_direct

```
tar_dx         N=    704 NaN=    0
  mean=7.1031 med=2.1103 std=127.45 s68=75.73
  |err| q68=75.849 q95=315.43 q99=412.25 tail |err|>400: 11 (1.562%)
tar_dy         N=    704 NaN=    0
  mean=-4.5078 med=-0.81066 std=131.47 s68=76.995
  |err| q68=74.543 q95=338.57 q99=409.64 tail |err|>400: 11 (1.562%)
tar_dr         N=    704 NaN=    0
  mean=126.76 med=69.484 std=132.4 s68=134.06
  |err| q68=141.18 q95=420.64 q99=462.53 tail |err|>400: 51 (7.244%)
tar_phase      N=    704 NaN=    0
  mean=-2.856 med=-0.71753 std=57.725 s68=38.464
  |err| q68=37.434 q95=143.63 q99=173.09 tail |err|>90: 95 (13.494%)
tar_center     N=    704 NaN=    0
  mean=6.2109 med=4.659 std=6.2332 s68=3.7926
  |err| q68=6.5954 q95=14.685 q99=32.303 tail |err|>40: 6 (0.852%)
tar_radius     N=    704 NaN=    0
  mean=-1.7398 med=-0.66239 std=5.3825 s68=2.8633
  |err| q68=2.5922 q95=9.4111 q99=26.569 tail |err|>20: 12 (1.705%)
tar_dp         N=    704 NaN=    0
  mean=-0.34225 med=-0.04975 std=1.0154 s68=0.36747
  |err| q68=0.235 q95=1.4656 q99=4.8966 tail |err|>2: 30 (4.261%)
tar_dpz        N=    704 NaN=    0
  mean=0.34997 med=0.069747 std=2.2924 s68=1.228
  |err| q68=1.188 q95=5.4989 q99=9.023 tail |err|>2: 123 (17.472%)
tar_pitch      N=    704 NaN=    0
  mean=-8.4674 med=-2.0392 std=45.805 s68=25.229
  |err| q68=24.537 q95=108.94 q99=181.17 tail |err|>20: 277 (39.347%)
tar_dt         N=    704 NaN=    0
  mean=0.22109 med=0.044128 std=1.1487 s68=0.56814
```

C: target production plane / truth_direct

```
  |err| q68=0.53708 q95=2.6855 q99=4.7988 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=    704 NaN=    0
  mean=0.19327 med=0.028849 std=1.2744 s68=0.88456
  |err| q68=0.83036 q95=2.682 q99=4.6355 tail |err|>10: 0 (0.000%)
```

C: target production plane / truth_reflected

```
tar_dx         N=    607 NaN=    0
  mean=-1.6575 med=1.1425 std=213.49 s68=233.49
  |err| q68=234.43 q95=402.14 q99=440.56 tail |err|>400: 35 (5.766%)
tar_dy         N=    607 NaN=    0
  mean=-5.1983 med=-1.7453 std=215.66 s68=239.27
  |err| q68=237.13 q95=400.32 q99=431.49 tail |err|>400: 31 (5.107%)
tar_dr         N=    607 NaN=    0
  mean=270.01 med=293.37 std=138.6 s68=162.97
  |err| q68=374.38 q95=447.72 q99=469.13 tail |err|>400: 145 (23.888%)
tar_phase      N=    607 NaN=    0
  mean=10.509 med=18.658 std=100.9 s68=119.39
  |err| q68=117.64 q95=168.87 q99=177.59 tail |err|>90: 288 (47.446%)
tar_center     N=    607 NaN=    0
  mean=9.5974 med=7.5608 std=7.7479 s68=5.8878
  |err| q68=10.135 q95=25.689 q99=39.198 tail |err|>40: 6 (0.988%)
tar_radius     N=    607 NaN=    0
  mean=-2.5667 med=-1.2073 std=6.8396 s68=4.5445
  |err| q68=4.8835 q95=15.373 q99=30.388 tail |err|>20: 16 (2.636%)
tar_dp         N=    607 NaN=    0
  mean=-0.50187 med=-0.13843 std=1.3921 s68=0.60663
  |err| q68=0.56301 q95=2.9237 q99=6.4168 tail |err|>2: 61 (10.049%)
tar_dpz        N=    607 NaN=    0
  mean=-0.4558 med=-0.20004 std=2.7578 s68=2.42
  |err| q68=2.4403 q95=5.8738 q99=8.527 tail |err|>2: 229 (37.727%)
tar_pitch      N=    607 NaN=    0
  mean=11.486 med=5.1829 std=54.421 s68=47.668
  |err| q68=50.577 q95=116.9 q99=167.26 tail |err|>20: 350 (57.661%)
tar_dt         N=    607 NaN=    0
  mean=-0.39223 med=-0.24717 std=3.0101 s68=2.3599
```

C: target production plane / truth_reflected

```
  |err| q68=2.3763 q95=6.3287 q99=9.2561 tail |err|>10: 4 (0.659%)
tar_dt_beam    N=    607 NaN=    0
  mean=-0.404 med=-0.37173 std=3.0216 s68=2.3527
  |err| q68=2.3904 q95=6.5771 q99=9.1482 tail |err|>10: 4 (0.659%)
```

C: phase statistics, wrapping-aware

```
baseline_direct        N=741
  circular mean=-2.2658 deg, R=0.60177, circular std=57.745 deg
baseline_reflected     N=589
  circular mean=99.706 deg, R=0.05767, circular std=136.86 deg
control_direct         N=692
  circular mean=-1.8642 deg, R=0.57762, circular std=60.029 deg
control_reflected      N=595
  circular mean=89.203 deg, R=0.0591, circular std=136.28 deg
momentum_direct        N=704
  circular mean=-0.11809 deg, R=0.65641, circular std=52.573 deg
momentum_reflected     N=609
  circular mean=49.573 deg, R=0.095079, circular std=124.29 deg
position_direct        N=685
  circular mean=-2.1409 deg, R=0.56954, circular std=60.795 deg
position_reflected     N=591
  circular mean=105.41 deg, R=0.029521, circular std=152.08 deg
truth_direct           N=704
  circular mean=-1.3193 deg, R=0.66313, circular std=51.933 deg
truth_reflected        N=607
  circular mean=56.61 deg, R=0.07696, circular std=129.76 deg
```

C: common-cohort configuration comparison

```
Same tracks valid in every enabled configuration; N=1183
This sensitivity cohort supplements, and never replaces, failure-inclusive efficiencies.
baseline_direct
  tar_dr N=615 med=111.93 s68=155.41 |err|q95=426.4
  tar_phase N=615 med=-1.5228 s68=56.036 |err|q95=154.21
  tar_dt N=615 med=0.48417 s68=1.093 |err|q95=3.8617
baseline_reflected
  tar_dr N=568 med=306.7 s68=153.47 |err|q95=450.65
  tar_phase N=568 med=3.121 s68=124.3 |err|q95=172.81
  tar_dt N=568 med=-0.13736 s68=2.788 |err|q95=7.5091
control_direct
  tar_dr N=615 med=112.45 s68=154.61 |err|q95=426.36
  tar_phase N=615 med=-1.3605 s68=55.887 |err|q95=154.12
  tar_dt N=615 med=0.48418 s68=1.0918 |err|q95=3.862
control_reflected
  tar_dr N=568 med=307.1 s68=156.7 |err|q95=451.2
  tar_phase N=568 med=3.8699 s68=124.73 |err|q95=172.64
  tar_dt N=568 med=-0.12197 s68=2.7767 |err|q95=7.5047
momentum_direct
  tar_dr N=615 med=73.058 s68=133.64 |err|q95=415.77
  tar_phase N=615 med=0.43993 s68=40.528 |err|q95=136
  tar_dt N=615 med=0.44856 s68=0.85834 |err|q95=3.4722
momentum_reflected
  tar_dr N=568 med=294.78 s68=161.8 |err|q95=447.43
  tar_phase N=568 med=14.037 s68=115.6 |err|q95=169.78
  tar_dt N=568 med=0.056575 s68=2.4772 |err|q95=6.2171
position_direct
  tar_dr N=615 med=117.68 s68=155.5 |err|q95=424.28
  tar_phase N=615 med=-1.7413 s68=57.58 |err|q95=149.07
```

C: common-cohort configuration comparison

```
  tar_dt N=615 med=0.46125 s68=1.1327 |err|q95=3.8935
position_reflected
  tar_dr N=568 med=319.22 s68=162.85 |err|q95=457.07
  tar_phase N=568 med=1.3045 s68=123.56 |err|q95=172.57
  tar_dt N=568 med=-0.14508 s68=2.8459 |err|q95=7.7225
truth_direct
  tar_dr N=615 med=72.375 s68=132.04 |err|q95=415.92
  tar_phase N=615 med=-0.34422 s68=38.131 |err|q95=142.22
  tar_dt N=615 med=0.057342 s68=0.58218 |err|q95=2.8107
truth_reflected
  tar_dr N=568 med=290.35 s68=161.41 |err|q95=447.91
  tar_phase N=568 med=19.494 s68=118.08 |err|q95=168.57
  tar_dt N=568 med=-0.26883 s68=2.3435 |err|q95=6.2692
```

B: event crossing efficiencies

```
baseline: read=1000, common denominator=1000
  any_plane                883 /     1000  fraction=0.883
  any_annulus              383 /     1000  fraction=0.383
  both_plane               506 /     1000  fraction=0.506
  both_annulus              32 /     1000  fraction=0.032
  both_compatible            0 /     1000  fraction=0
control: read=1000, common denominator=1000
  any_plane                794 /     1000  fraction=0.794
  any_annulus              313 /     1000  fraction=0.313
  both_plane               496 /     1000  fraction=0.496
  both_annulus              31 /     1000  fraction=0.031
  both_compatible            0 /     1000  fraction=0
momentum: read=1000, common denominator=1000
  any_plane                803 /     1000  fraction=0.803
  any_annulus              362 /     1000  fraction=0.362
  both_plane               513 /     1000  fraction=0.513
  both_annulus              38 /     1000  fraction=0.038
  both_compatible            0 /     1000  fraction=0
position: read=1000, common denominator=1000
  any_plane                785 /     1000  fraction=0.785
  any_annulus              311 /     1000  fraction=0.311
  both_plane               495 /     1000  fraction=0.495
  both_annulus              30 /     1000  fraction=0.03
  both_compatible            0 /     1000  fraction=0
truth: read=1000, common denominator=1000
  any_plane                804 /     1000  fraction=0.804
  any_annulus              375 /     1000  fraction=0.375
  both_plane               509 /     1000  fraction=0.509
  both_annulus              38 /     1000  fraction=0.038
```

B: event crossing efficiencies

```
both_compatible             1 /     1000  fraction=0.001
```

E: event availability and failure flags, including zero-track events

```
baseline
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
control
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
momentum
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
position
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
truth
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
```

E: status, missing states and covariance failures

```
baseline_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        165
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  116
  MissingRecoEntrance      1
  MissingTruth             0
  MultipleMCEntrances      3
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      857
  branch_missing           112
  covariance_invalid       165
  ent_valid                853
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
```

E: status, missing states and covariance failures

```
  propagated_covariance_invalid 164
  propagation_complete     853
  selected                 853
  status_0                 596
  status_16384             94
  status_16640             2
  status_20480             18
  status_20736             1
  status_20992             1
  status_4096              145
  tar_valid                741
baseline_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        144
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  99
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      9
  MultipleRecoEntrances    0
  NotElectron              0
```

E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      688
  branch_missing           90
  covariance_invalid       144
  ent_valid                679
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 143
  propagation_complete     679
  selected                 679
  status_0                 459
  status_16384             78
  status_16640             7
  status_20480             12
  status_20736             2
  status_4096              130
  tar_valid                589
baseline_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
```

E: status, missing states and covariance failures

```
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              1
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      1
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_2                 1
control_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
```

E: status, missing states and covariance failures

```
Exception                0
InvalidCovariance        264
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  68
MissingRecoEntrance      1
MissingTruth             0
MultipleMCEntrances      3
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       93
UnsupportedTrajectory    0
all                      857
branch_missing           68
covariance_invalid       264
ent_valid                853
input_covariance_invalid 0
partial_target_available 93
product_missing          0
propagated_covariance_invalid 262
propagation_complete     760
selected                 853
status_0                 448
status_16384             48
status_20480             20
```

E: status, missing states and covariance failures

```
  status_256               3
  status_4096              244
  status_512               1
  status_8192              93
  tar_valid                692
control_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        248
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  84
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      9
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       6
  UnsupportedTrajectory    0
  all                      688
  branch_missing           84
```

E: status, missing states and covariance failures

```
  covariance_invalid       248
  ent_valid                679
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 246
  propagation_complete     673
  selected                 679
  status_0                 366
  status_16384             59
  status_20480             19
  status_24576             6
  status_256               9
  status_4096              229
  tar_valid                595
control_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
```

E: status, missing states and covariance failures

```
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              1
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      1
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_2                 1
momentum_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        252
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
```

E: status, missing states and covariance failures

MissingProductionBranch	54
MissingRecoEntrance	1
MissingTruth	0
MultipleMCEntrances	3
MultipleRecoEntrances	0
NotElectron	0
NotPrimary	0
OutsideTarget	0
PropagationFailure	97
UnsupportedTrajectory	0
all	857
branch_missing	54
covariance_invalid	252
ent_valid	853
input_covariance_invalid	0
partial_target_available	95
product_missing	0
propagated_covariance_invalid	248
propagation_complete	756
selected	853
status_0	469
status_16384	35
status_20480	17
status_24576	2
status_256	3
status_4096	235
status_512	1
status_8192	95
tar_valid	704

E: status, missing states and covariance failures

```
momentum_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        267
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  70
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      9
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       7
  UnsupportedTrajectory    0
  all                      688
  branch_missing           70
  covariance_invalid       267
  ent_valid                679
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
```

E: status, missing states and covariance failures

```
  propagated_covariance_invalid 262
  propagation_complete     672
  selected                 679
  status_0                 360
  status_16384             46
  status_20480             17
  status_24576             6
  status_256               9
  status_28672             1
  status_4096              249
  tar_valid                609
momentum_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              1
```

E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      1
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_2                 1
position_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        275
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  73
  MissingRecoEntrance      1
  MissingTruth             0
  MultipleMCEntrances      3
```

E: status, missing states and covariance failures

```
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       96
  UnsupportedTrajectory    0
  all                      857
  branch_missing           73
  covariance_invalid       275
  ent_valid                853
  input_covariance_invalid 0
  partial_target_available 95
  product_missing          0
  propagated_covariance_invalid 271
  propagation_complete     757
  selected                 853
  status_0                 438
  status_16384             44
  status_20480             28
  status_24576             1
  status_256               3
  status_4096              247
  status_512               1
  status_8192              95
  tar_valid                685
position_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
```

E: status, missing states and covariance failures

```
ConfigurationDisabled    0
CrossingSearchFailure    0
Exception                0
InvalidCovariance        262
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  88
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      9
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       5
UnsupportedTrajectory    0
all                      688
branch_missing           88
covariance_invalid       262
ent_valid                679
input_covariance_invalid 0
partial_target_available 0
product_missing          0
propagated_covariance_invalid 258
propagation_complete     674
selected                 679
status_0                 353
```

E: status, missing states and covariance failures

```
  status_16384             59
  status_20480             24
  status_24576             5
  status_256               9
  status_4096              238
  tar_valid                591
position_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              1
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      1
```

E: status, missing states and covariance failures

```
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_2                 1
truth_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        267
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  56
  MissingRecoEntrance      1
  MissingTruth             0
  MultipleMCEntrances      3
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       93
```

E: status, missing states and covariance failures

```
  UnsupportedTrajectory    0
  all                      857
  branch_missing           56
  covariance_invalid       267
  ent_valid                853
  input_covariance_invalid 0
  partial_target_available 93
  product_missing          0
  propagated_covariance_invalid 264
  propagation_complete     760
  selected                 853
  status_0                 451
  status_16384             42
  status_20480             14
  status_256               3
  status_4096              253
  status_512               1
  status_8192              93
  tar_valid                704
truth_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        255
  MissingAssociation       0
  MissingClock             0
```

E: status, missing states and covariance failures

```
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  72
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      9
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       5
UnsupportedTrajectory    0
all                      688
branch_missing           72
covariance_invalid       255
ent_valid                679
input_covariance_invalid 0
partial_target_available 0
product_missing          0
propagated_covariance_invalid 255
propagation_complete     674
selected                 679
status_0                 376
status_16384             43
status_20480             24
status_24576             5
status_256               9
status_4096              231
tar_valid                607
```

E: status, missing states and covariance failures

```
truth_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              1
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      1
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
```

E: status, missing states and covariance failures

```
propagation_complete     0
status_2                 1
```

D: two-electron timing and vertex / baseline

```
pair rows=1000 denominator=606 valid=471
time_difference: N=471, NaN=0
  mean=0.78107 med=0.49039 std=4.0629 s68=3.4246
  |err| q68=3.5271 q95=8.8094; tail >10: 3.397%
time_difference_error: N=471, NaN=0
  mean=0.78107 med=0.49039 std=4.0629 s68=3.4246
  |err| q68=3.5271 q95=8.8094; tail >10: 3.397%
separation: N=471, NaN=0
  mean=392.36 med=399.5 std=202.63 s68=212.78
  |err| q68=462.43 q95=755.1; tail >400: 49.894%
opening_error_deg: N=471, NaN=0
  mean=79.401 med=84.24 std=45.781 s68=54.146
  |err| q68=110.01 q95=147.82; tail >30: 79.618%
max_vertex_dr: N=471, NaN=0
  mean=311.38 med=348.25 std=121.25 s68=133.01
  |err| q68=397.19 q95=452.3; tail >400: 30.786%
max_vertex_dt: N=471, NaN=0
  mean=2.8404 med=2.1129 std=2.4234 s68=1.9795
  |err| q68=3.1756 q95=7.8891; tail >10: 1.699%
```

D: two-electron timing and vertex / control

```
pair rows=1000 denominator=606 valid=490
time_difference: N=490, NaN=0
  mean=0.7728 med=0.48798 std=4.0265 s68=3.3943
  |err| q68=3.505 q95=8.7121; tail >10: 3.265%
time_difference_error: N=490, NaN=0
  mean=0.7728 med=0.48798 std=4.0265 s68=3.3943
  |err| q68=3.505 q95=8.7121; tail >10: 3.265%
separation: N=490, NaN=0
  mean=388.28 med=393.96 std=201.8 s68=211.07
  |err| q68=456.8 q95=752.2; tail >400: 48.776%
opening_error_deg: N=490, NaN=0
  mean=78.89 med=82.528 std=45.873 s68=54.688
  |err| q68=110.25 q95=147.8; tail >30: 79.184%
max_vertex_dr: N=490, NaN=0
  mean=309.62 med=342.73 std=122.52 s68=137.4
  |err| q68=398.58 q95=452.19; tail >400: 31.224%
max_vertex_dt: N=490, NaN=0
  mean=2.8101 med=2.1006 std=2.4066 s68=1.9614
  |err| q68=3.1714 q95=7.8676; tail >10: 1.633%
```

D: two-electron timing and vertex / momentum

```
pair rows=1000 denominator=606 valid=509
time_difference: N=509, NaN=0
  mean=0.59378 med=0.32929 std=3.4178 s68=2.8048
  |err| q68=2.6878 q95=7.5067; tail >10: 1.572%
time_difference_error: N=509, NaN=0
  mean=0.59378 med=0.32929 std=3.4178 s68=2.8048
  |err| q68=2.6878 q95=7.5067; tail >10: 1.572%
separation: N=509, NaN=0
  mean=358.17 med=367.04 std=193.69 s68=192.67
  |err| q68=440.59 q95=729.38; tail >400: 42.436%
opening_error_deg: N=509, NaN=0
  mean=77.213 med=78.302 std=45.794 s68=53.722
  |err| q68=106.57 q95=148.49; tail >30: 79.371%
max_vertex_dr: N=509, NaN=0
  mean=298.25 med=334.07 std=126.44 s68=142.06
  |err| q68=387.85 q95=455.48; tail >400: 28.094%
max_vertex_dt: N=509, NaN=0
  mean=2.4178 med=1.7869 std=2.0174 s68=1.7069
  |err| q68=2.5889 q95=6.3082; tail >10: 0.589%
```

D: two-electron timing and vertex / position

```
pair rows=1000 denominator=606 valid=489
time_difference: N=489, NaN=0
  mean=0.67683 med=0.45159 std=4.1226 s68=3.5413
  |err| q68=3.5857 q95=8.8576; tail >10: 3.067%
time_difference_error: N=489, NaN=0
  mean=0.67683 med=0.45159 std=4.1226 s68=3.5413
  |err| q68=3.5857 q95=8.8576; tail >10: 3.067%
separation: N=489, NaN=0
  mean=391.45 med=403.04 std=205.33 s68=218.1
  |err| q68=467.81 q95=773.67; tail >400: 50.716%
opening_error_deg: N=489, NaN=0
  mean=78.716 med=81.773 std=45.014 s68=53.35
  |err| q68=108.75 q95=145.38; tail >30: 79.755%
max_vertex_dr: N=489, NaN=0
  mean=311.43 med=348.48 std=123.57 s68=139.86
  |err| q68=400.62 q95=458.88; tail >400: 32.311%
max_vertex_dt: N=489, NaN=0
  mean=2.87 med=2.1673 std=2.4619 s68=2.0076
  |err| q68=3.2005 q95=8.0023; tail >10: 1.636%
```

D: two-electron timing and vertex / truth

```
pair rows=1000 denominator=606 valid=505
time_difference: N=505, NaN=0
  mean=0.42288 med=0.21876 std=3.2605 s68=2.4844
  |err| q68=2.491 q95=7.1986; tail >10: 1.584%
time_difference_error: N=505, NaN=0
  mean=0.42288 med=0.21876 std=3.2605 s68=2.4844
  |err| q68=2.491 q95=7.1986; tail >10: 1.584%
separation: N=505, NaN=0
  mean=355.54 med=368.9 std=192.67 s68=198.16
  |err| q68=432.84 q95=701.09; tail >400: 43.366%
opening_error_deg: N=505, NaN=0
  mean=78.975 med=79.827 std=46.127 s68=53.982
  |err| q68=106.98 q95=148.38; tail >30: 80.000%
max_vertex_dr: N=505, NaN=0
  mean=297.81 med=333.35 std=127.56 s68=142.25
  |err| q68=392.39 q95=451.15; tail >400: 28.515%
max_vertex_dt: N=505, NaN=0
  mean=2.2669 med=1.6475 std=2.0571 s68=1.7748
  |err| q68=2.4095 q95=6.3586; tail >10: 0.594%
```

E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
baseline_direct:ent_dpz:tar_phase
  n=741, circular_linear_rho=0.15141
baseline_direct:ent_dr:tar_dr
  n=741, pearson_r=0.15102
baseline_direct:ent_dt:tar_dt
  n=741, pearson_r=0.52569
baseline_direct:ent_pitch:tar_phase
  n=741, circular_linear_rho=0.27714
baseline_direct:tar_phase:tar_dr
  n=741, circular_linear_rho=0.96001
baseline_reflected:ent_dpz:tar_phase
  n=589, circular_linear_rho=0.031564
baseline_reflected:ent_dr:tar_dr
  n=589, pearson_r=0.044209
baseline_reflected:ent_dt:tar_dt
  n=589, pearson_r=0.23034
baseline_reflected:ent_pitch:tar_phase
  n=589, circular_linear_rho=0.080257
baseline_reflected:tar_phase:tar_dr
  n=589, circular_linear_rho=0.96267
control_direct:ent_dpz:tar_phase
  n=692, circular_linear_rho=0.13868
control_direct:ent_dr:tar_dr
  n=692, pearson_r=0.15063
control_direct:ent_dt:tar_dt
  n=692, pearson_r=0.45863
control_direct:ent_pitch:tar_phase
```

E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=692, circular_linear_rho=0.27252
control_direct:tar_phase:tar_dr
  n=692, circular_linear_rho=0.96023
control_reflected:ent_dpz:tar_phase
  n=595, circular_linear_rho=0.031728
control_reflected:ent_dr:tar_dr
  n=595, pearson_r=0.043693
control_reflected:ent_dt:tar_dt
  n=595, pearson_r=0.23058
control_reflected:ent_pitch:tar_phase
  n=595, circular_linear_rho=0.082768
control_reflected:tar_phase:tar_dr
  n=595, circular_linear_rho=0.96318
momentum_direct:ent_dpz:tar_phase
  n=704, circular_linear_rho=0.025425
momentum_direct:ent_dr:tar_dr
  n=704, pearson_r=0.047222
momentum_direct:ent_dt:tar_dt
  n=704, pearson_r=0.48729
momentum_direct:ent_pitch:tar_phase
  n=704, circular_linear_rho=0.053773
momentum_direct:tar_phase:tar_dr
  n=704, circular_linear_rho=0.96048
momentum_reflected:ent_dpz:tar_phase
  n=609, circular_linear_rho=0.071542
momentum_reflected:ent_dr:tar_dr
  n=609, pearson_r=-0.032436
```

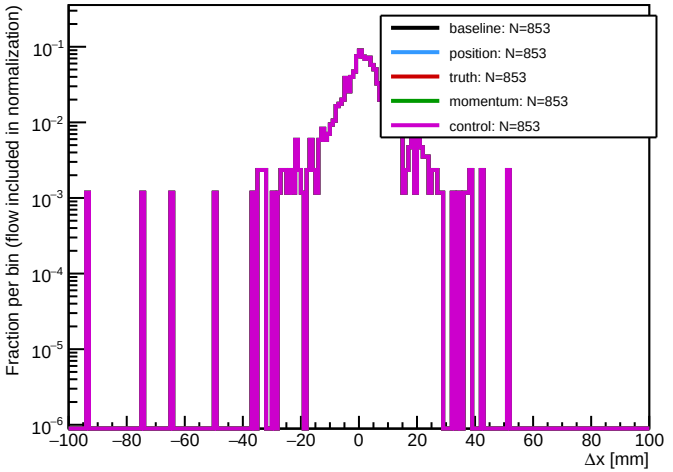
E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
momentum_reflected:ent_dt:tar_dt
  n=609, pearson_r=0.28874
momentum_reflected:ent_pitch:tar_phase
  n=609, circular_linear_rho=0.052302
momentum_reflected:tar_phase:tar_dr
  n=609, circular_linear_rho=0.96194
position_direct:ent_dpz:tar_phase
  n=685, circular_linear_rho=0.13534
position_direct:ent_dr:tar_dr
  n=685, pearson_r=0.10335
position_direct:ent_dt:tar_dt
  n=685, pearson_r=0.45225
position_direct:ent_pitch:tar_phase
  n=685, circular_linear_rho=0.30661
position_direct:tar_phase:tar_dr
  n=685, circular_linear_rho=0.95853
position_reflected:ent_dpz:tar_phase
  n=591, circular_linear_rho=0.041765
position_reflected:ent_dr:tar_dr
  n=591, pearson_r=0.043933
position_reflected:ent_dt:tar_dt
  n=591, pearson_r=0.21694
position_reflected:ent_pitch:tar_phase
  n=591, circular_linear_rho=0.050605
position_reflected:tar_phase:tar_dr
  n=591, circular_linear_rho=0.95902
truth_direct:ent_dpz:tar_phase
```

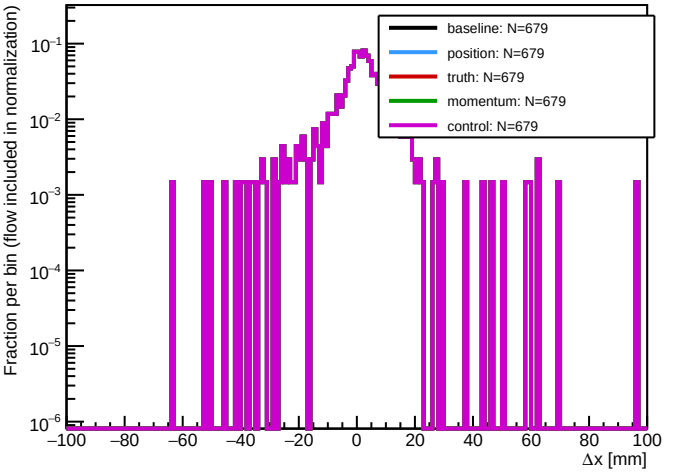
E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=704, circular_linear_rho=0.037326
truth_direct:ent_dr:tar_dr
  n=704, pearson_r=-0.0094023
truth_direct:ent_dt:tar_dt
  n=704, pearson_r=0.031332
truth_direct:ent_pitch:tar_phase
  n=704, circular_linear_rho=0.035741
truth_direct:tar_phase:tar_dr
  n=704, circular_linear_rho=0.96209
truth_reflected:ent_dpz:tar_phase
  n=607, circular_linear_rho=0.063025
truth_reflected:ent_dr:tar_dr
  n=607, pearson_r=0.052067
truth_reflected:ent_dt:tar_dt
  n=607, pearson_r=0.060185
truth_reflected:ent_pitch:tar_phase
  n=607, circular_linear_rho=0.035145
truth_reflected:tar_phase:tar_dr
  n=607, circular_linear_rho=0.96684
```

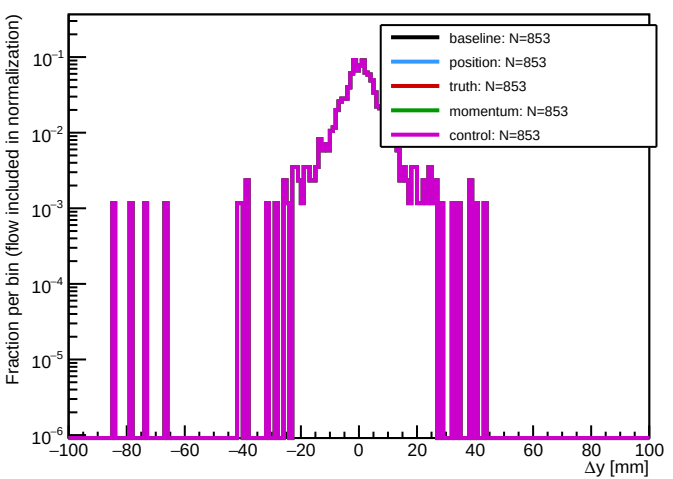
Entrance dx [mm] / direct



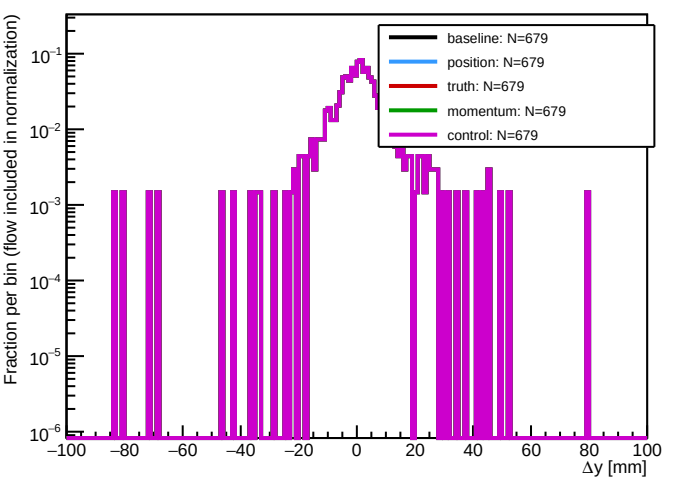
Entrance dx [mm] / reflected



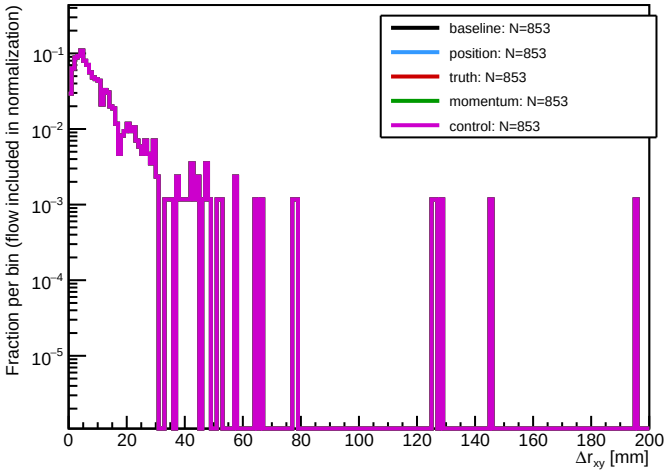
Entrance dy [mm] / direct



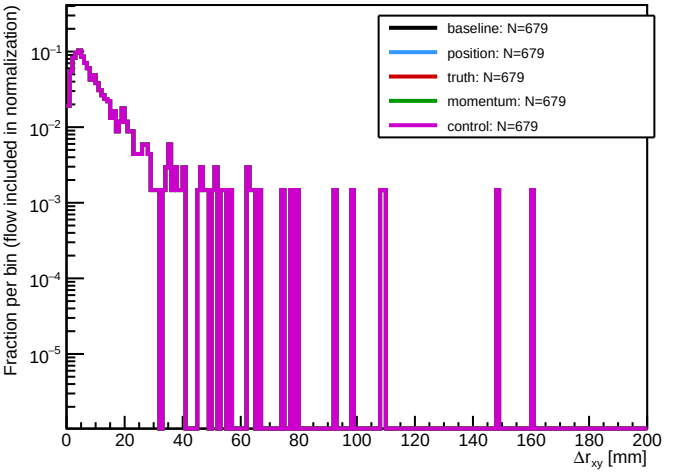
Entrance dy [mm] / reflected



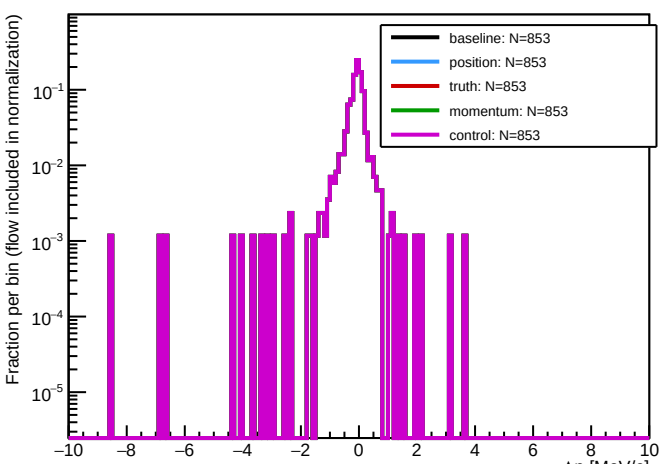
Entrance transverse error [mm] / direct



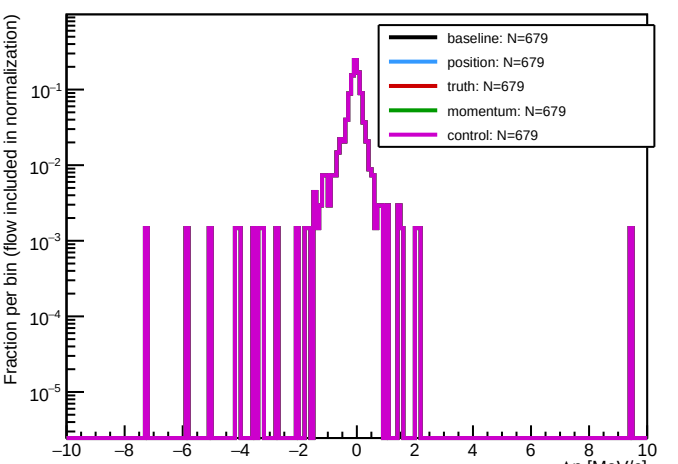
Entrance transverse error [mm] / reflected



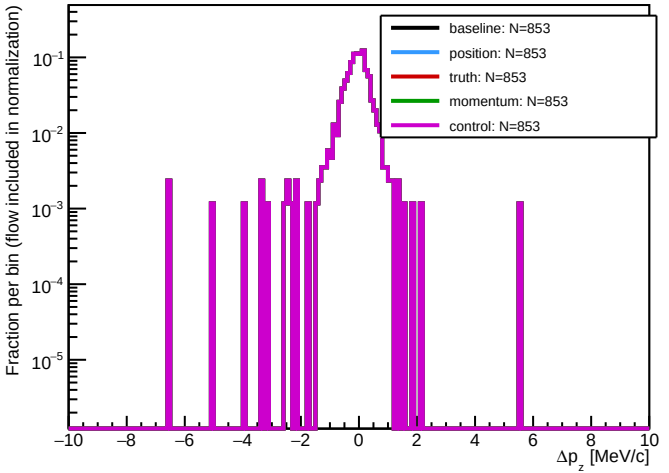
Entrance momentum error [MeV/c] / direct



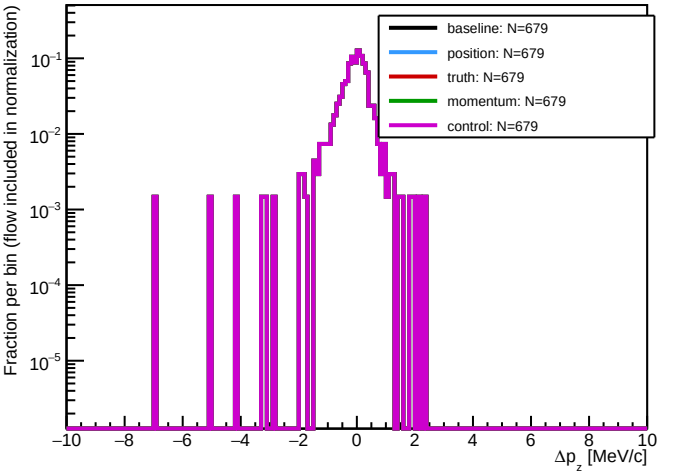
Entrance momentum error [MeV/c] / reflected



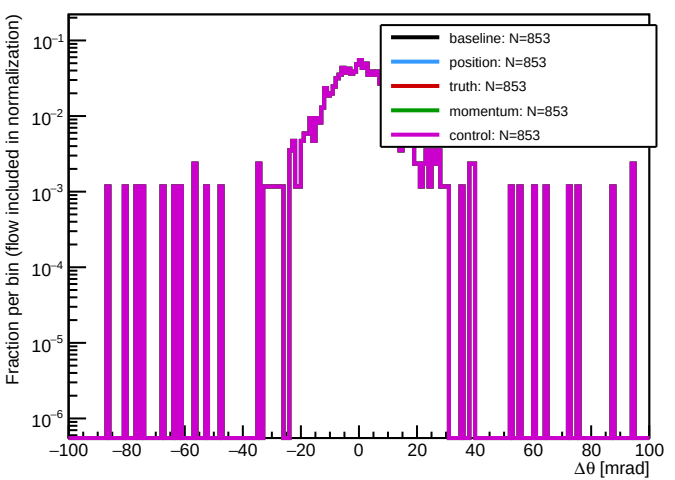
Entrance pz error [MeV/c] / direct



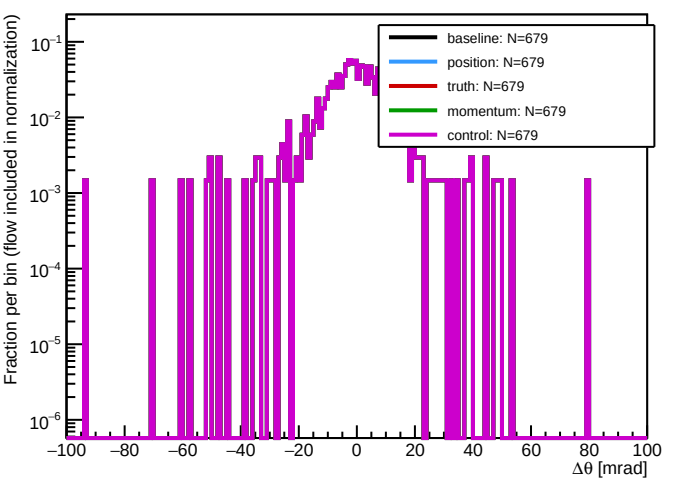
Entrance pz error [MeV/c] / reflected



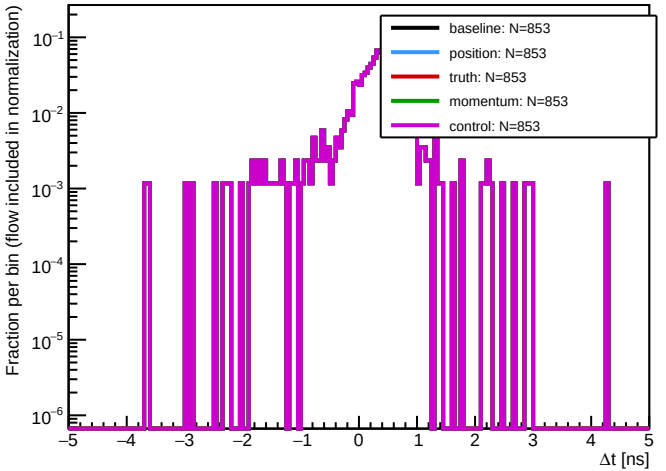
Entrance pitch error [mrad] / direct



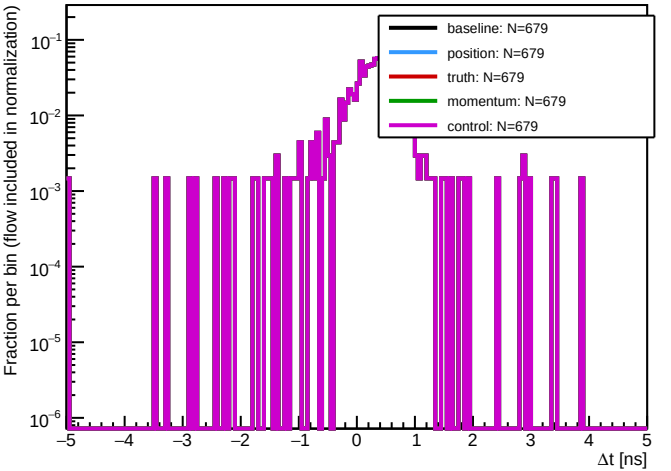
Entrance pitch error [mrad] / reflected



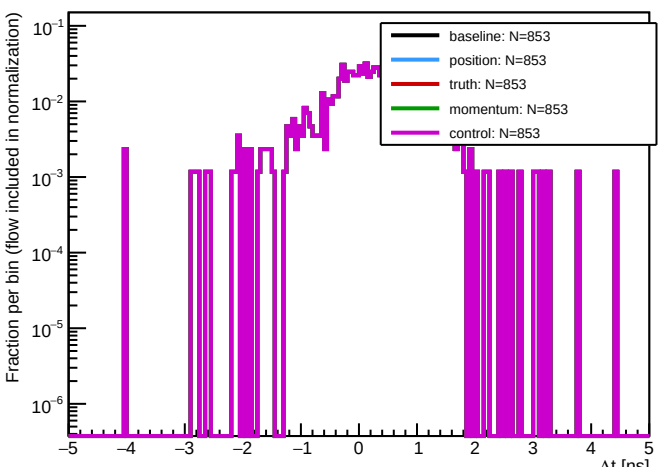
Entrance time error, clock corrected [ns] / direct



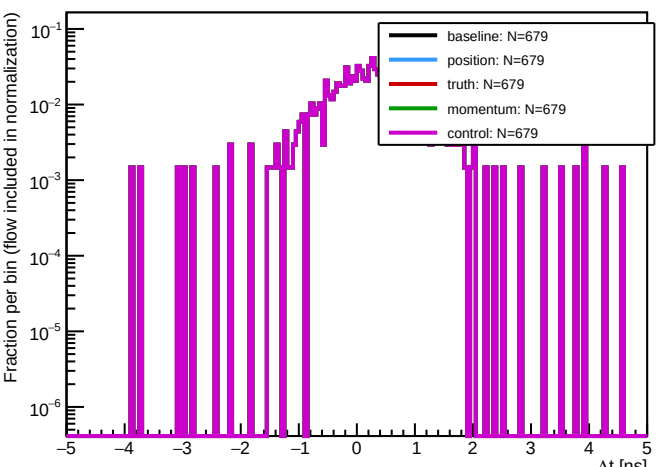
Entrance time error, clock corrected [ns] / reflected

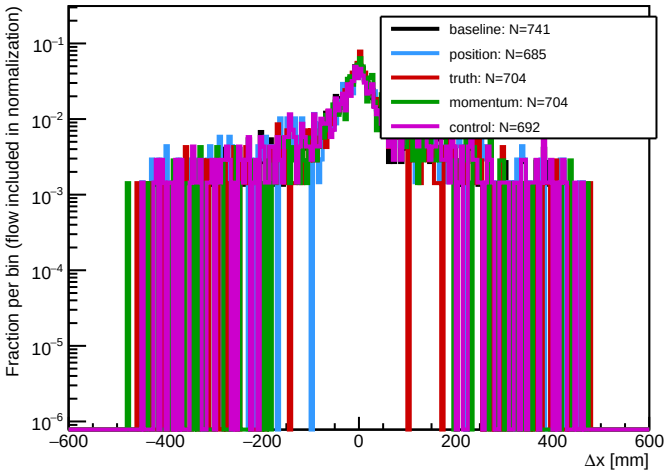


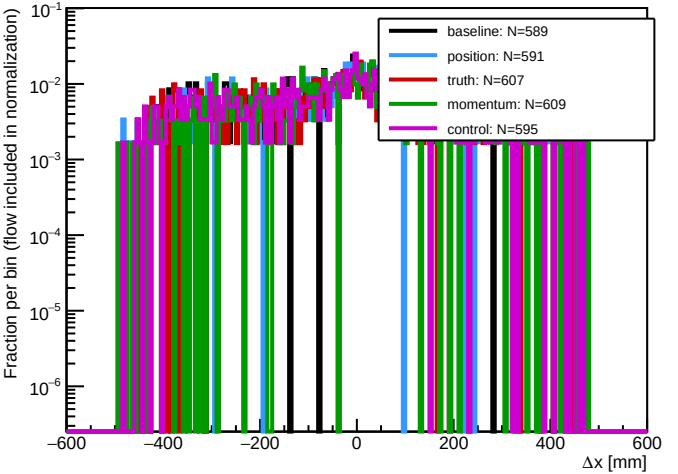
Entrance time error, bunch clock included [ns] / direct

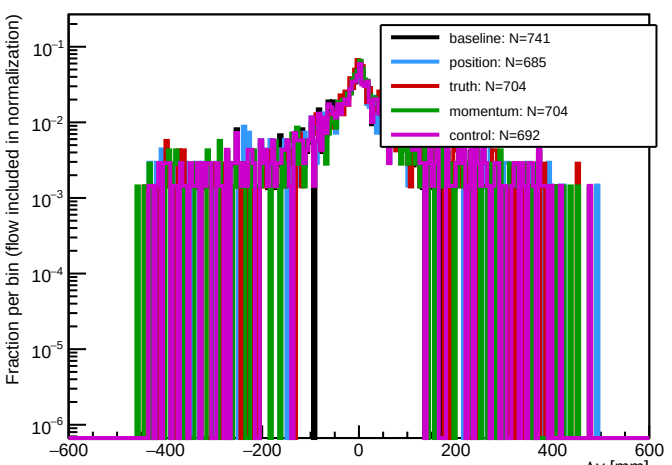


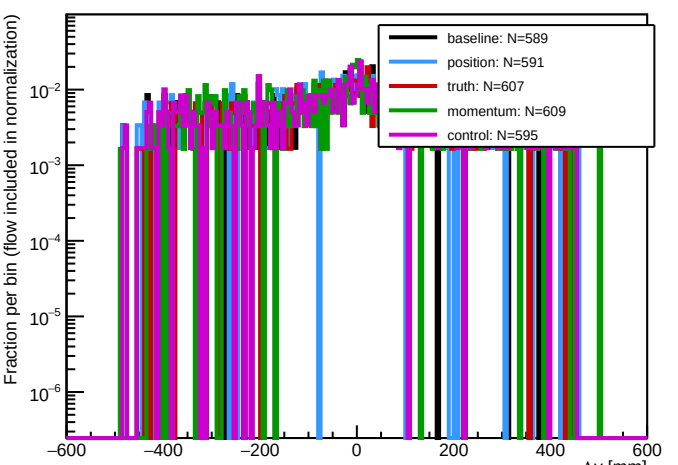
Entrance time error, bunch clock included [ns] / reflected



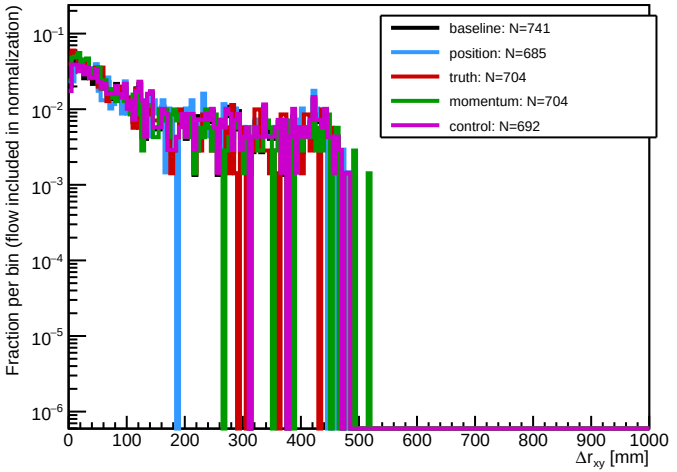

Target dx [mm] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
Δx [mm]


Target dx [mm] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
Δx [mm]

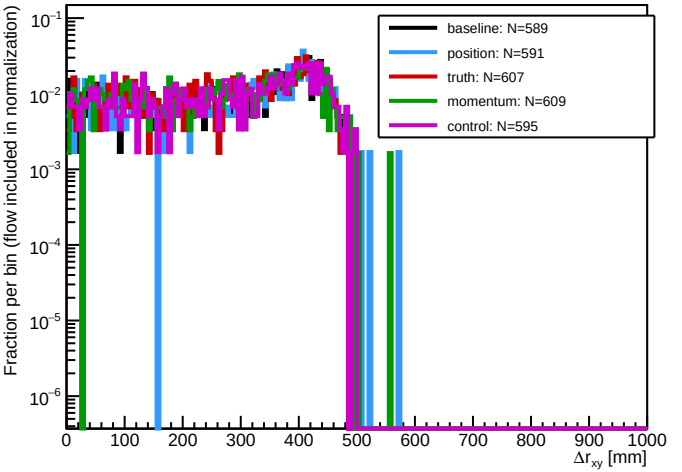

Target dy [mm] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
Δy [mm]


Target dy [mm] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
Δy [mm]

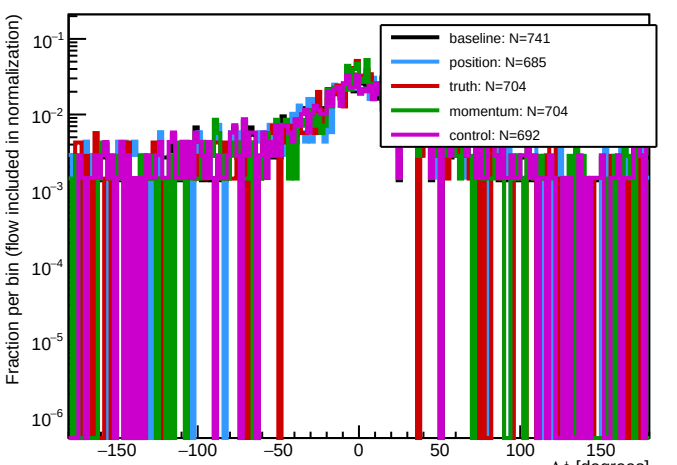
Target transverse error [mm] / direct



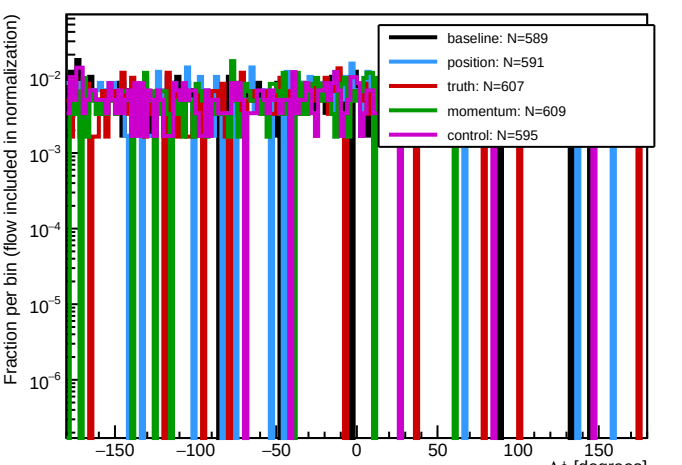
Target transverse error [mm] / reflected

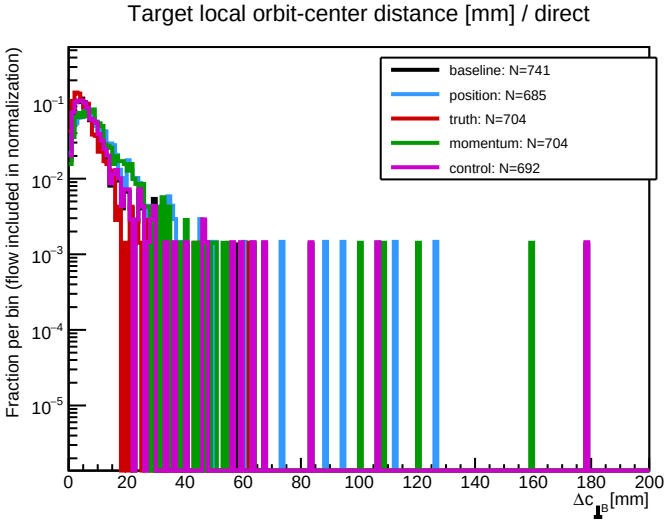


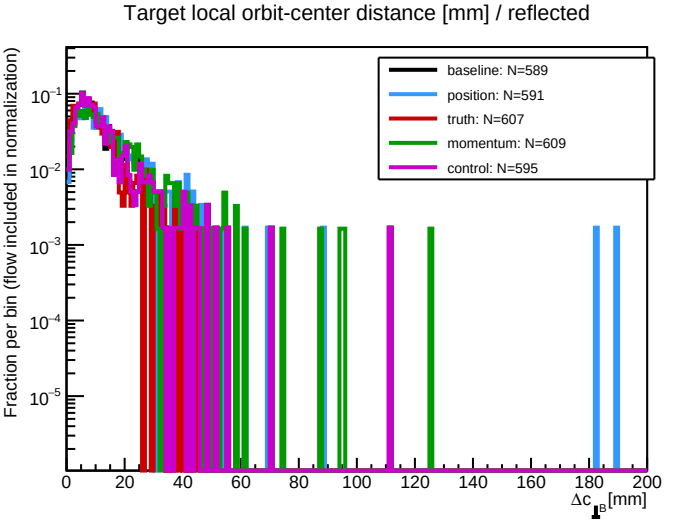
Target wrapped phase error [degrees] / direct

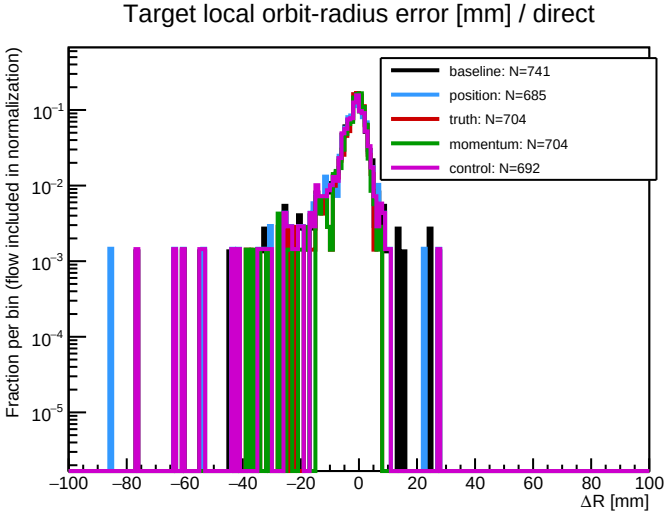


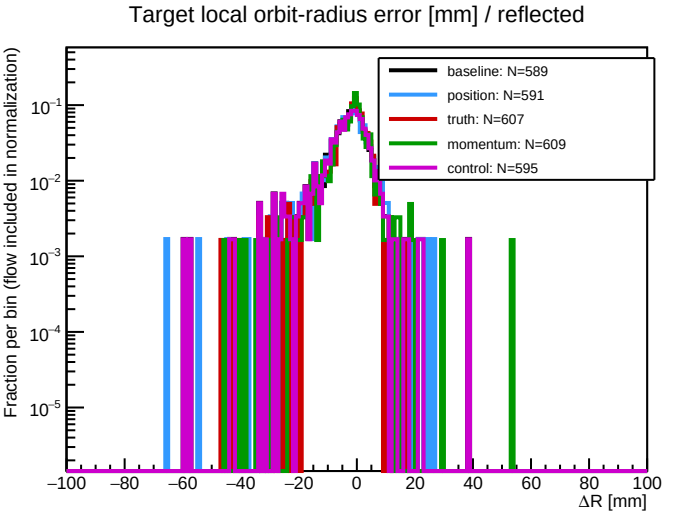
Target wrapped phase error [degrees] / reflected

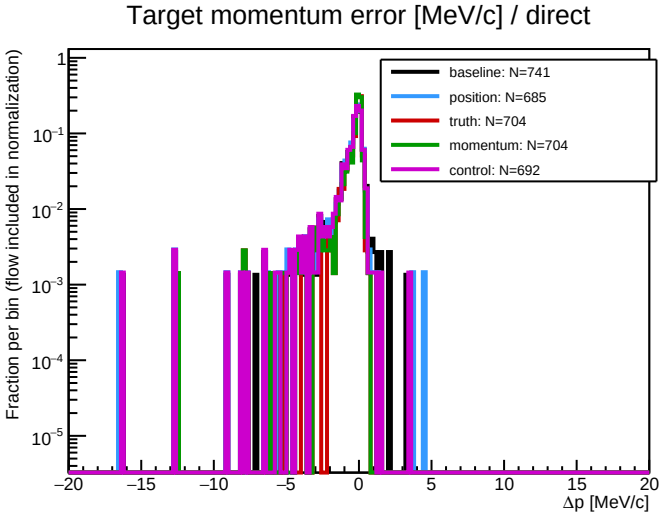


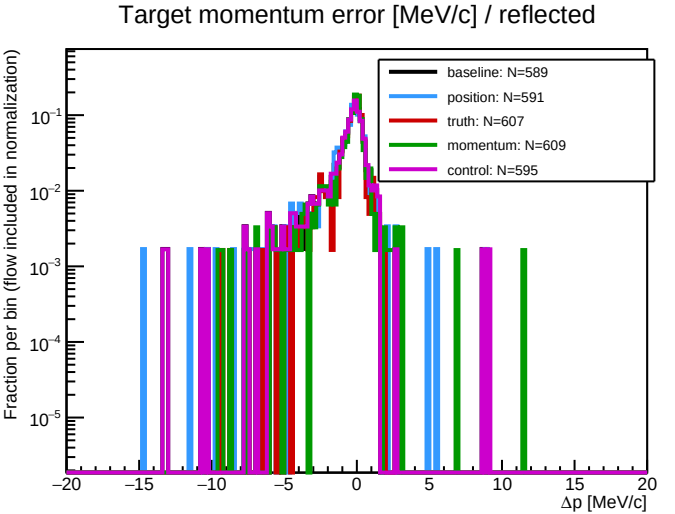

Target local orbit-center distance [mm] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
$\Delta c_{\perp}^{B}$ [mm]

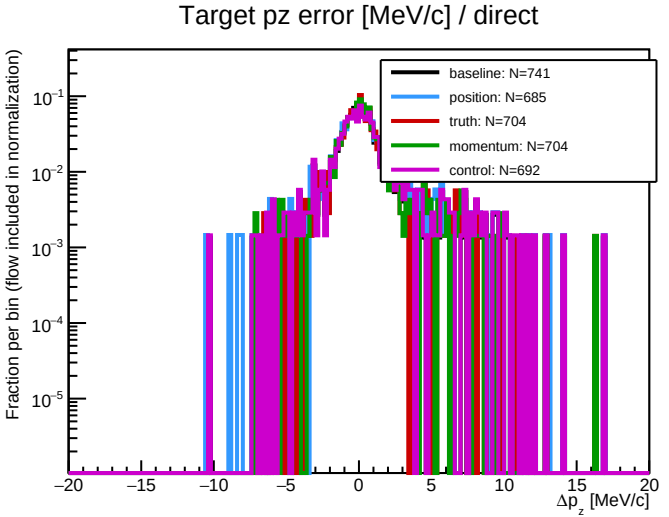

Target local orbit-center distance [mm] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
$\Delta c_{\perp}^{B}$ [mm]

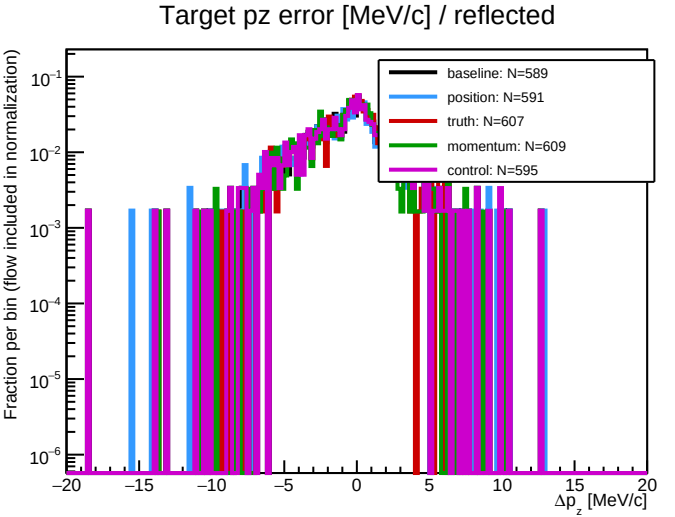

Target local orbit-radius error [mm] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
ΔR [mm]

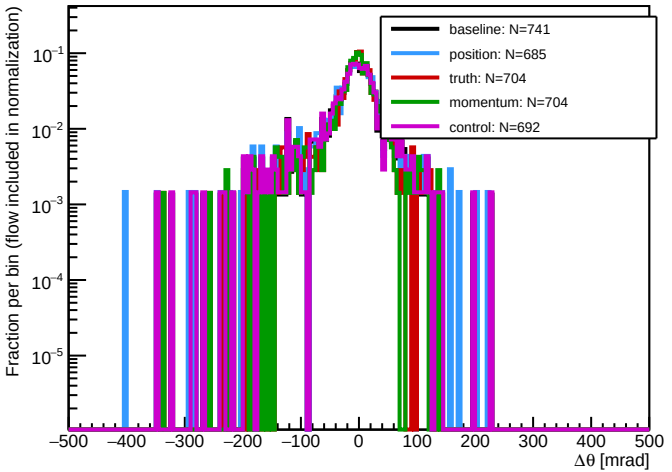

Target local orbit-radius error [mm] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
ΔR [mm]

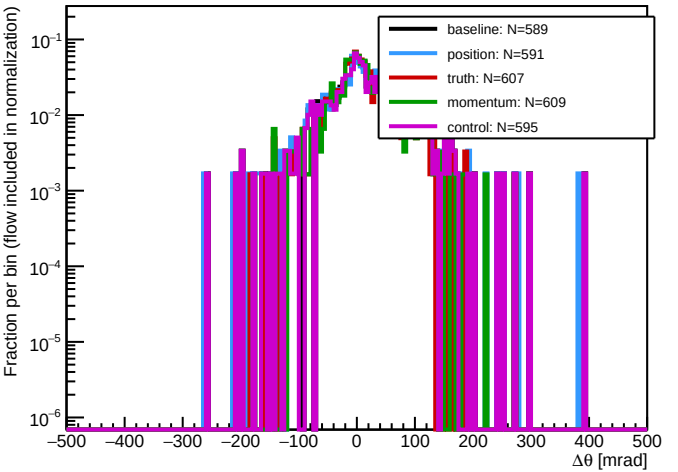

Target momentum error [MeV/c] / direct
Fraction per bin (flow included in normalization)
Δp [MeV/c]
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692

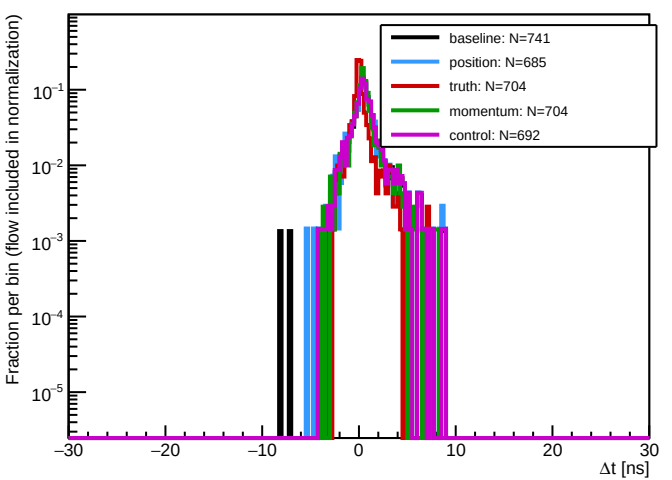

Target momentum error [MeV/c] / reflected
Fraction per bin (flow included in normalization)
Δp [MeV/c]
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595

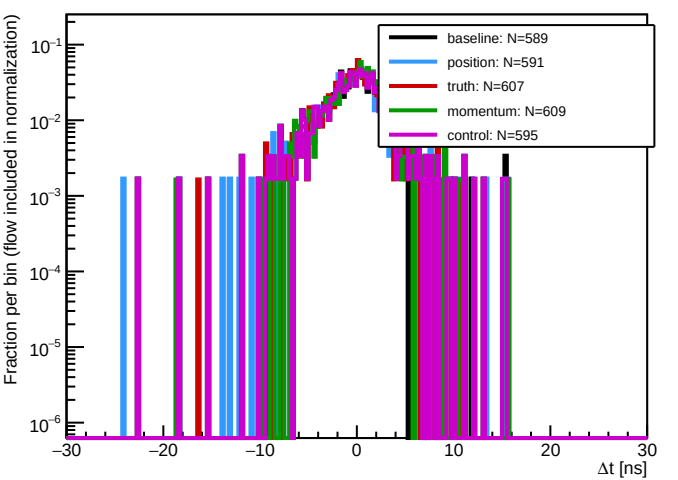

Target pz error [MeV/c] / direct
Fraction per bin (flow included in normalization)
Δp_z [MeV/c]
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692

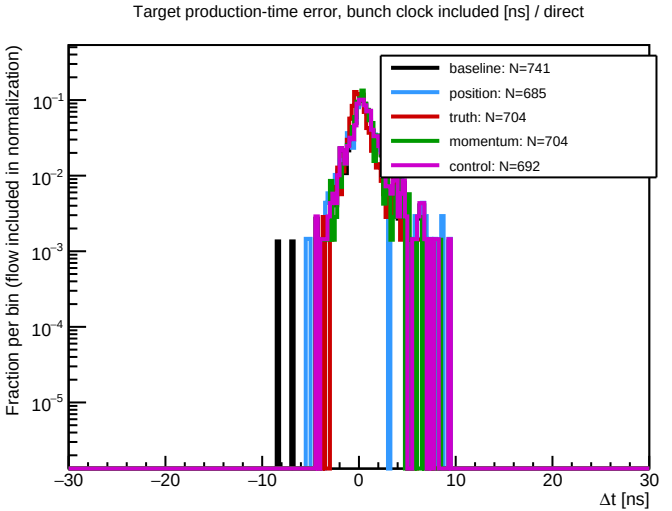

Target pz error [MeV/c] / reflected
Fraction per bin (flow included in normalization)
Δp_z [MeV/c]
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595

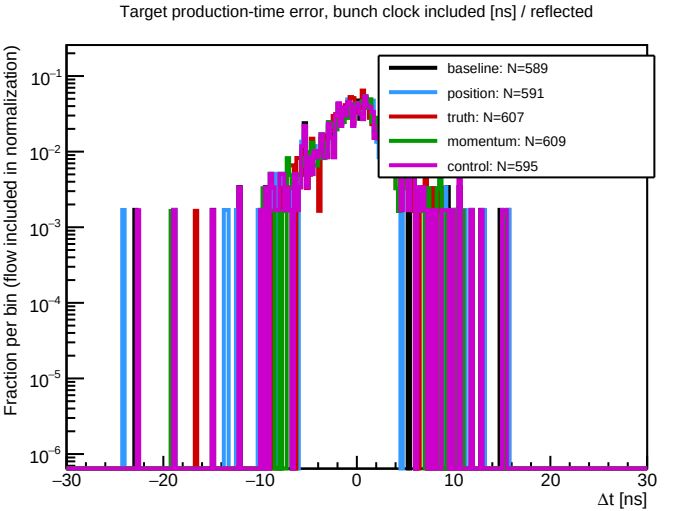

Target pitch error [mrad] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
Δθ [mrad]

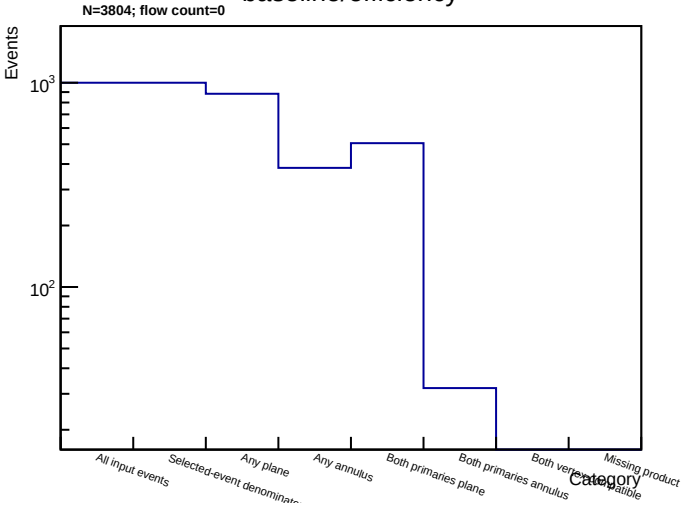

Target pitch error [mrad] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
Δθ [mrad]

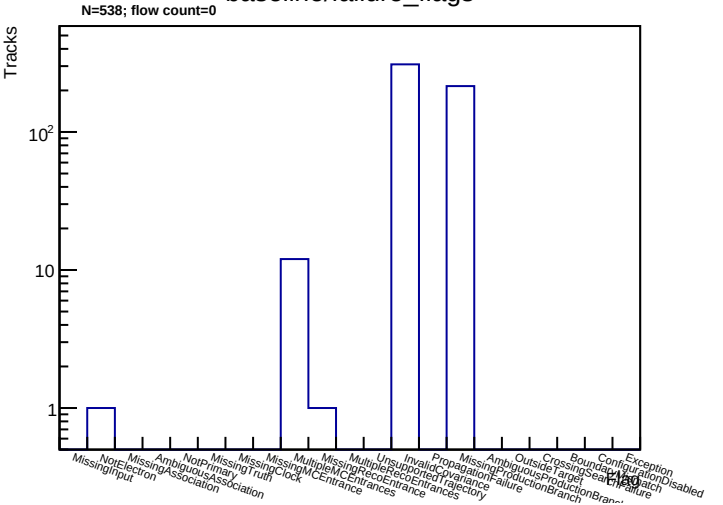

Target production-time error, clock corrected [ns] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
Δt [ns]

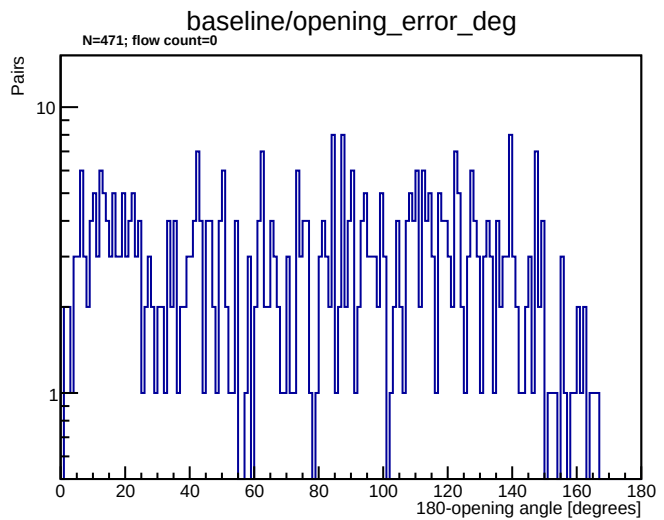

Target production-time error, clock corrected [ns] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
Δt [ns]

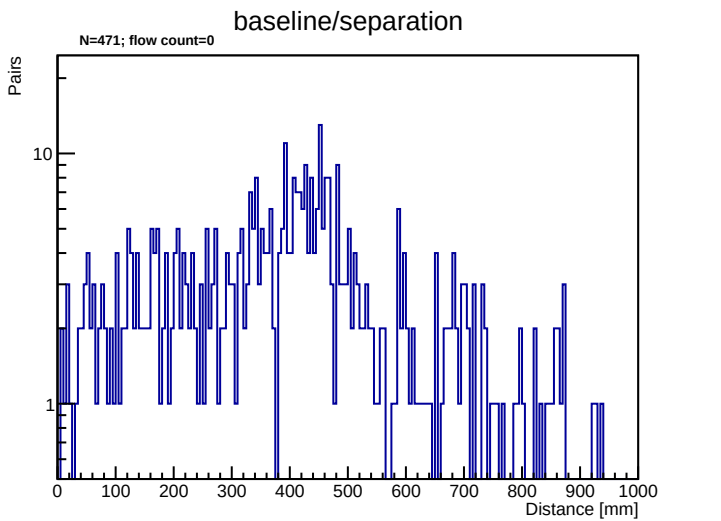

Target production-time error, bunch clock included [ns] / direct
baseline: N=741
position: N=685
truth: N=704
momentum: N=704
control: N=692
Fraction per bin (flow included in normalization)
Δt [ns]

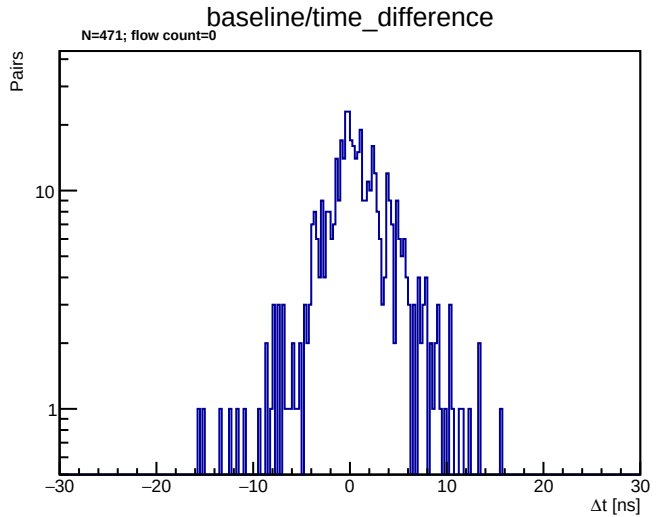

Target production-time error, bunch clock included [ns] / reflected
baseline: N=589
position: N=591
truth: N=607
momentum: N=609
control: N=595
Fraction per bin (flow included in normalization)
Δt [ns]

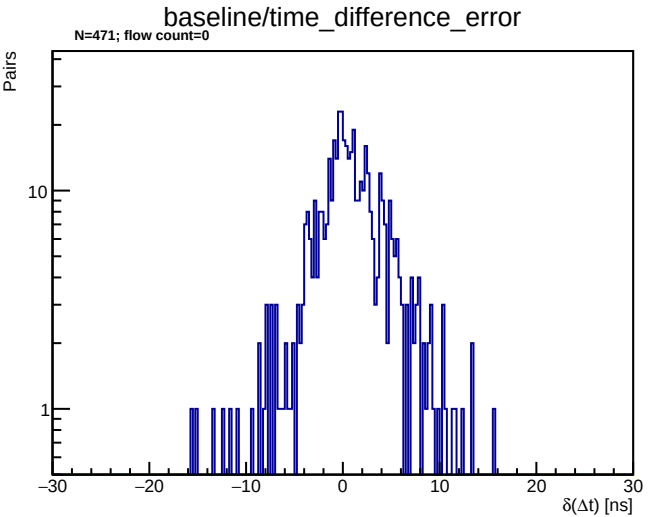

baseline/efficiency
N=3804; flow count=0
Events
10^3
10^2
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

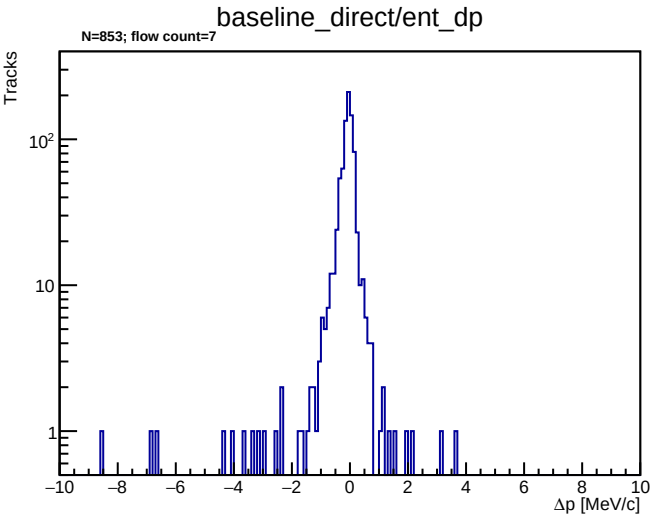

baseline/failure_flags
N=538; flow count=0
Tracks
10^2
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrances
MissingRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeed
BoundaryMismatch
ConfigurationDisabled
Exception
Failed

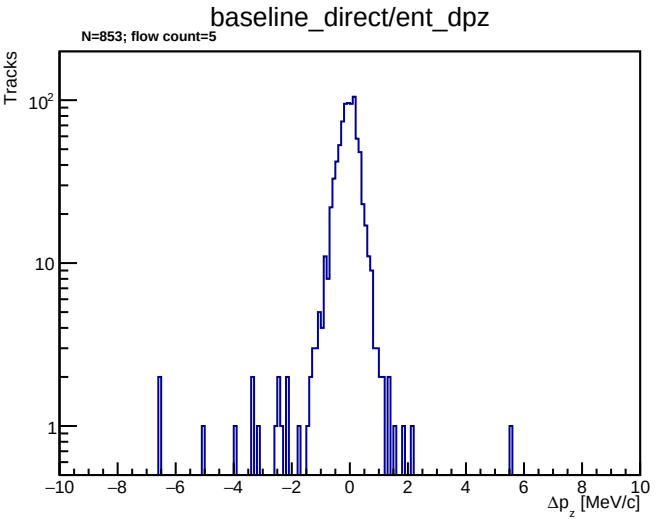

baseline/opening_error_deg
N=471; flow count=0
Pairs
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

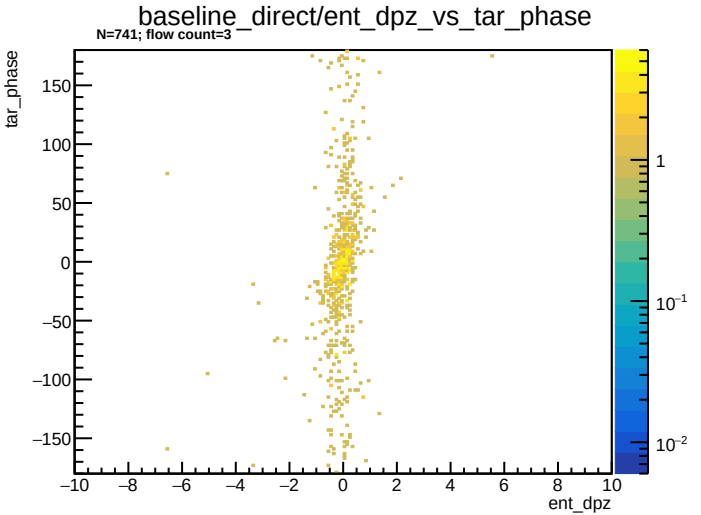

baseline/separation
N=471; flow count=0
Pairs
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

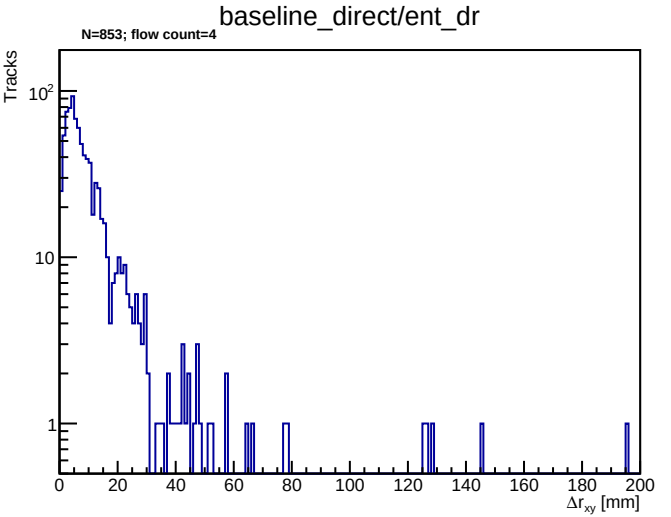

baseline/time_difference
N=471; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

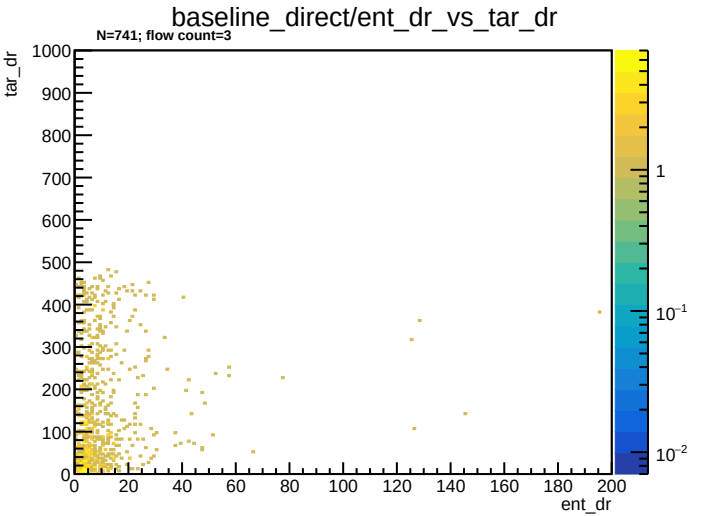

baseline/time_difference_error
N=471; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

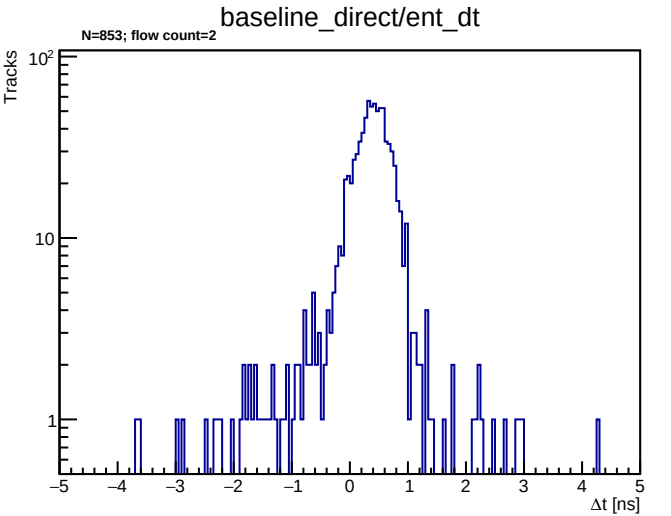

baseline_direct/ent_dp
N=853; flow count=7
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp [MeV/c]

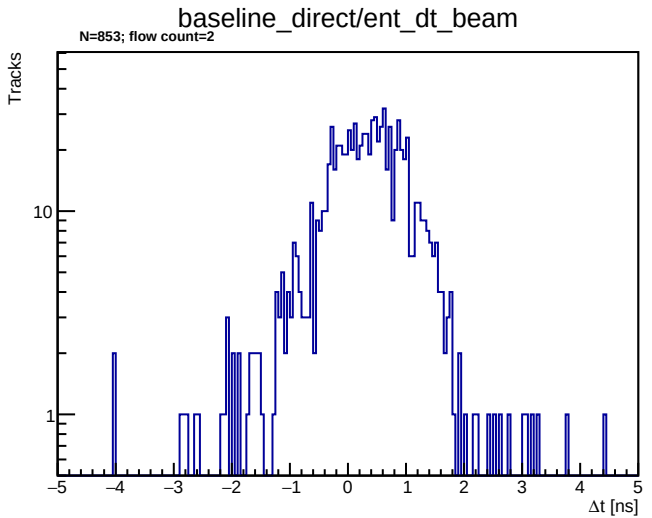

baseline_direct/ent_dpz
N=853; flow count=5
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp_z [MeV/c]

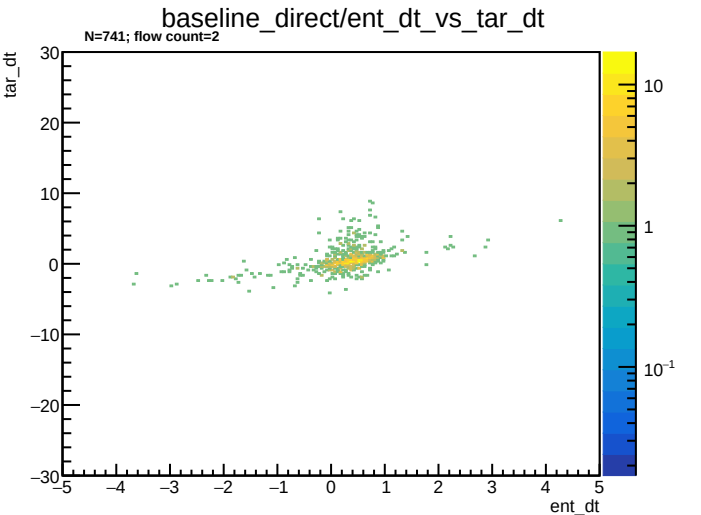

baseline_direct/ent_dpz_vs_tar_phase
N=741; flow count=3
tar_phase
ent_dpz

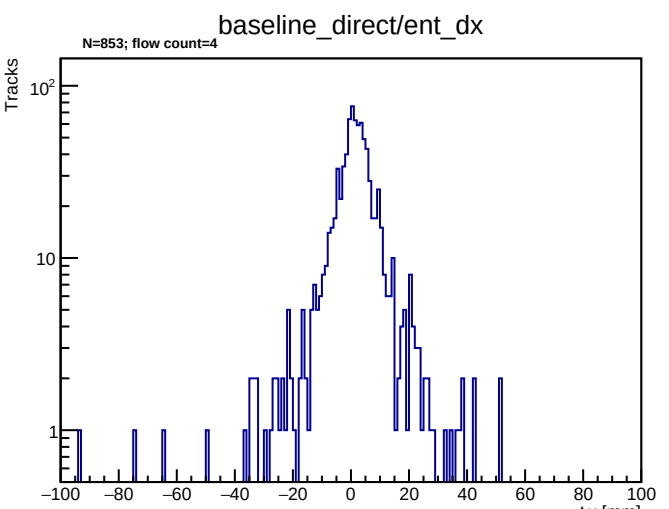

baseline_direct/ent_dr
N=853; flow count=4
Tracks
Δr_xy [mm]

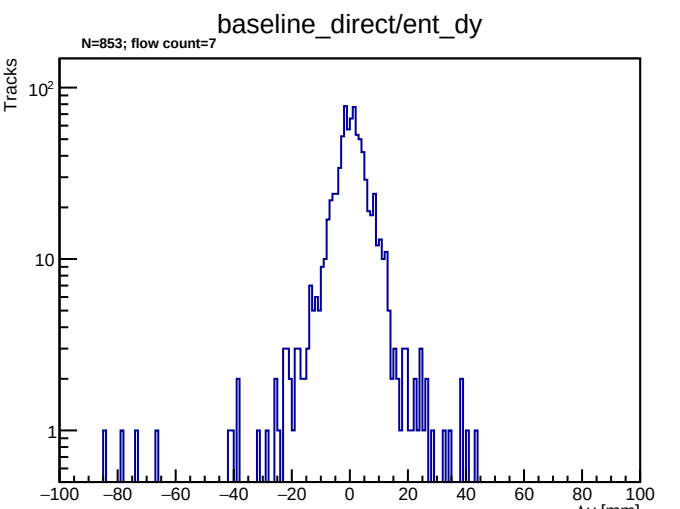

baseline_direct/ent_dr_vs_tar_dr
N=741; flow count=3
tar_dr
ent_dr

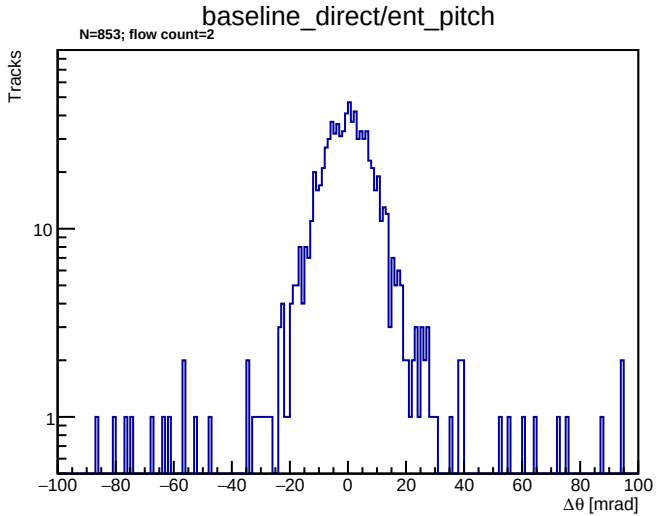

baseline_direct/ent_dt
N=853; flow count=2
Tracks
Δt [ns]

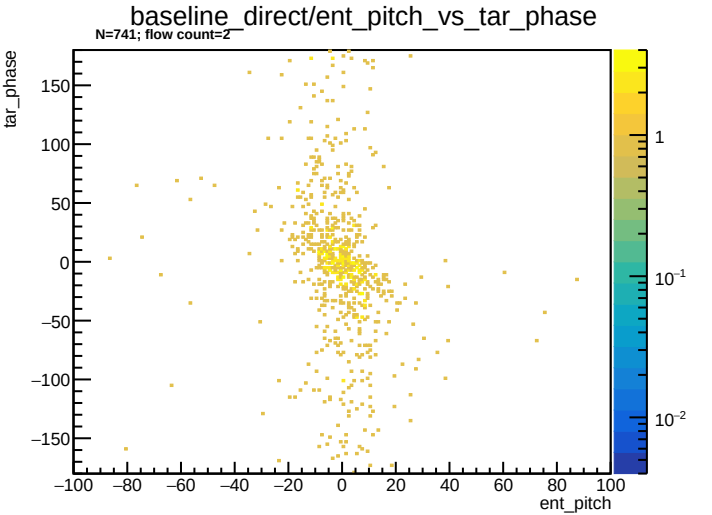

baseline_direct/ent_dt_beam
N=853; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

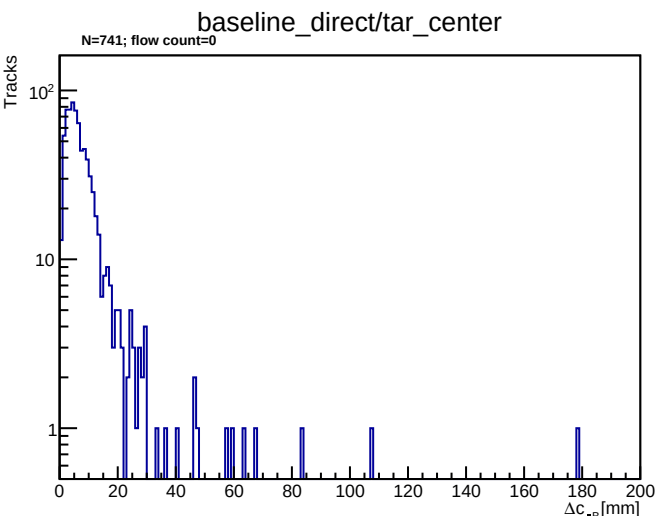

baseline_direct/ent_dt_vs_tar_dt
N=741; flow count=2
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
10
1
10⁻¹

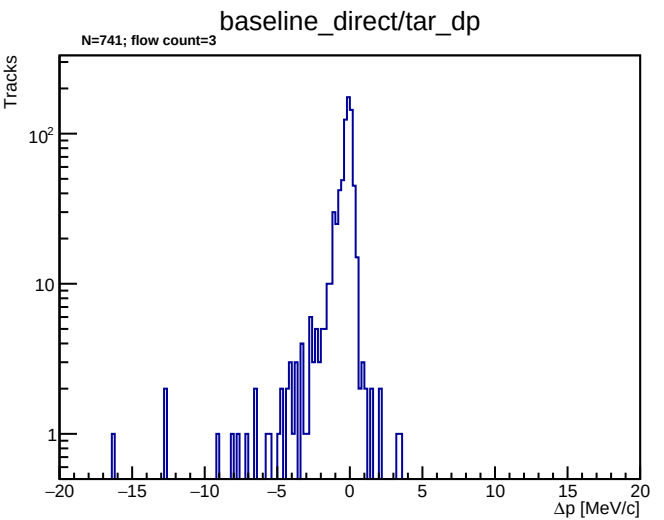

baseline_direct/ent_dx
N=853; flow count=4
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

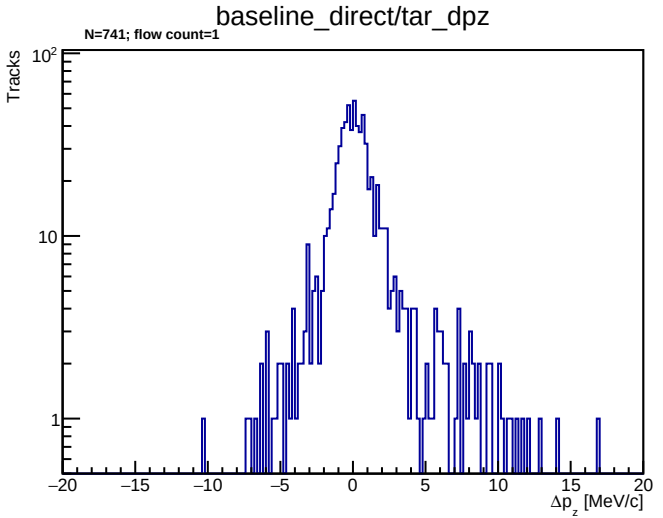

baseline_direct/ent_dy
N=853; flow count=7
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

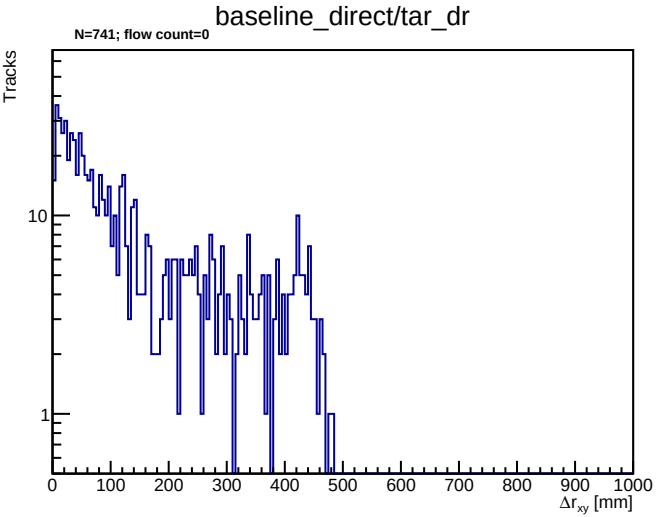

baseline_direct/ent_pitch
N=853; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

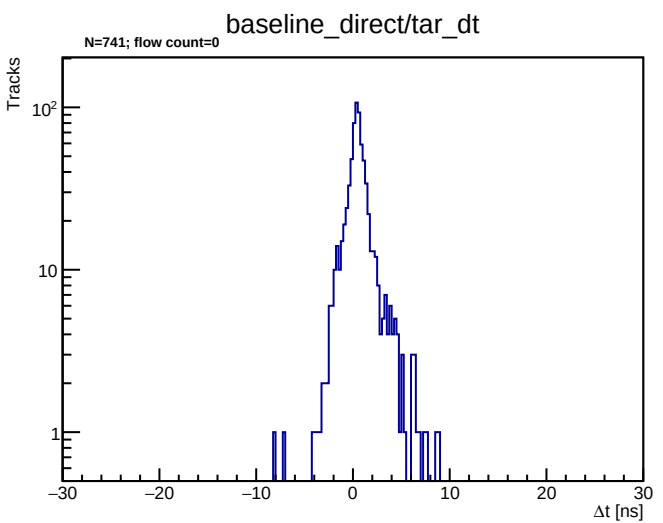

baseline_direct/ent_pitch_vs_tar_phase
N=741; flow count=2
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^−1
10^−2

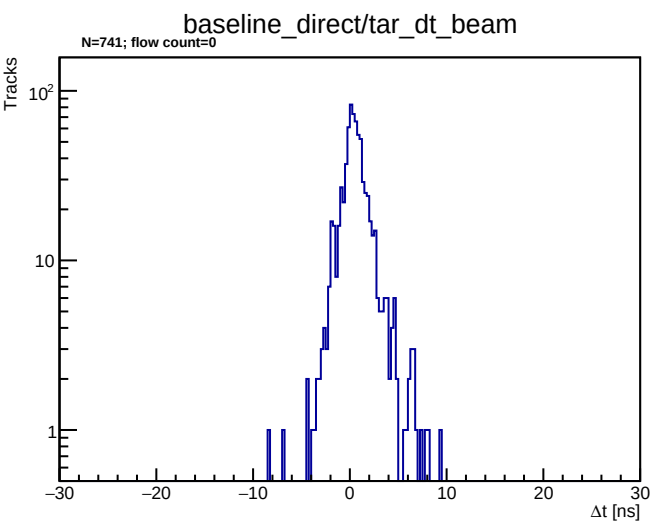

baseline_direct/tar_center
N=741; flow count=0
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_TB [mm]

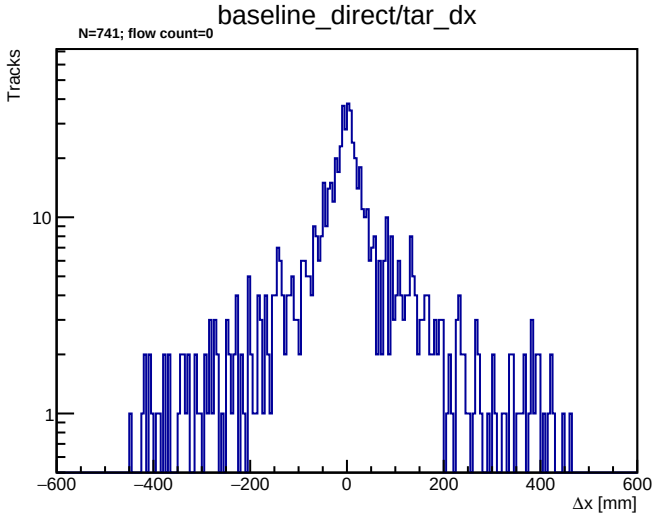

baseline_direct/tar_dp
N=741; flow count=3
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

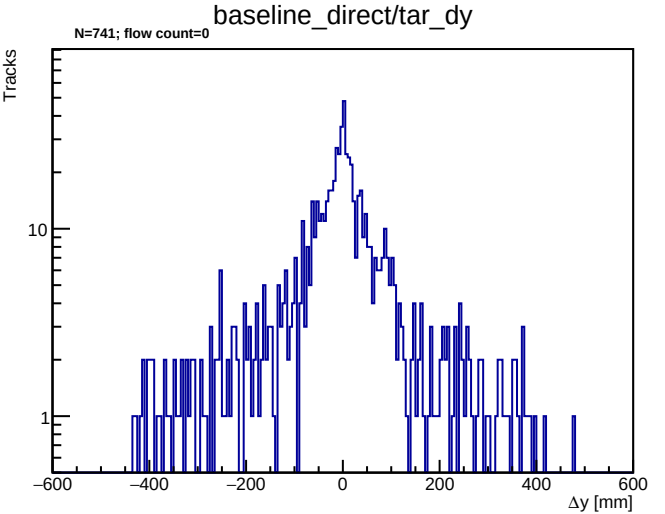

baseline_direct/tar_dpz
N=741; flow count=1
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

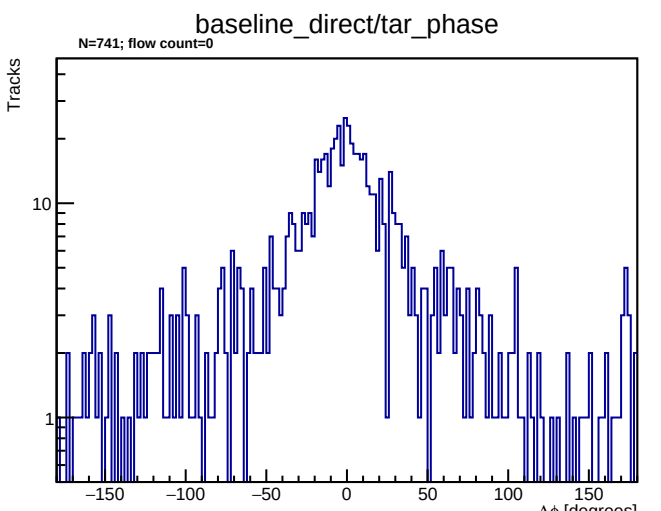

baseline_direct/tar_dr
N=741; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

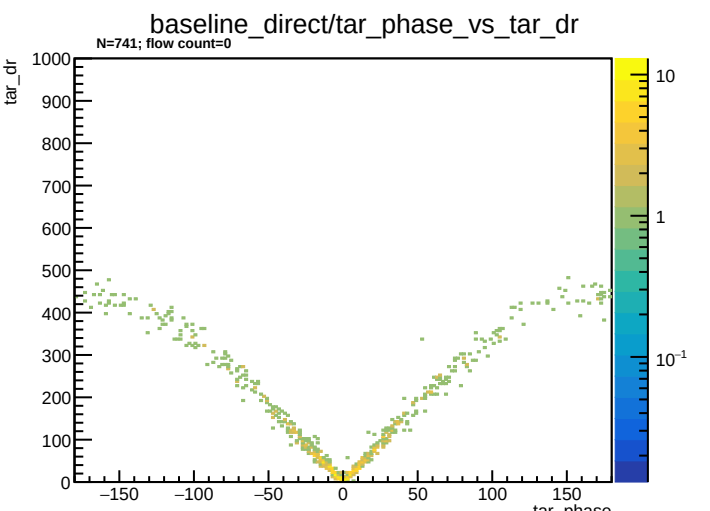

baseline_direct/tar_dt
N=741; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

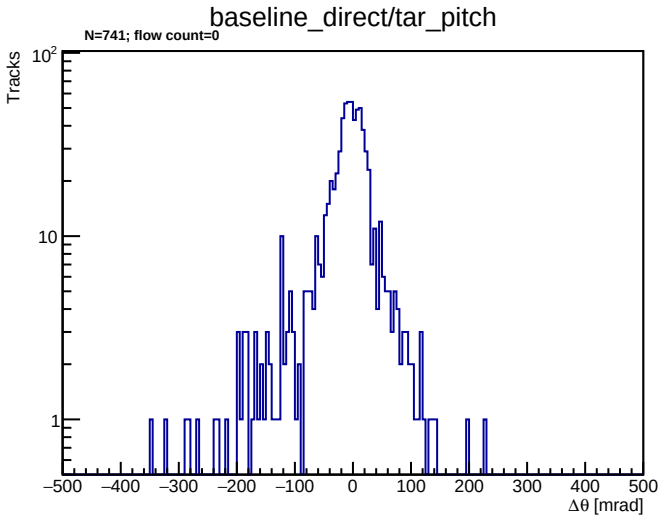

baseline_direct/tar_dt_beam
N=741; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

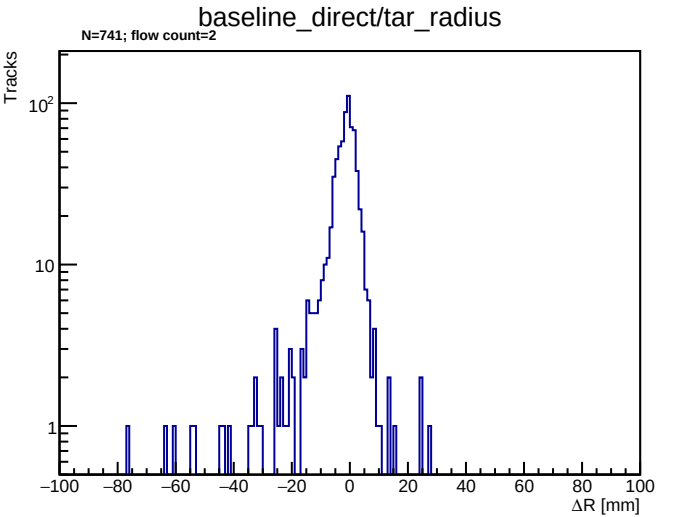

baseline_direct/tar_dx
N=741; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

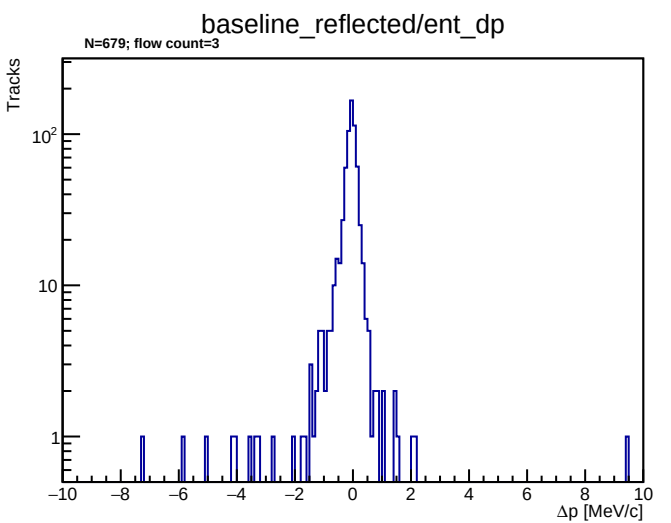

baseline_direct/tar_dy
N=741; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

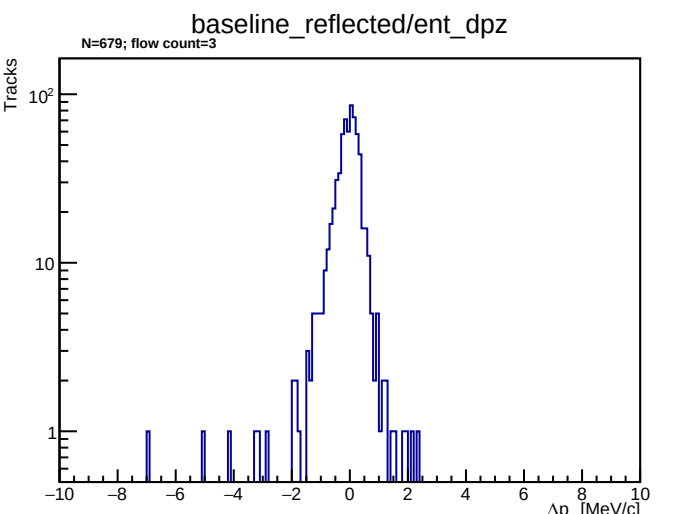

baseline_direct/tar_phase
N=741; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

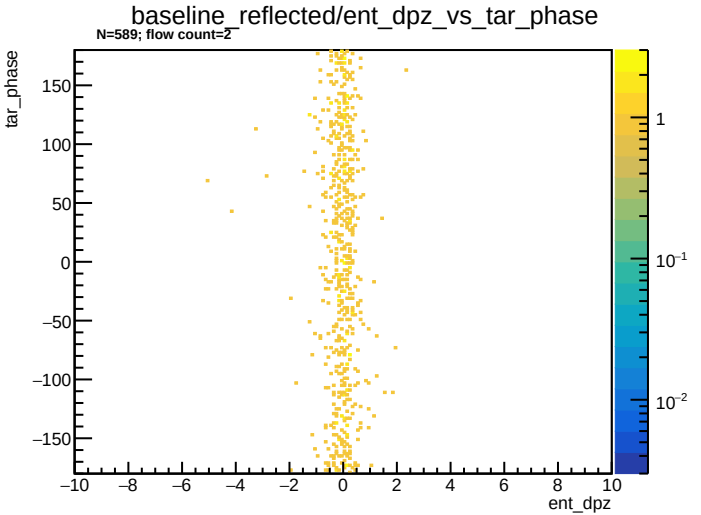

baseline_direct/tar_phase_vs_tar_dr
N=741; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
10
1
10⁻¹

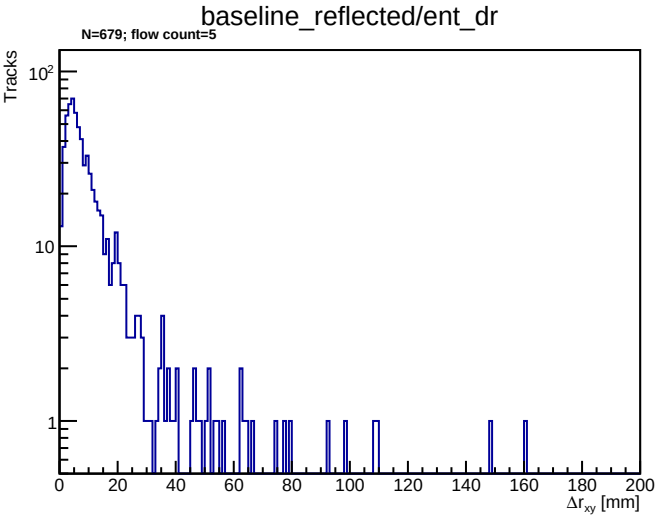

baseline_direct/tar_pitch
N=741; flow count=0
Tracks
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

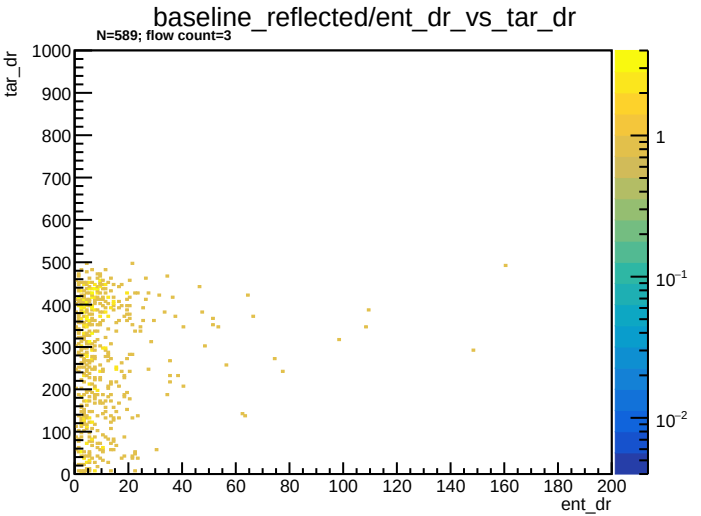

baseline_direct/tar_radius
N=741; flow count=2
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

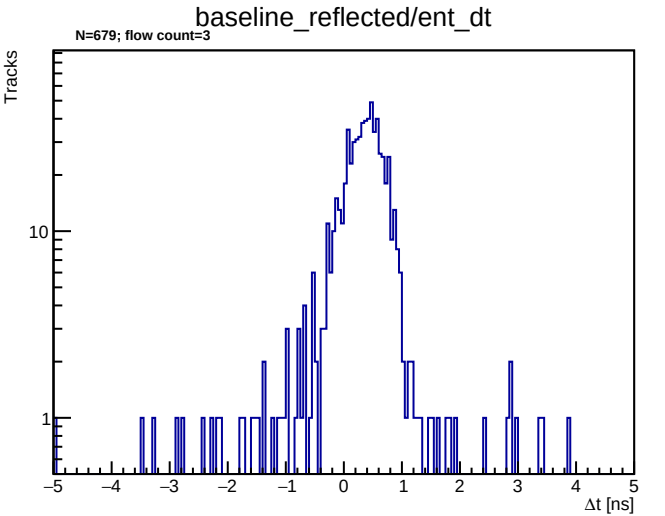

baseline_reflected/ent_dp
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

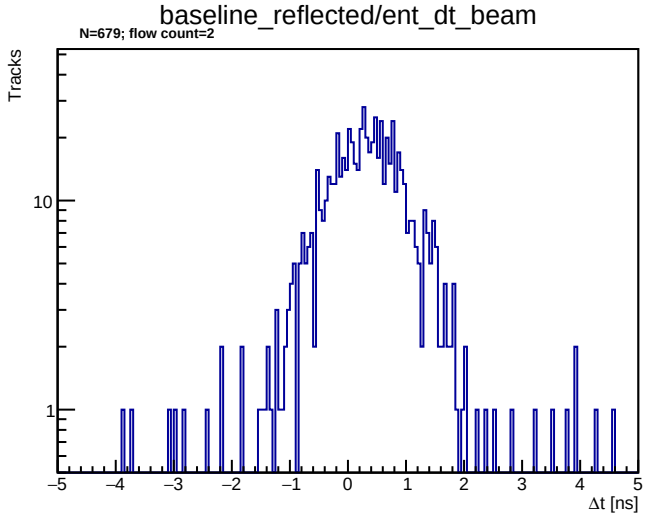

baseline_reflected/ent_dpz
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

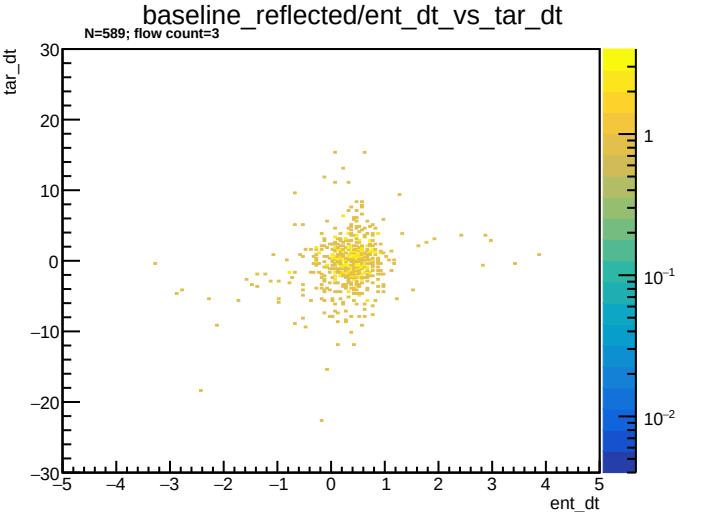

baseline_reflected/ent_dpz_vs_tar_phase
N=589; flow count=2
tar_phase
ent_dpz

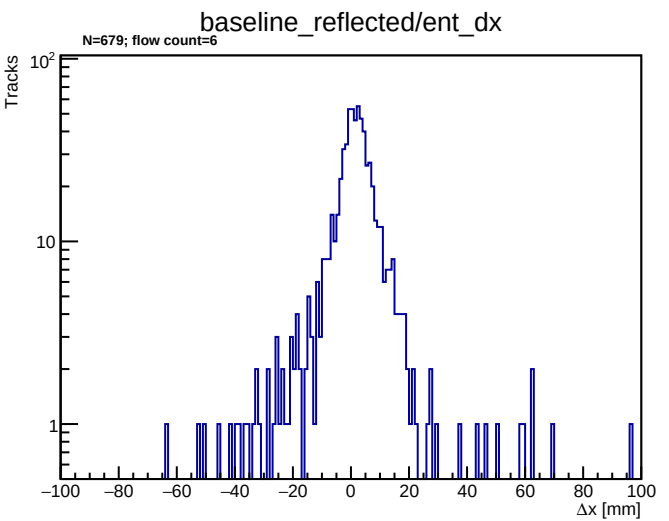

baseline_reflected/ent_dr
N=679; flow count=5
Tracks
Δr_xy [mm]

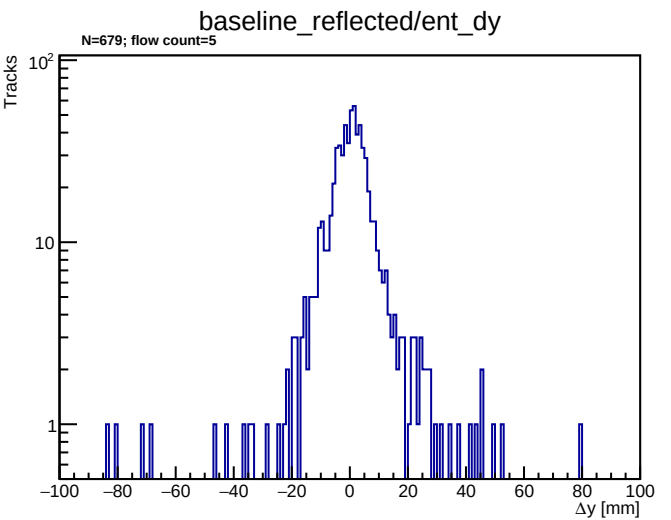

baseline_reflected/ent_dr_vs_tar_dr
N=589; flow count=3
tar_dr
ent_dr

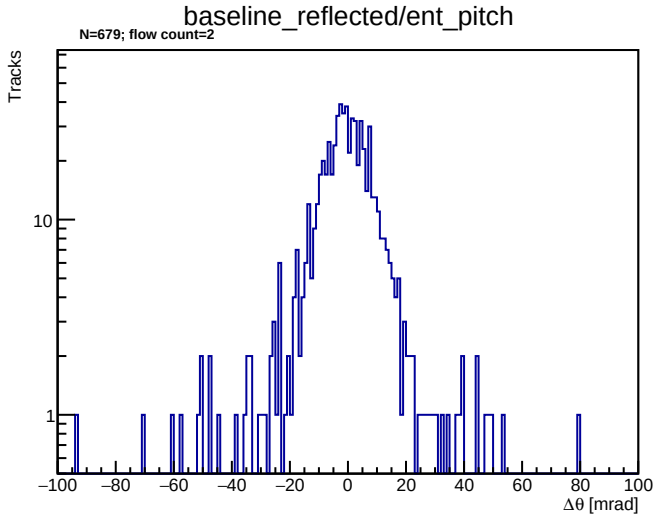

baseline_reflected/ent_dt
N=679; flow count=3
Tracks
Δt [ns]

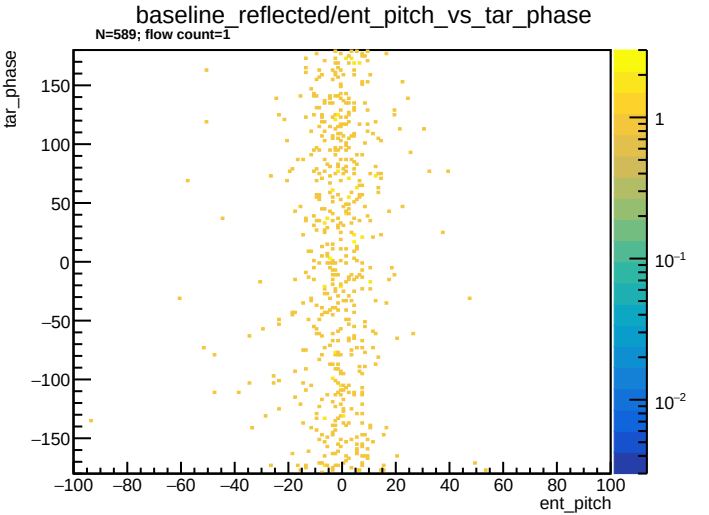

baseline_reflected/ent_dt_beam
N=679; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

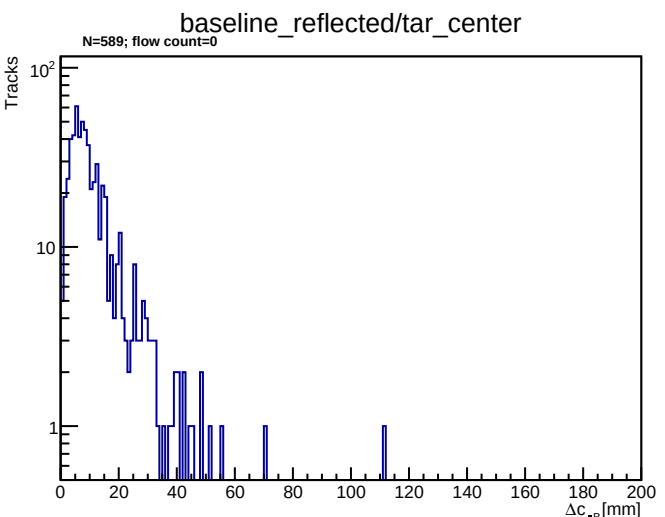

baseline_reflected/ent_dt_vs_tar_dt
N=589; flow count=3
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^{-1}
10^{-2}

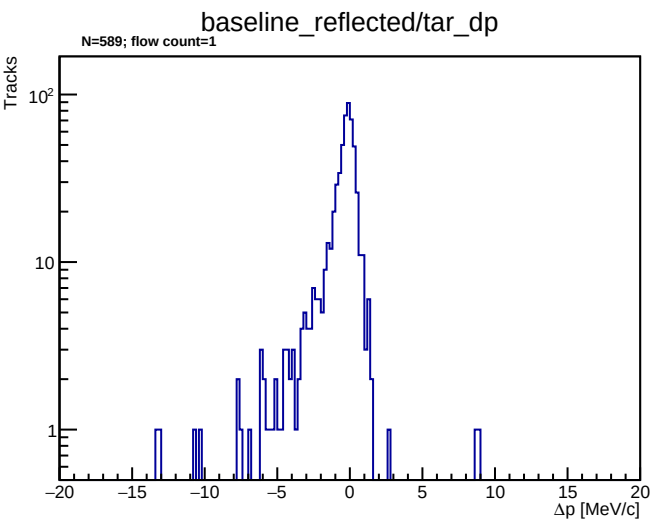

baseline_reflected/ent_dx
N=679; flow count=6
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

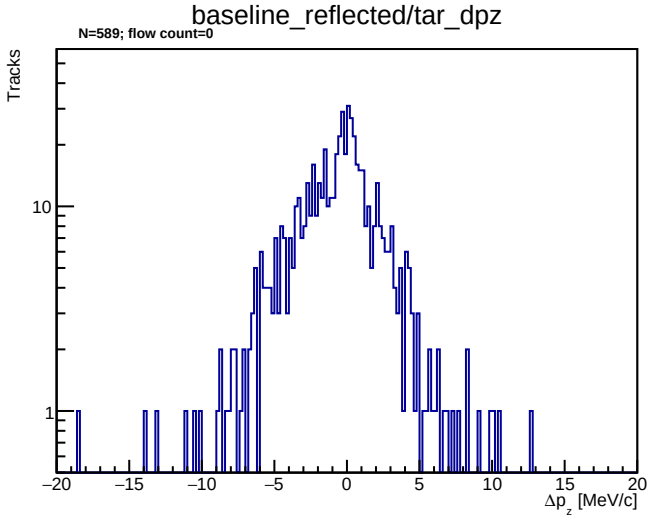

baseline_reflected/ent_dy
N=679; flow count=5
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

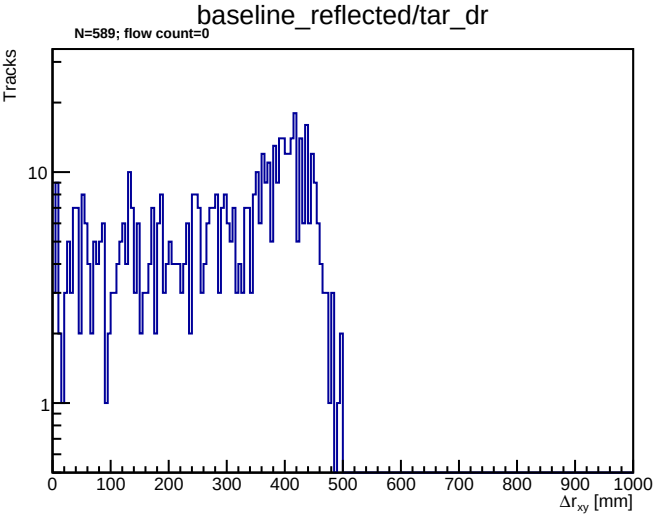

baseline_reflected/ent_pitch
N=679; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

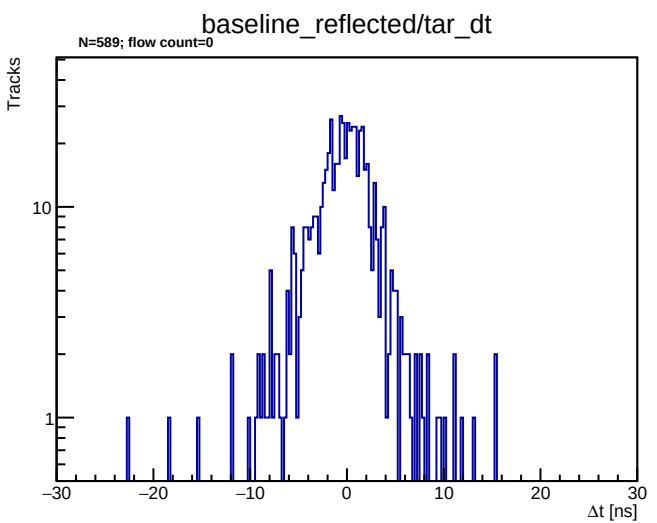

baseline_reflected/ent_pitch_vs_tar_phase
N=589; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²

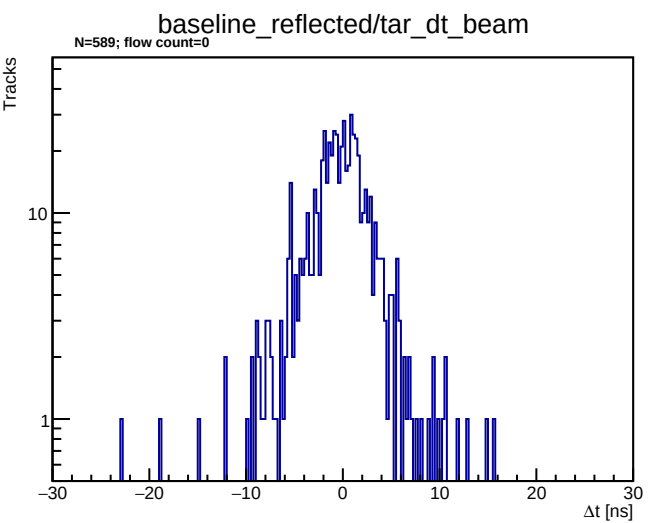

baseline_reflected/tar_center
N=589; flow count=0
Tracks
10²
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

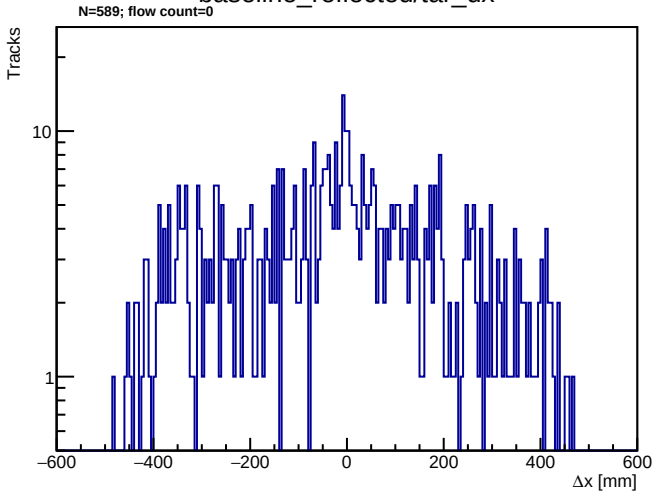

baseline_reflected/tar_dp
N=589; flow count=1
Tracks
10²
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

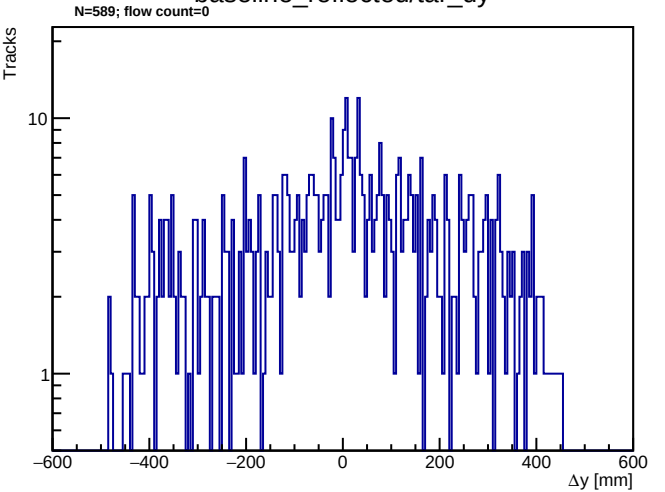

baseline_reflected/tar_dpz
N=589; flow count=0
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

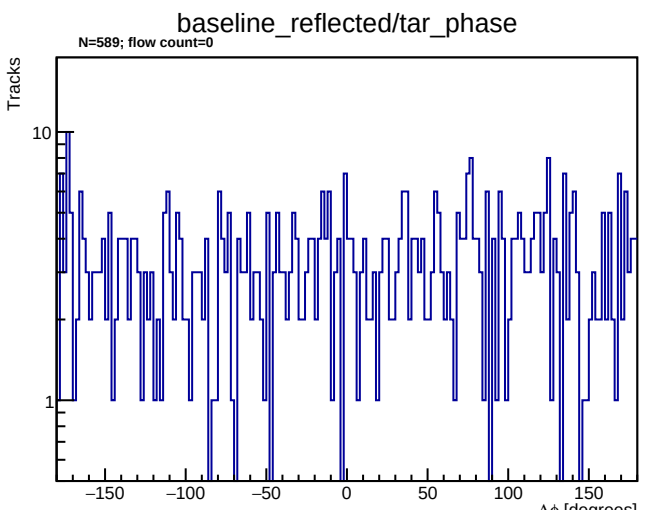

baseline_reflected/tar_dr
N=589; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

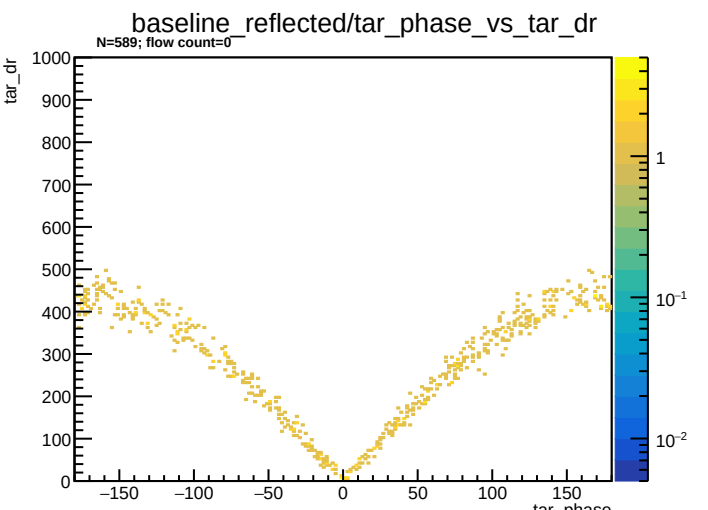

baseline_reflected/tar_dt
N=589; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

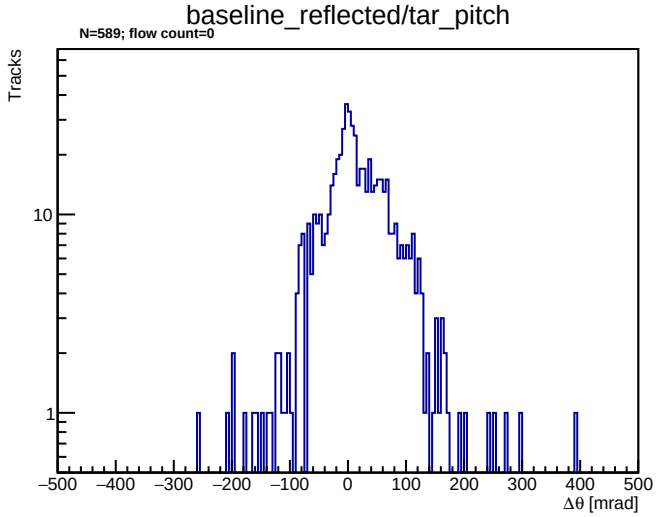

baseline_reflected/tar_dt_beam
N=589; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

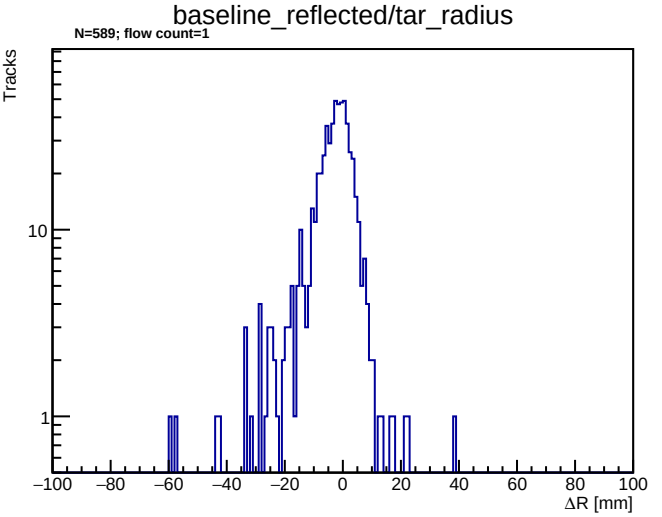

baseline_reflected/tar_dx
N=589; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

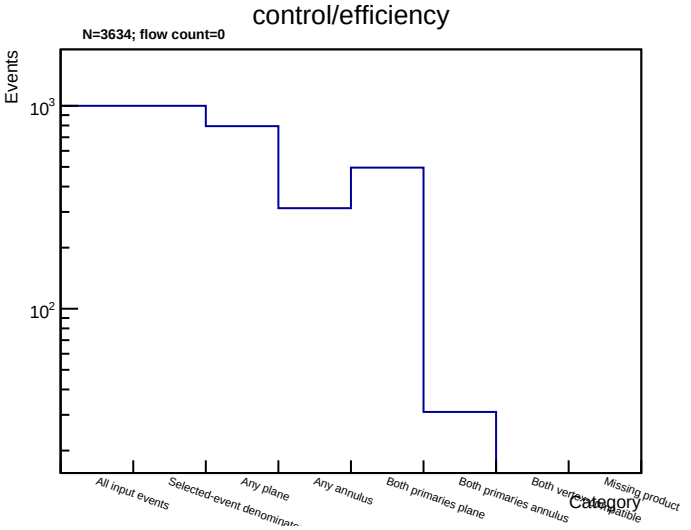

baseline_reflected/tar_dy
N=589; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

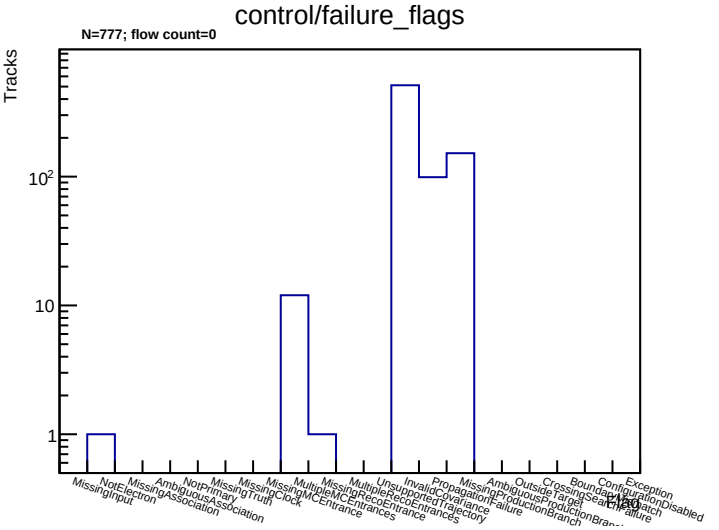

baseline_reflected/tar_phase
N=589; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

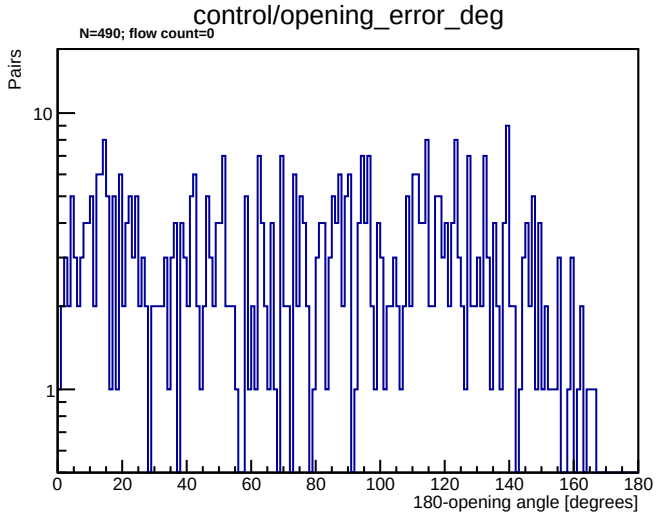

baseline_reflected/tar_phase_vs_tar_dr
N=589; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

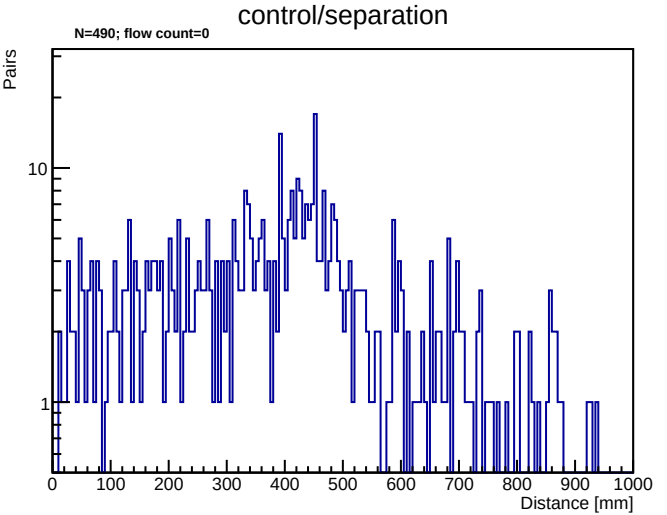

baseline_reflected/tar_pitch
N=589; flow count=0
Tracks
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

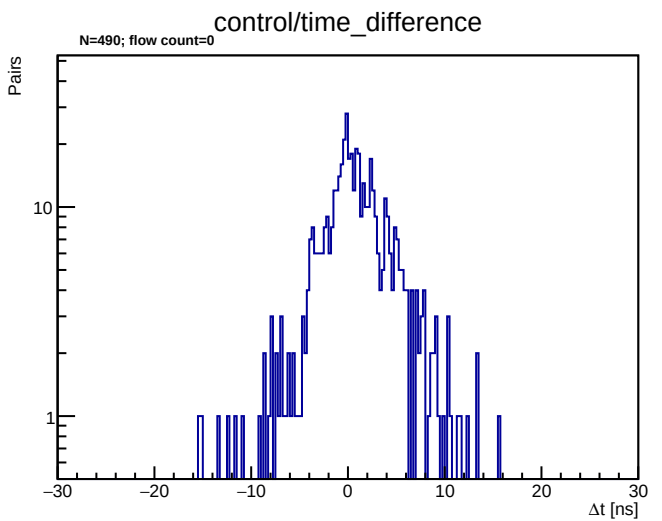

baseline_reflected/tar_radius
N=589; flow count=1
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

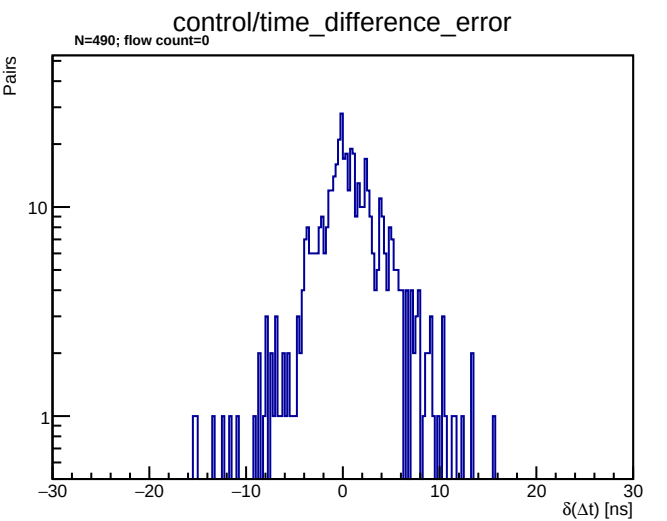

control/efficiency
N=3634; flow count=0
Events
10^3
10^2
All input events
Selected-event denominator
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both vertex compatible
Missing product
Category

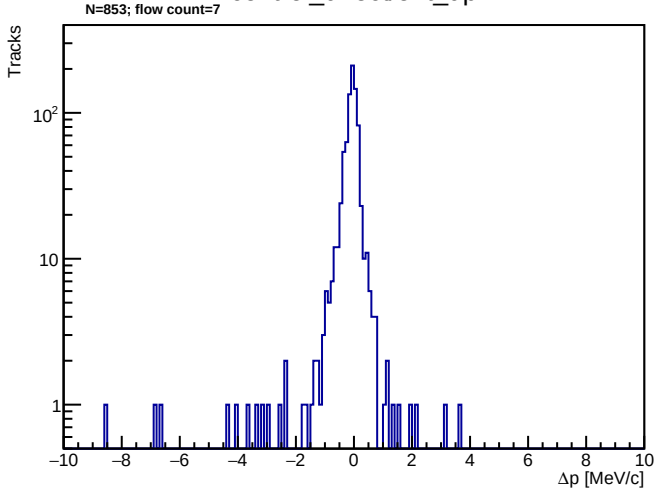

control/failure_flags
N=777; flow count=0
Tracks
10^2
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleReferenceEntrances
MissingReferenceEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProductionBranch
AmbiguousProductionBranch
OutsideTarget
CrossingSeamAtTarget
BoundaryAtTarget
ConfigurationBranch
ConfigurationDisabled
Exception
Failure
Flag

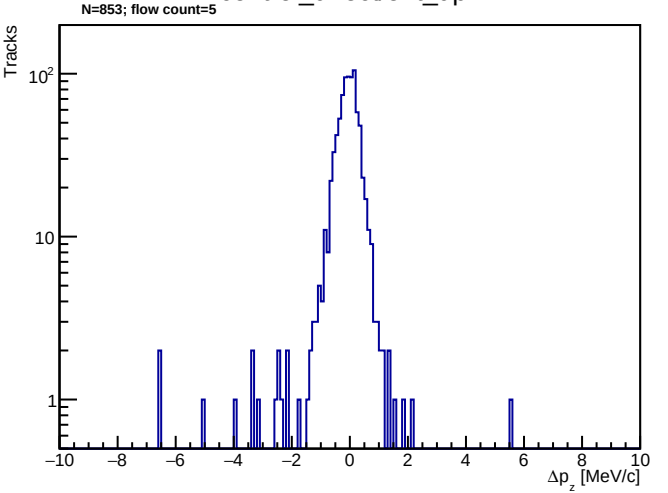

control/opening_error_deg
N=490; flow count=0
Pairs
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

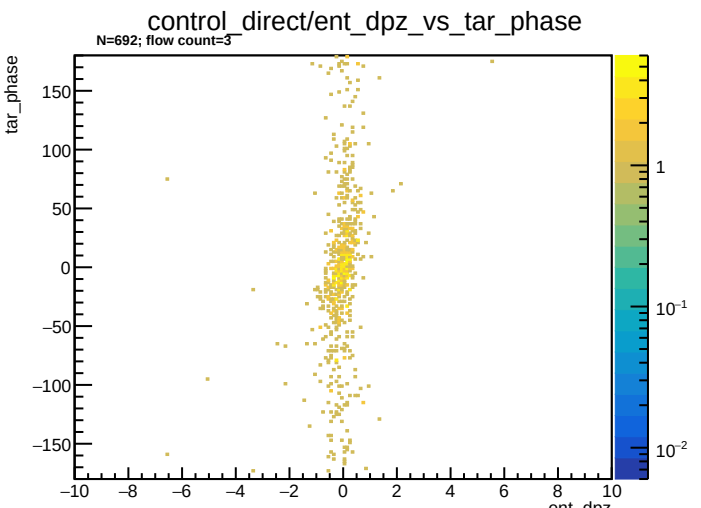

control/separation
N=490; flow count=0
Pairs
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

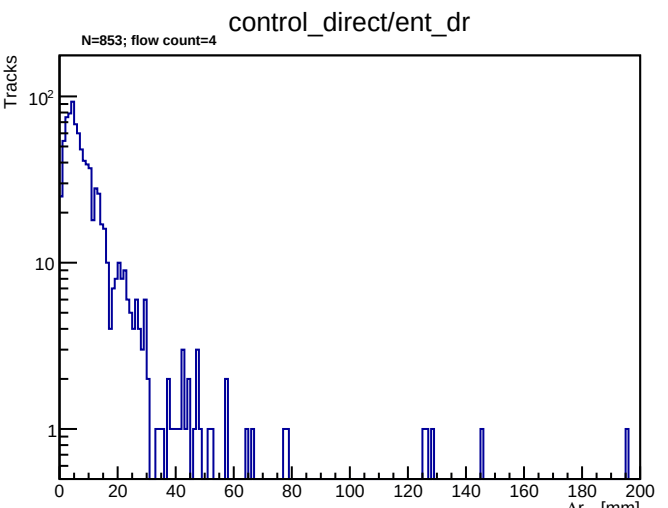

control/time_difference
N=490; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

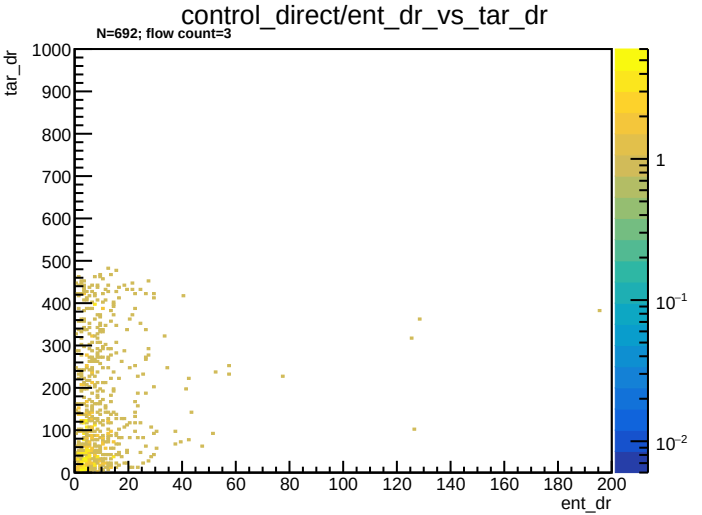

control/time_difference_error
N=490; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

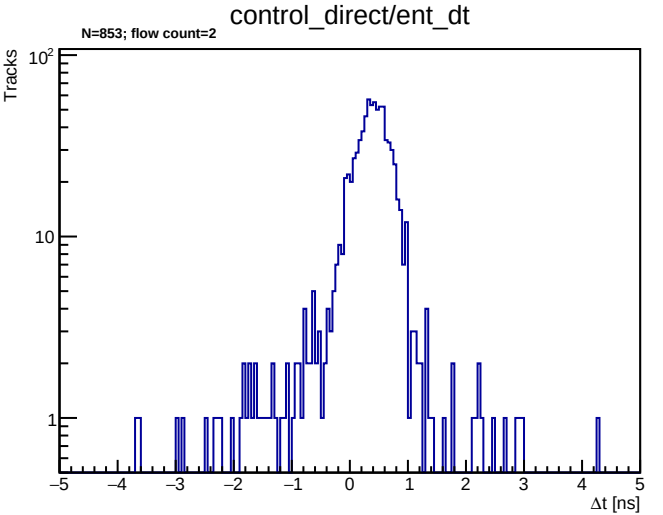

control_direct/ent_dp
N=853; flow count=7
Tracks
Δp [MeV/c]

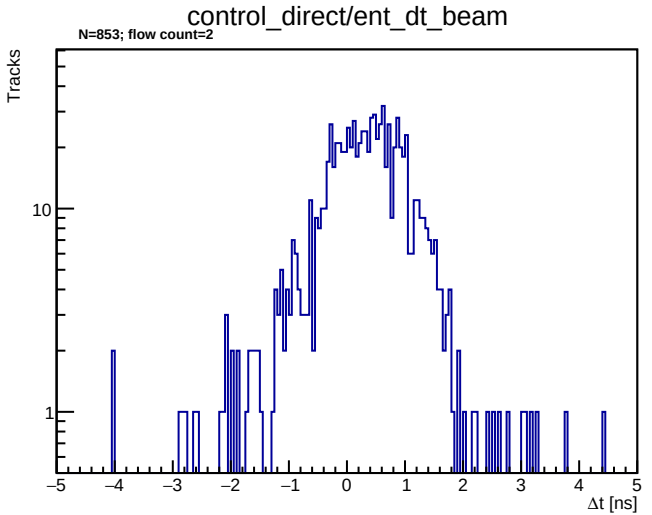

control_direct/ent_dpz
N=853; flow count=5
Tracks
Δp_z [MeV/c]

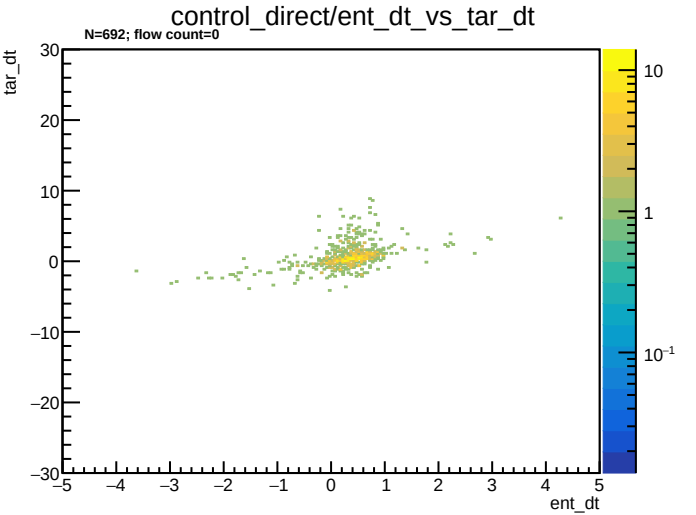

control_direct/ent_dpz_vs_tar_phase
N=692; flow count=3
tar_phase
ent_dpz

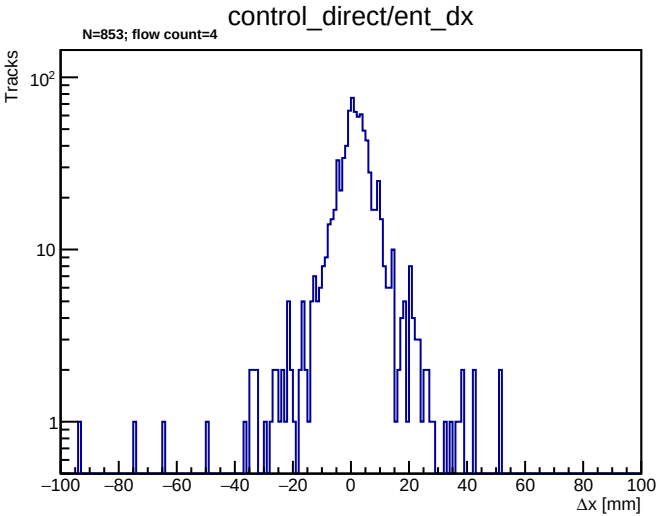

control_direct/ent_dr
N=853; flow count=4
Tracks
Δr_xy [mm]

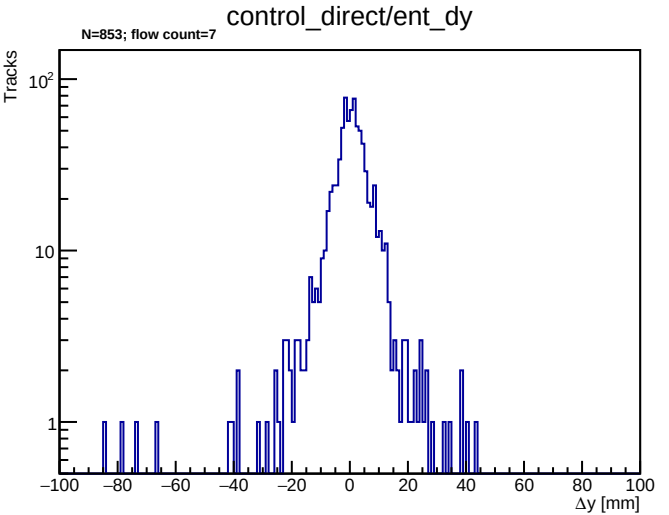

control_direct/ent_dr_vs_tar_dr
N=692; flow count=3
tar_dr
ent_dr

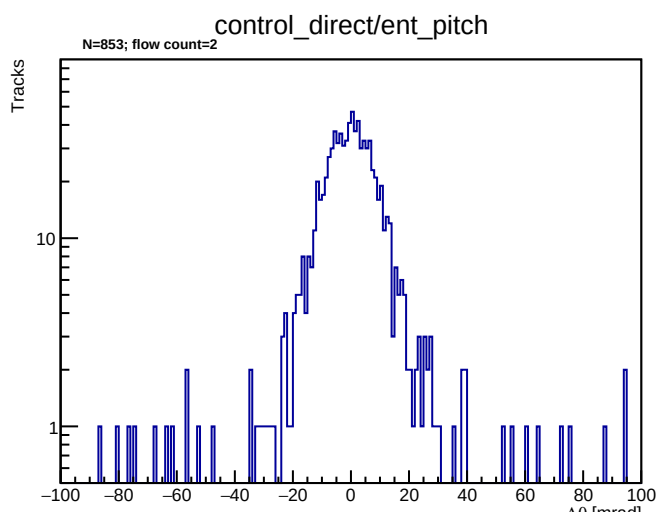

control_direct/ent_dt
N=853; flow count=2
Tracks
Δt [ns]

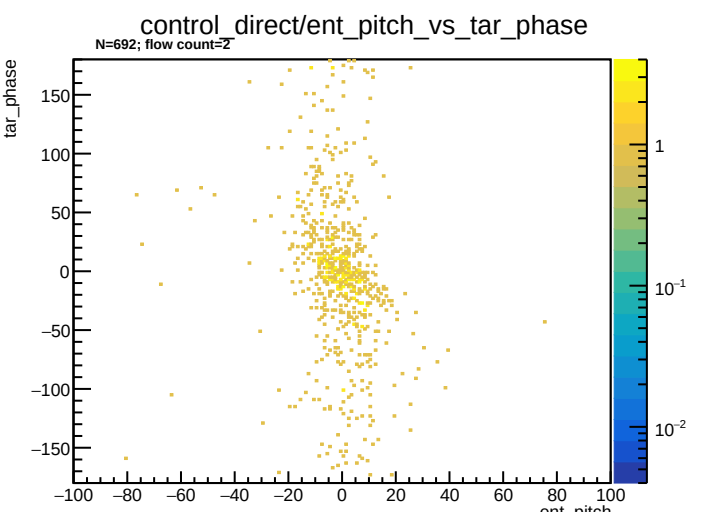

control_direct/ent_dt_beam
N=853; flow count=2
Tracks
Δt [ns]

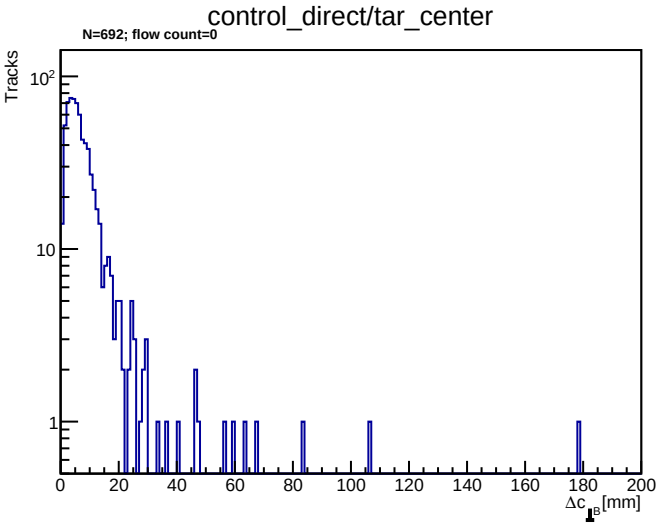

control_direct/ent_dt_vs_tar_dt
N=692; flow count=0
tar_dt
ent_dt

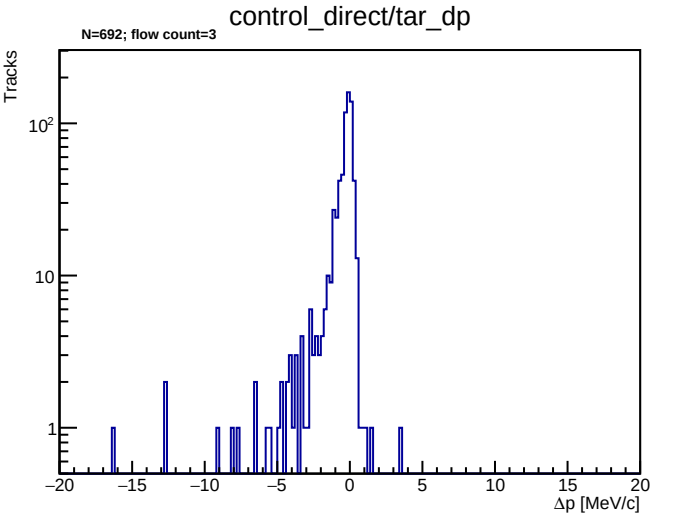

control_direct/ent_dx
N=853; flow count=4
Tracks
Δx [mm]

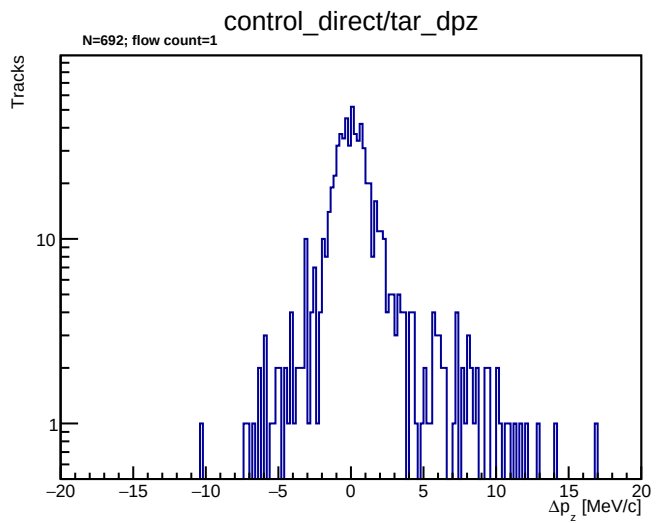

control_direct/ent_dy
N=853; flow count=7
Tracks
Δy [mm]

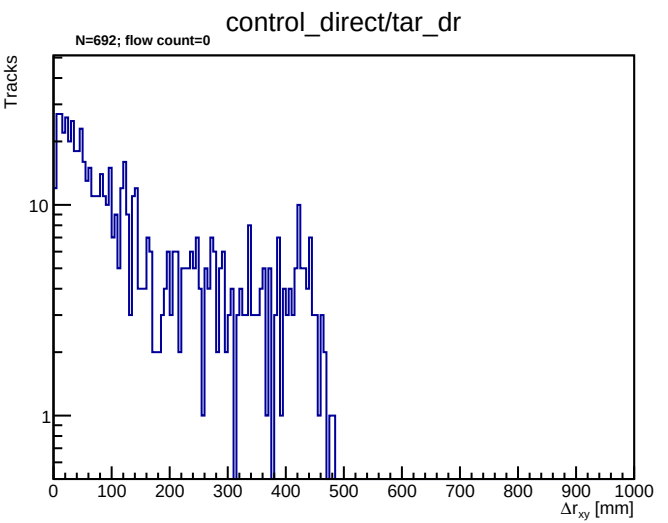

control_direct/ent_pitch
N=853; flow count=2
Tracks
Δθ [mrad]

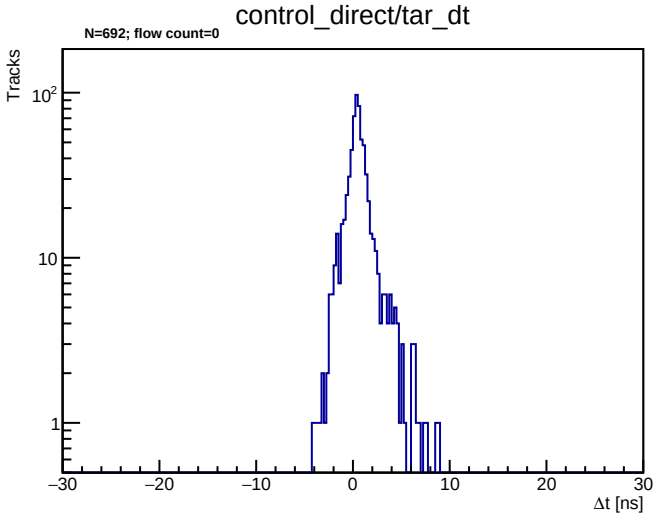

control_direct/ent_pitch_vs_tar_phase
N=692; flow count=2
tar_phase
ent_pitch

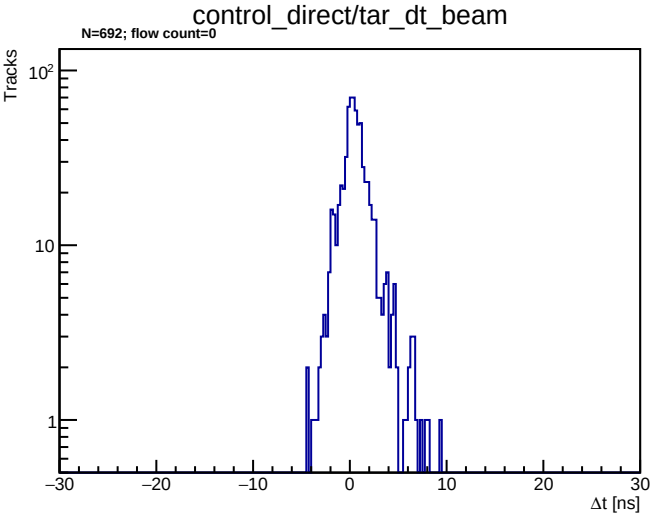

control_direct/tar_center
N=692; flow count=0
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

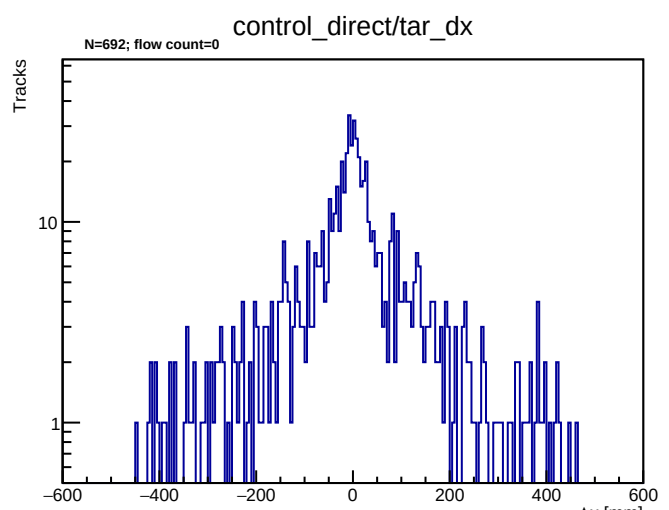

control_direct/tar_dp
N=692; flow count=3
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

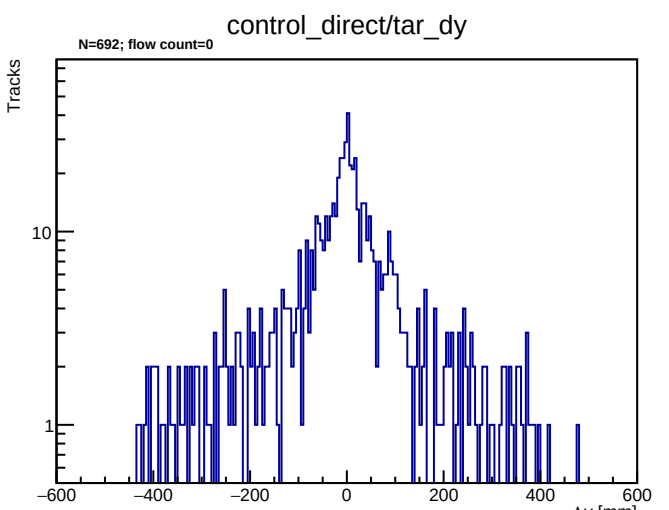

control_direct/tar_dpz
N=692; flow count=1
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

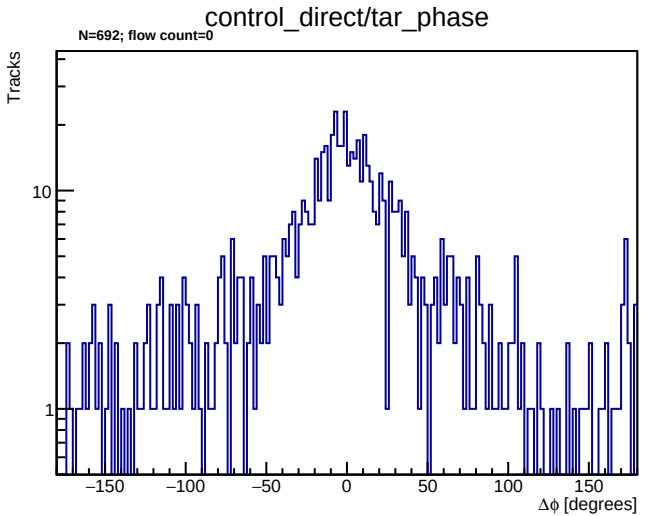

control_direct/tar_dr
N=692; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

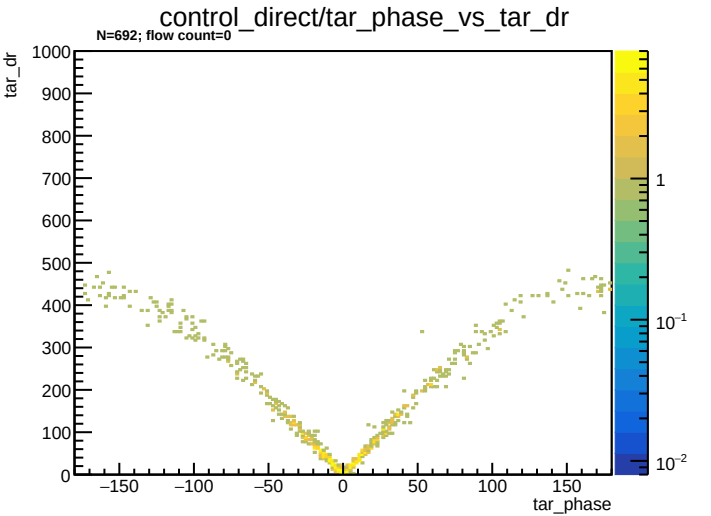

control_direct/tar_dt
N=692; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

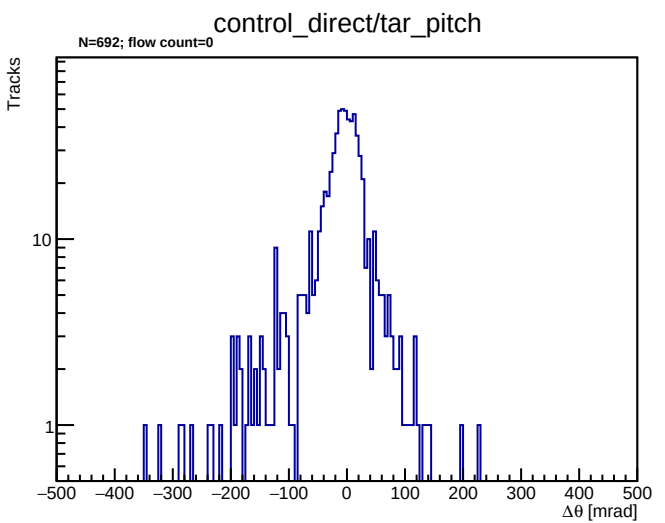

control_direct/tar_dt_beam
N=692; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

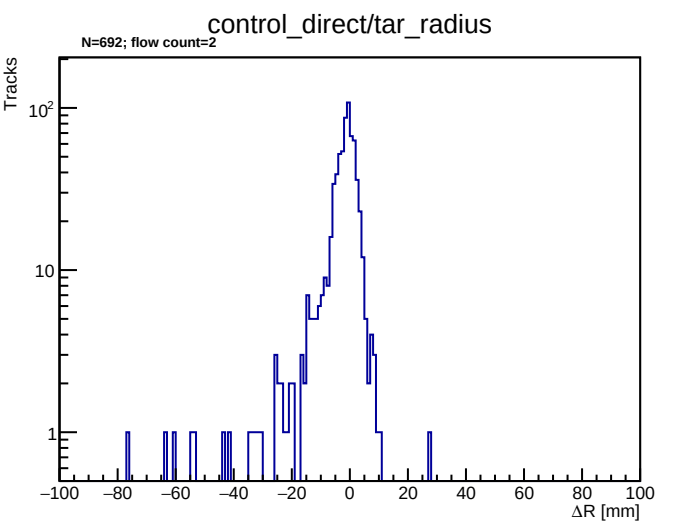

control_direct/tar_dx
N=692; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

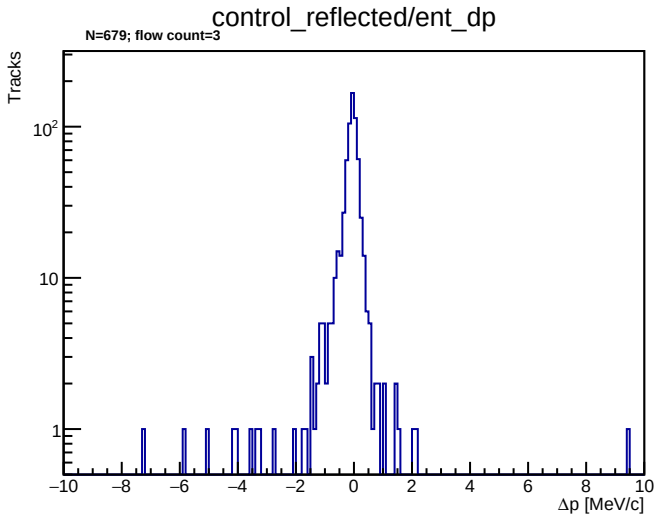

control_direct/tar_dy
N=692; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

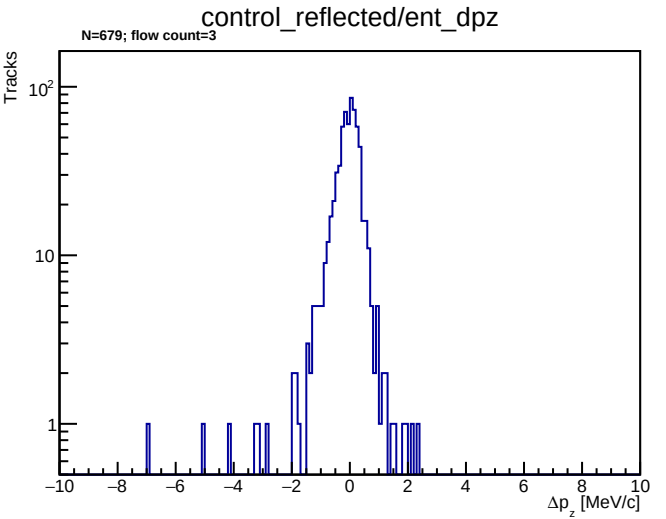

control_direct/tar_phase
N=692; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

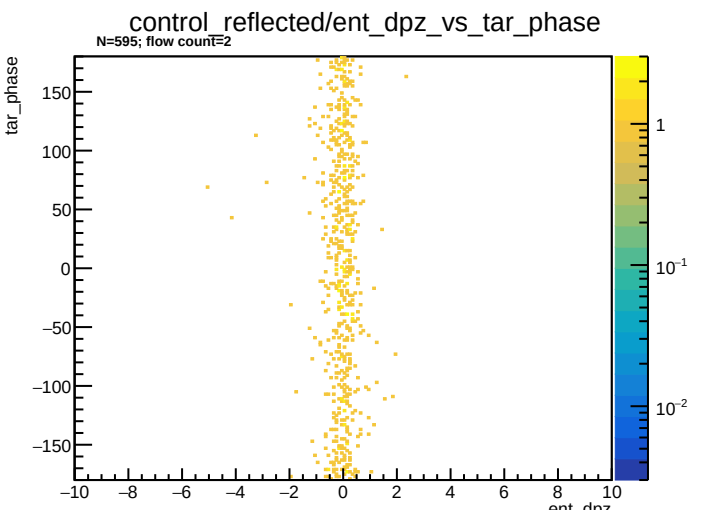

control_direct/tar_phase_vs_tar_dr
N=692; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10^{-1}
10^{-2}

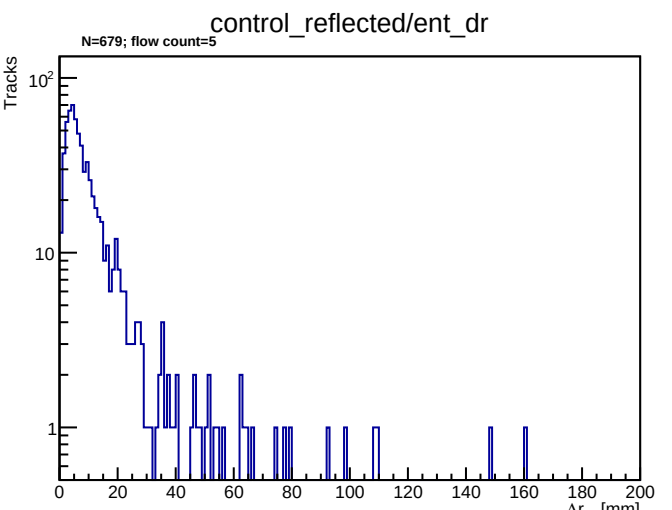

control_direct/tar_pitch
N=692; flow count=0
Tracks
10
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

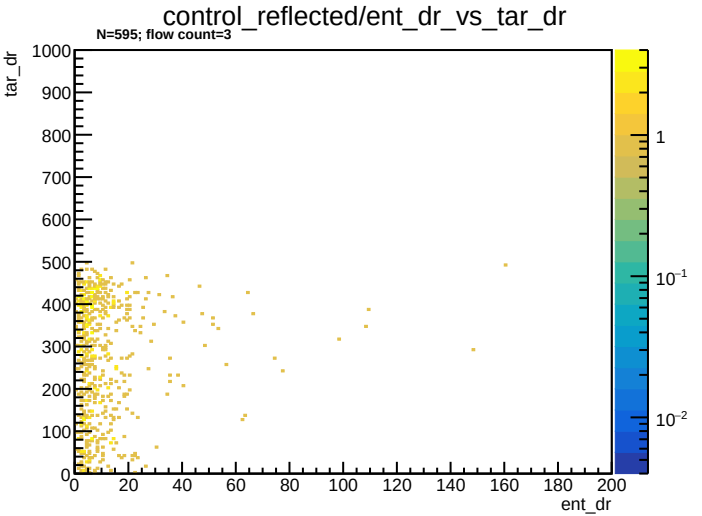

control_direct/tar_radius
N=692; flow count=2
Tracks
10^2
10
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

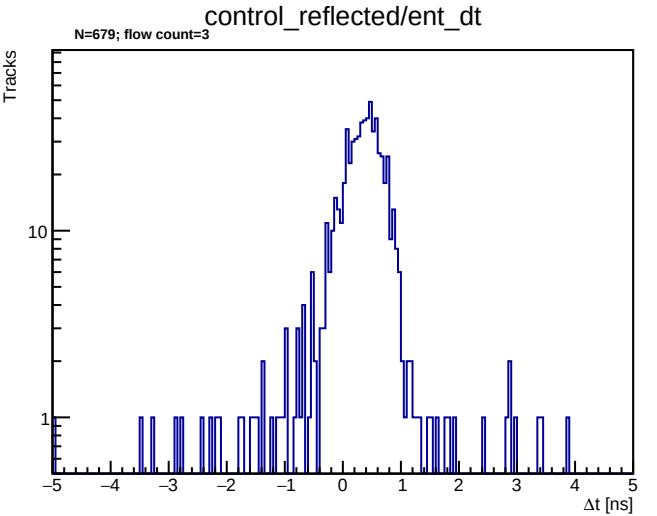

control_reflected/ent_dp
N=679; flow count=3
Tracks
Δp [MeV/c]

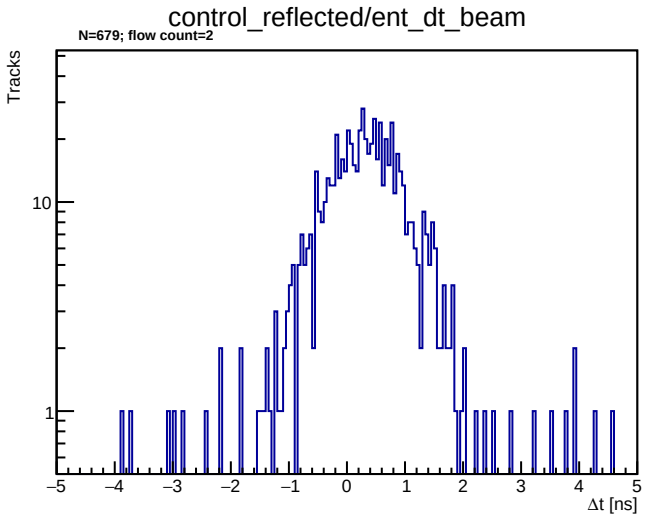

control_reflected/ent_dpz
N=679; flow count=3
Tracks
Δp_z [MeV/c]

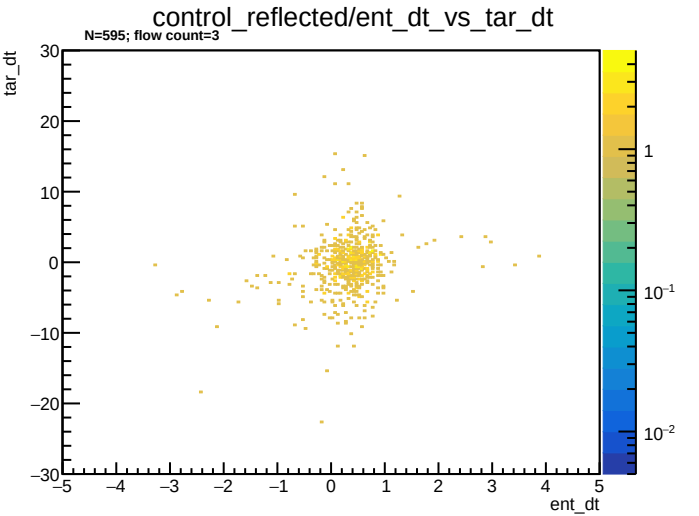

control_reflected/ent_dpz_vs_tar_phase
N=595; flow count=2
tar_phase
ent_dpz

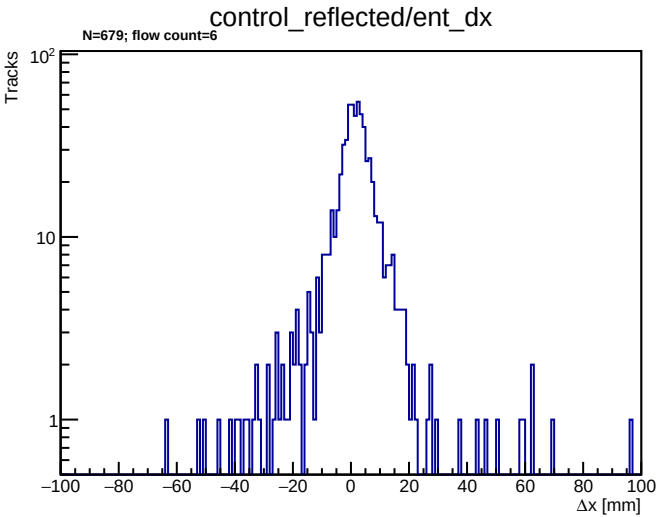

control_reflected/ent_dr
N=679; flow count=5
Tracks
Δr_xy [mm]

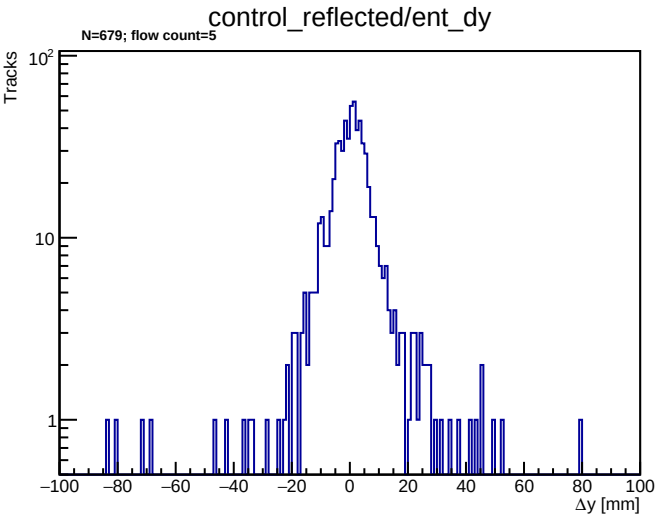

control_reflected/ent_dr_vs_tar_dr
N=595; flow count=3
tar_dr
ent_dr

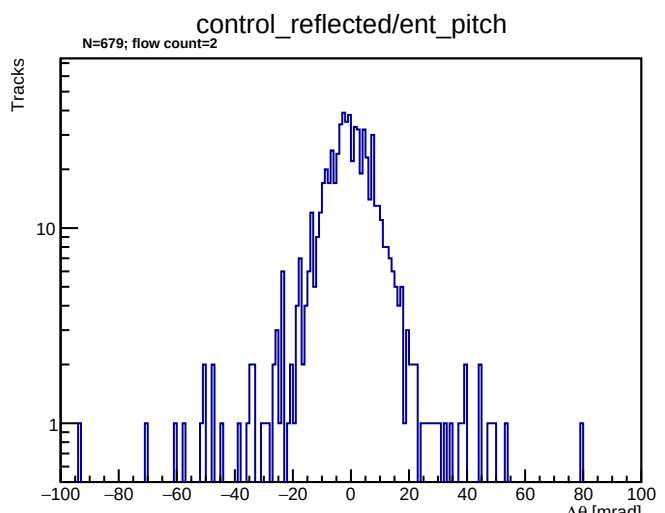

control_reflected/ent_dt
N=679; flow count=3
Tracks
Δt [ns]

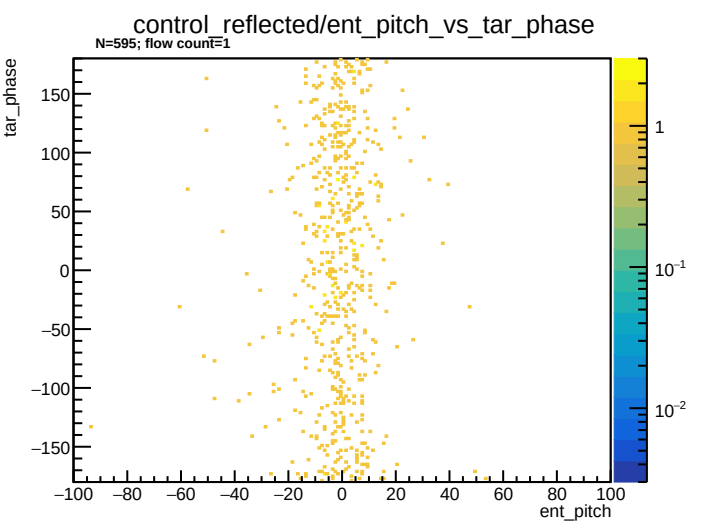

control_reflected/ent_dt_beam
N=679; flow count=2
Tracks
Δt [ns]

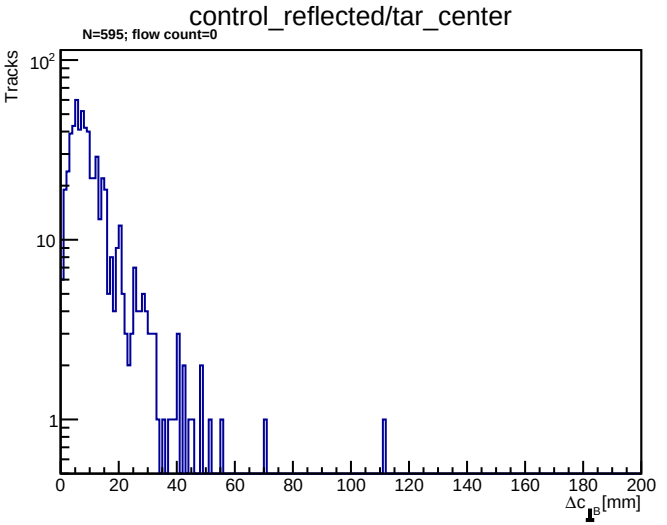

control_reflected/ent_dt_vs_tar_dt
N=595; flow count=3
tar_dt
ent_dt

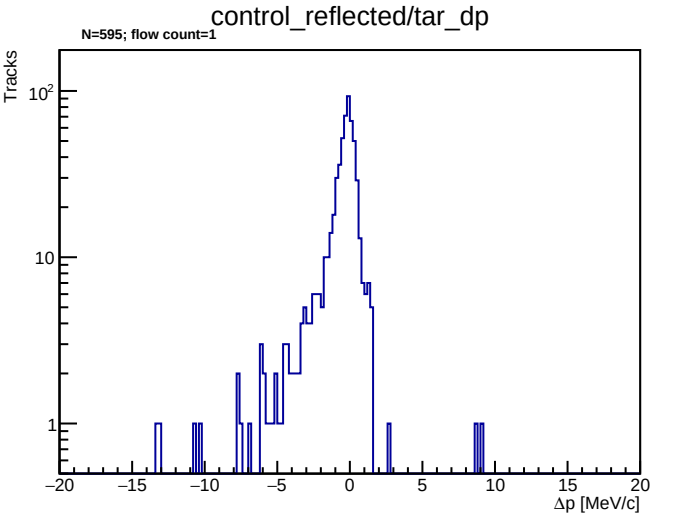

control_reflected/ent_dx
N=679; flow count=6
Tracks
Δx [mm]

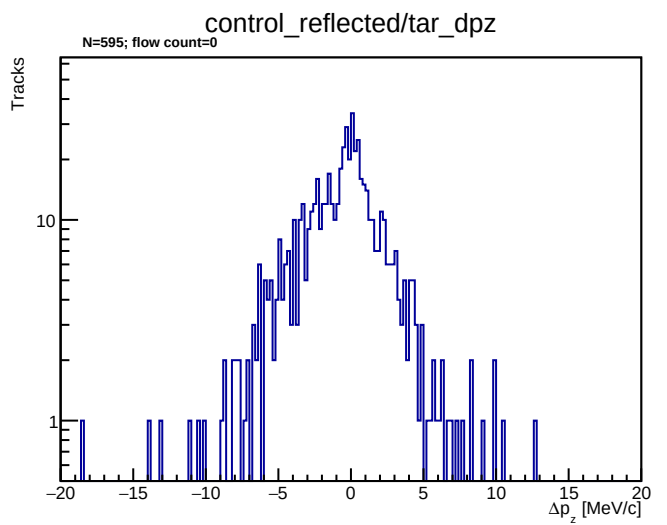

control_reflected/ent_dy
N=679; flow count=5
Tracks
Δy [mm]

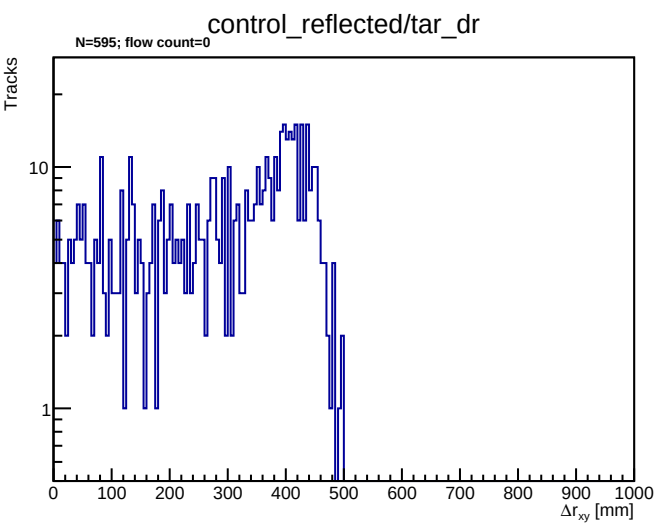

control_reflected/ent_pitch
N=679; flow count=2
Tracks
Δθ [mrad]

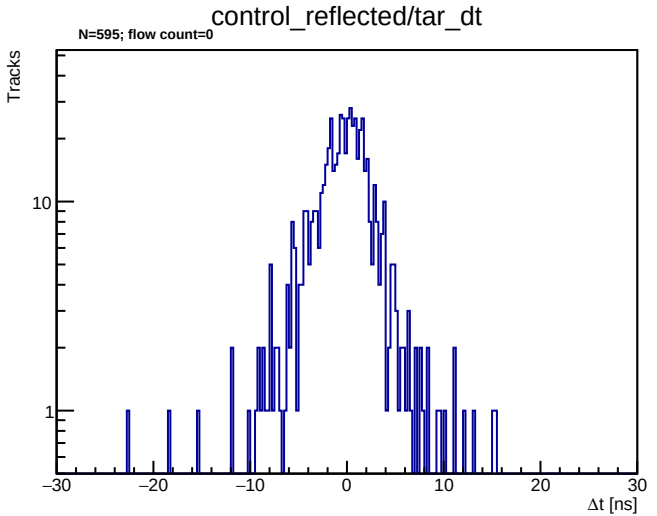

control_reflected/ent_pitch_vs_tar_phase
N=595; flow count=1
tar_phase
ent_pitch

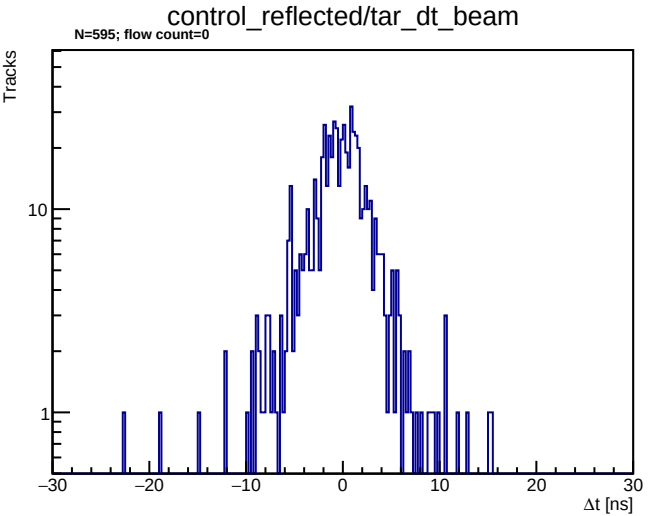

control_reflected/tar_center
N=595; flow count=0
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

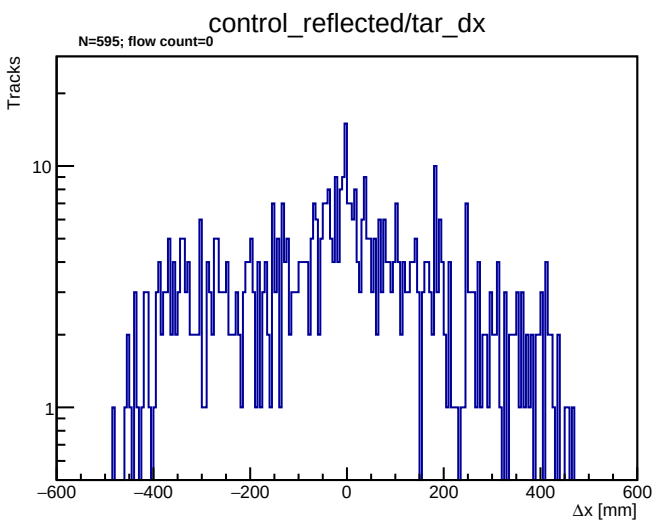

control_reflected/tar_dp
N=595; flow count=1
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

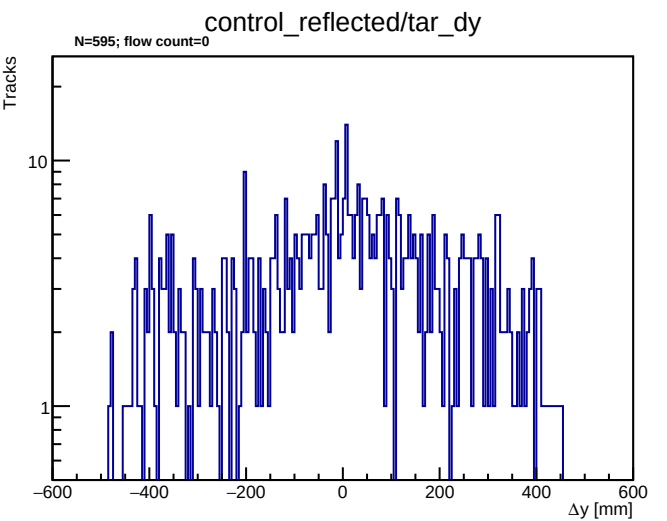

control_reflected/tar_dpz
N=595; flow count=0
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

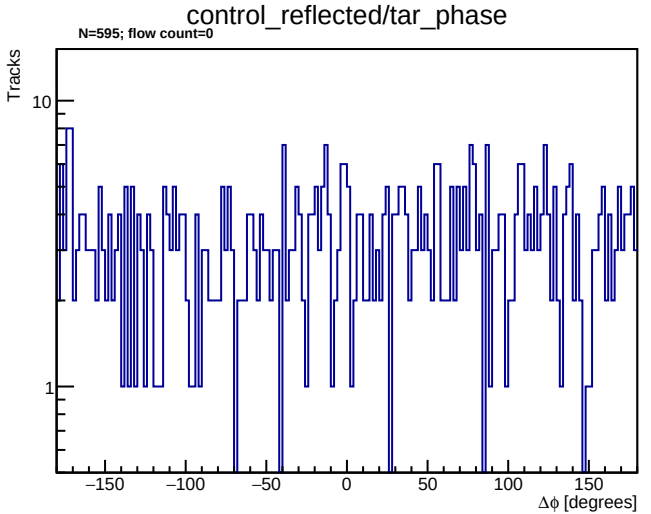

control_reflected/tar_dr
N=595; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

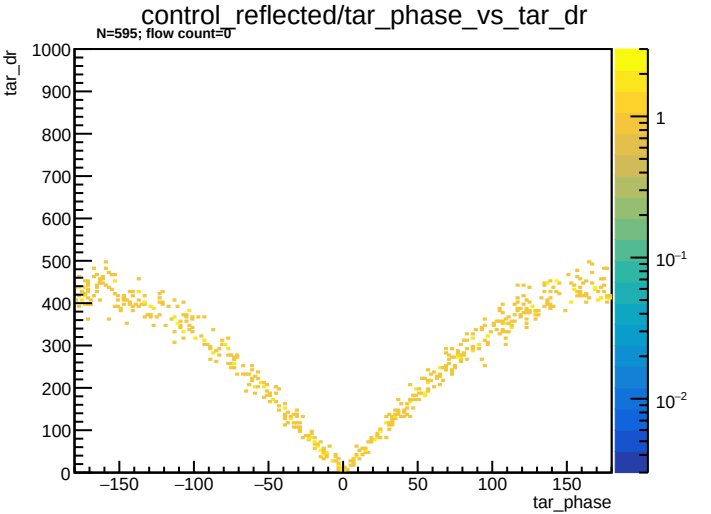

control_reflected/tar_dt
N=595; flow count=0
Tracks
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

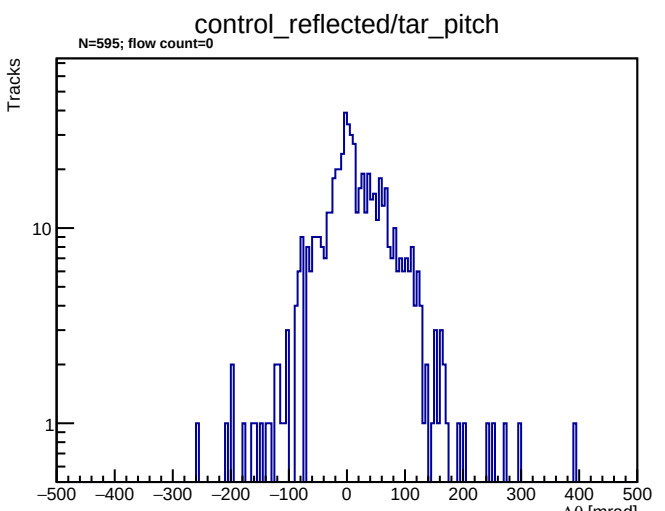

control_reflected/tar_dt_beam
N=595; flow count=0
Tracks
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

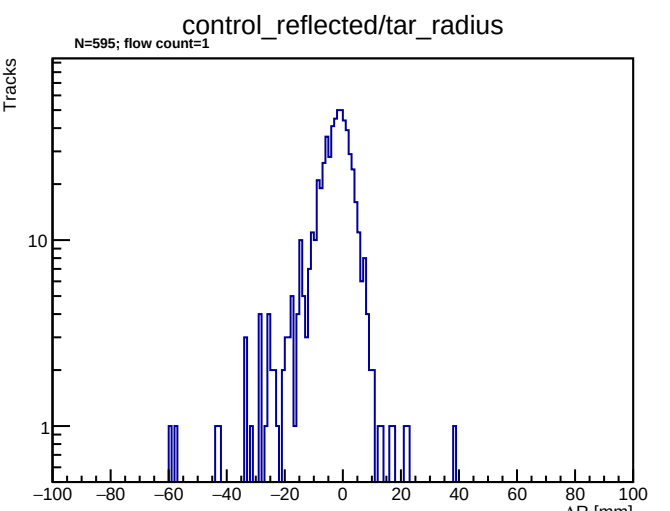

control_reflected/tar_dx
N=595; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

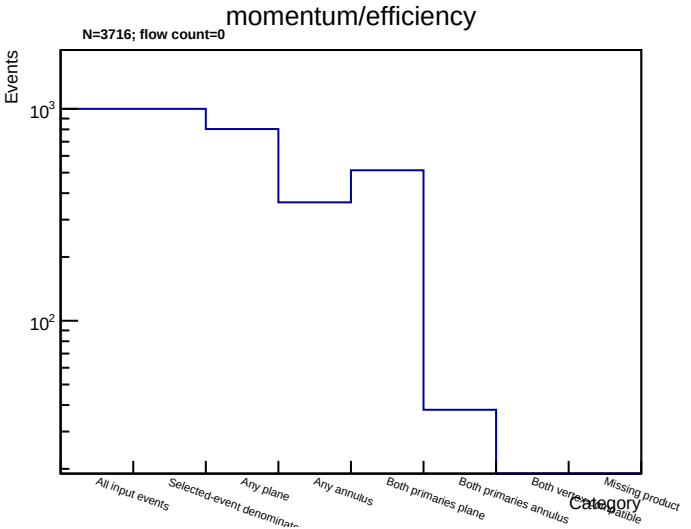

control_reflected/tar_dy
N=595; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

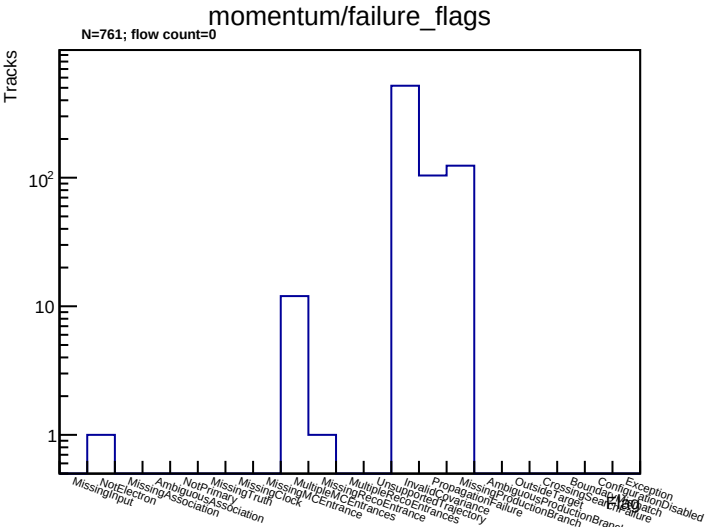

control_reflected/tar_phase
N=595; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

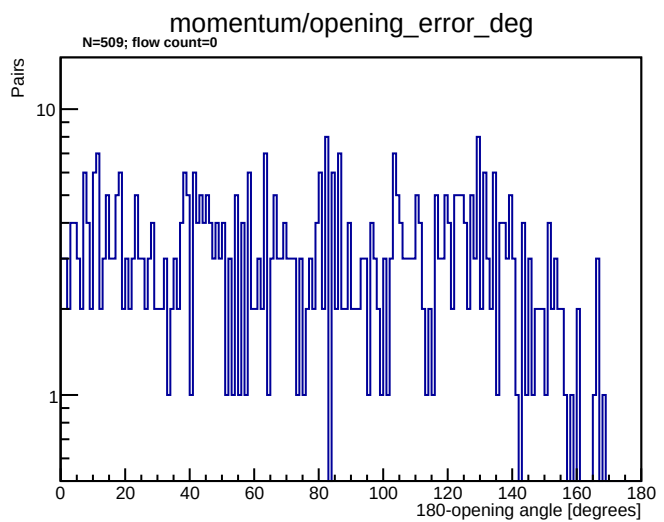

control_reflected/tar_phase_vs_tar_dr
N=595; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

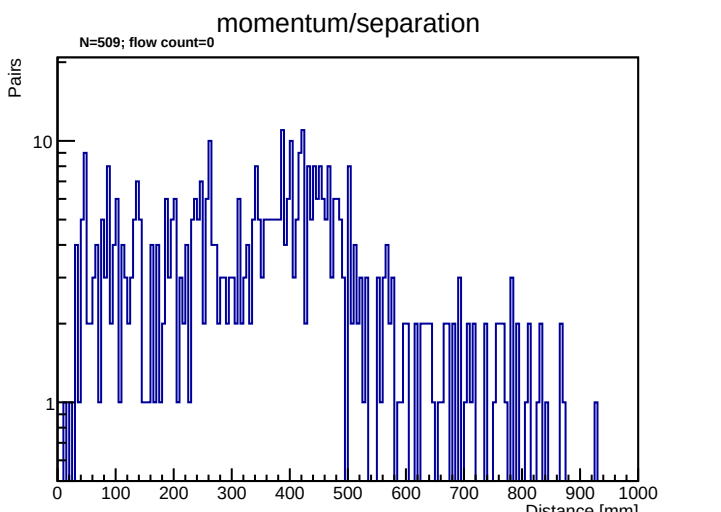

control_reflected/tar_pitch
N=595; flow count=0
Tracks
10
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

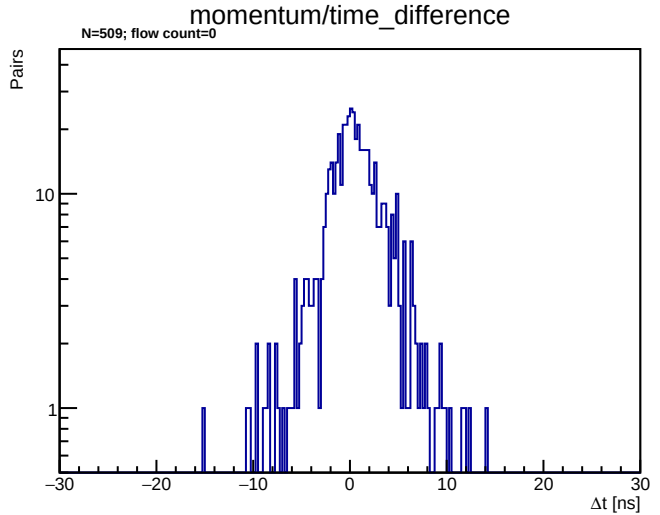

control_reflected/tar_radius
N=595; flow count=1
Tracks
10
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

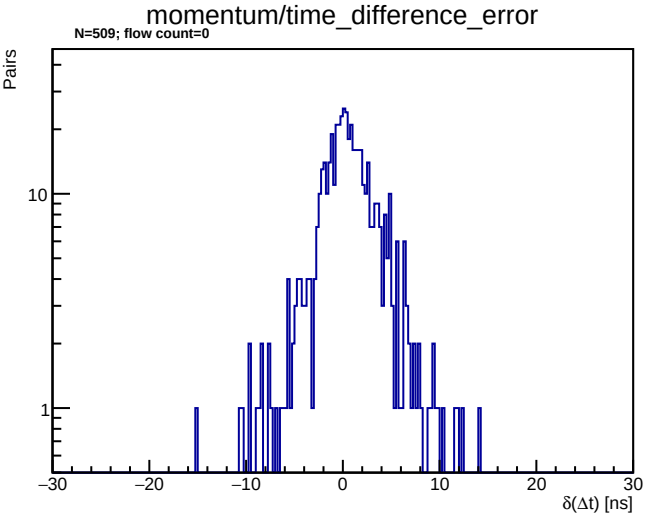

momentum/efficiency
N=3716; flow count=0
Events
10^3
10^2
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

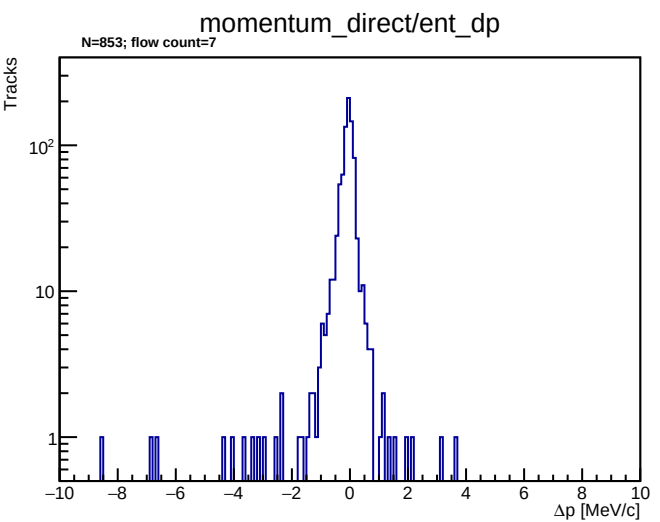

momentum/failure_flags
N=761; flow count=0
Tracks
10^2
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrances
MissingRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeed
BoundaryMatch
Configuration
Exception
Disabled
Failed

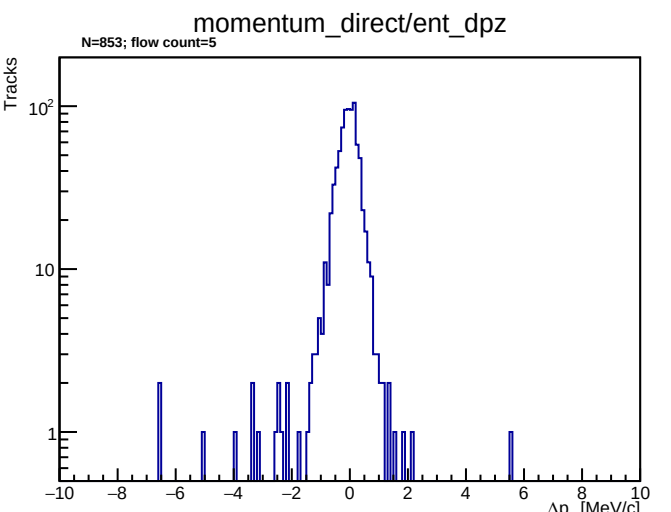

momentum/opening_error_deg
N=509; flow count=0
Pairs
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

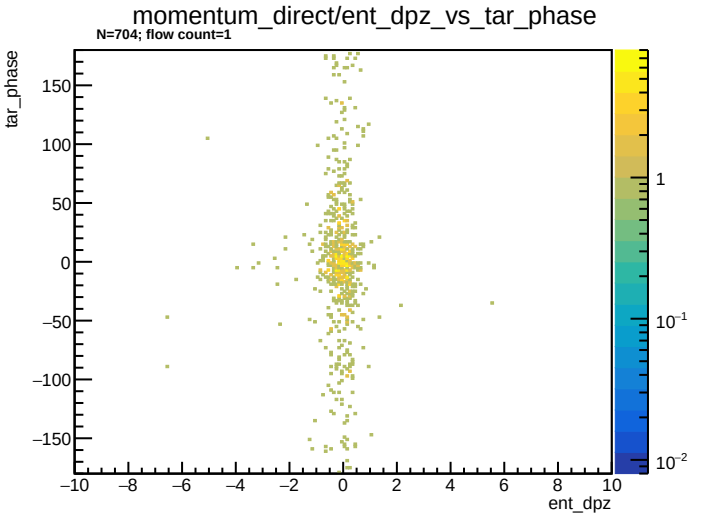

momentum/separation
N=509; flow count=0
Pairs
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

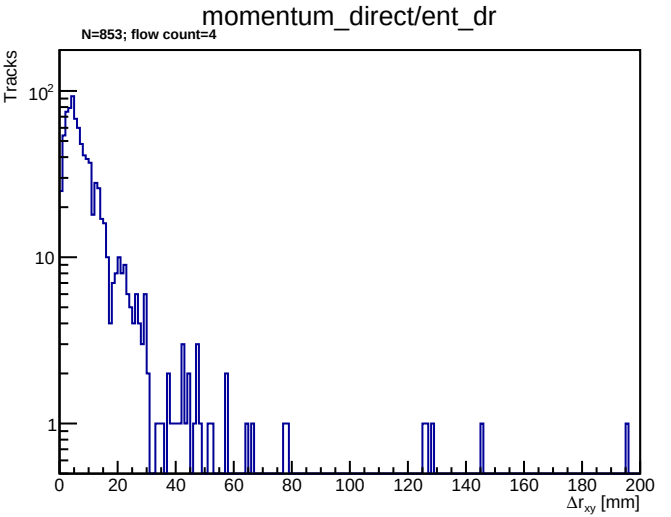

momentum/time_difference
N=509; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

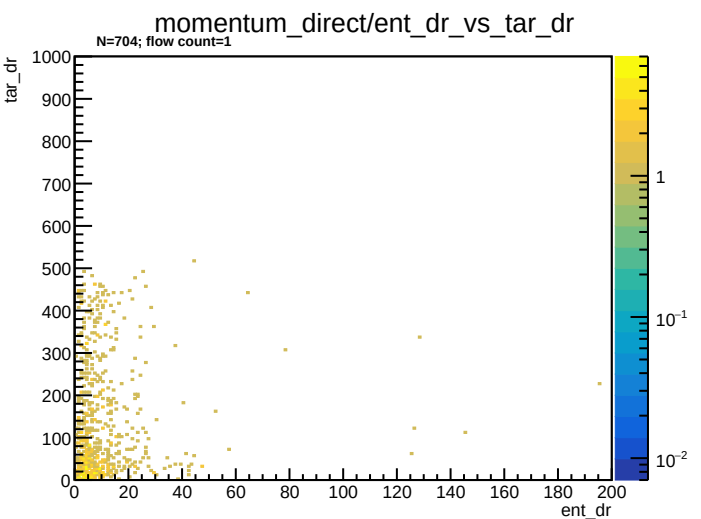

momentum/time_difference_error
N=509; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

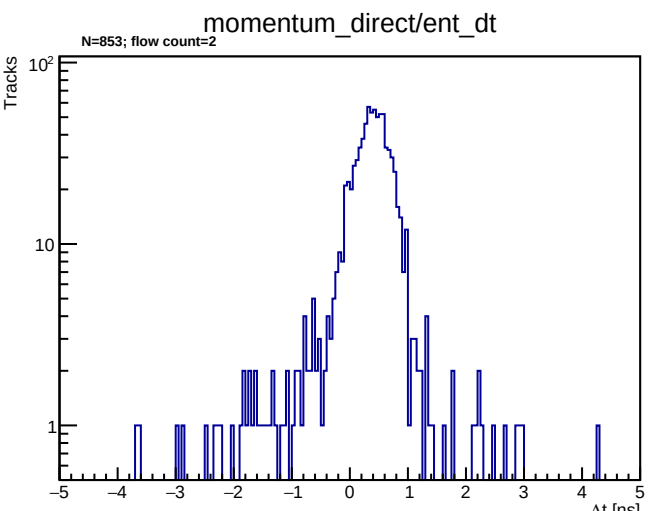

momentum_direct/ent_dp
N=853; flow count=7
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp [MeV/c]

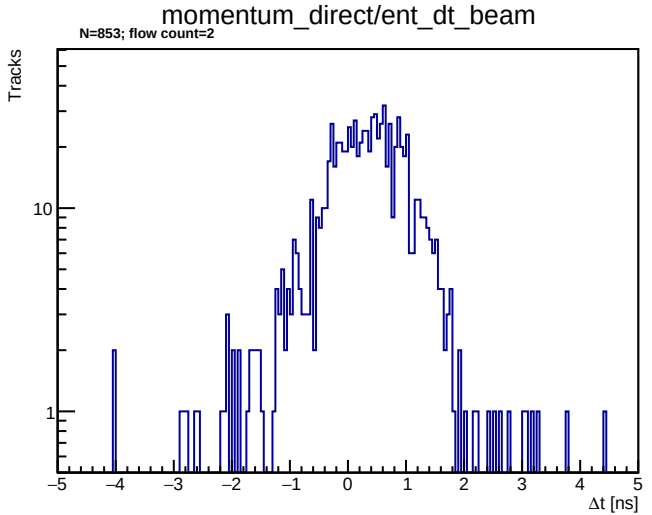

momentum_direct/ent_dpz
N=853; flow count=5
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp_z [MeV/c]

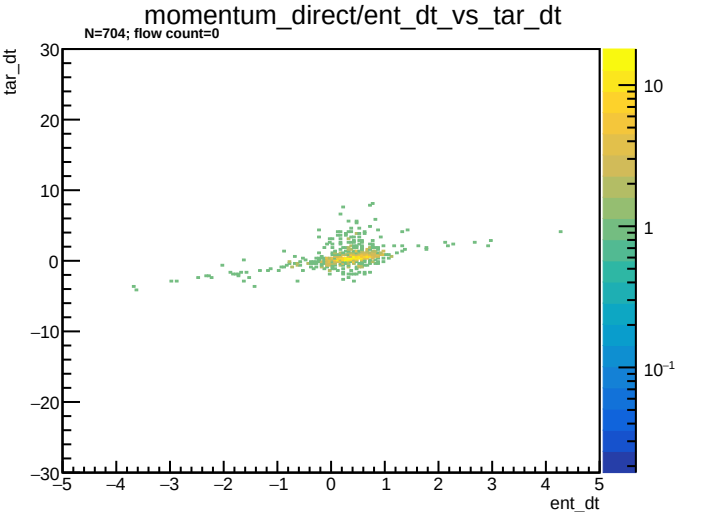

momentum_direct/ent_dpz_vs_tar_phase
N=704; flow count=1
tar_phase
ent_dpz

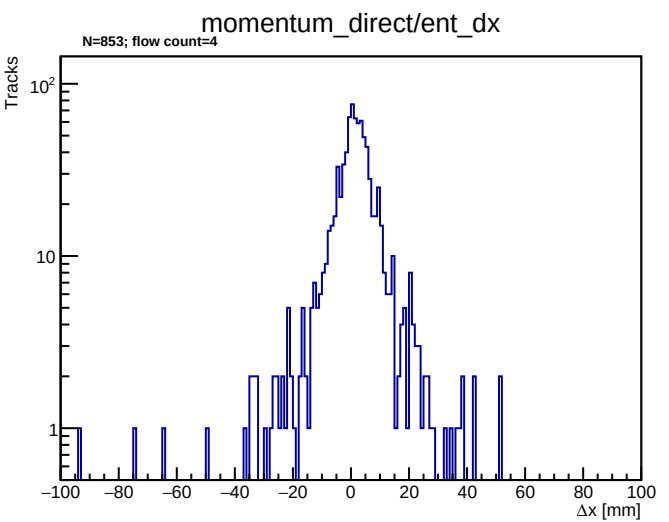

momentum_direct/ent_dr
N=853; flow count=4
Tracks
Δr_xy [mm]

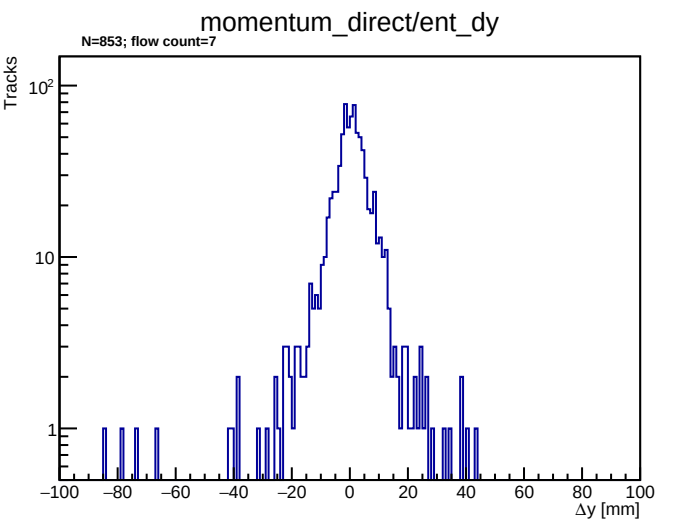

momentum_direct/ent_dr_vs_tar_dr
N=704; flow count=1
tar_dr
ent_dr

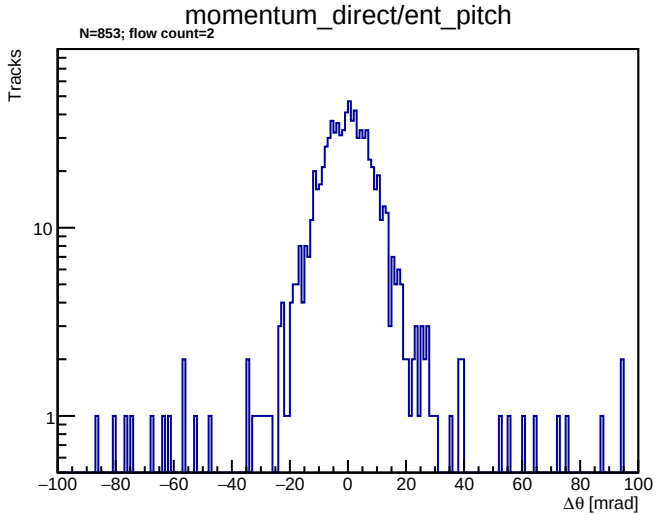

momentum_direct/ent_dt
N=853; flow count=2
Tracks
Δt [ns]

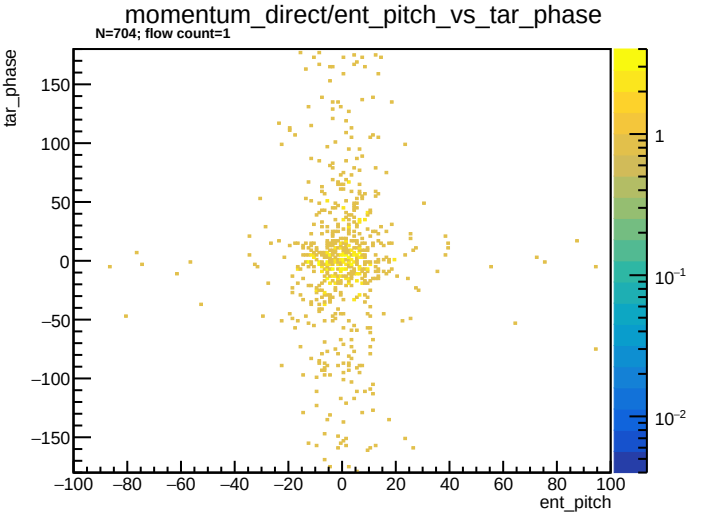

momentum_direct/ent_dt_beam
N=853; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

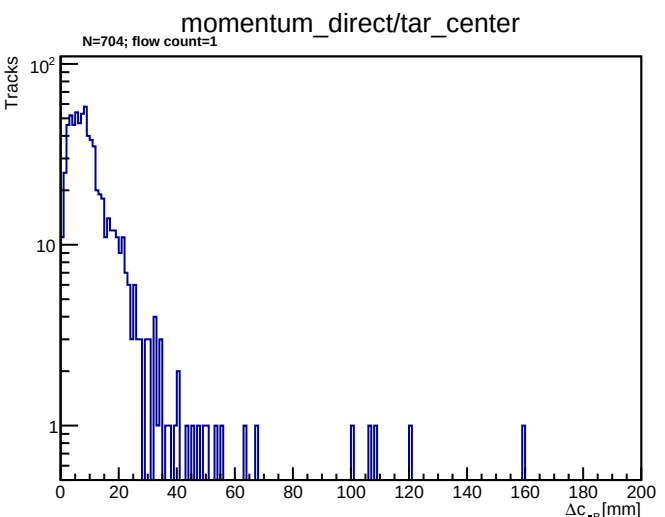

momentum_direct/ent_dt_vs_tar_dt
N=704; flow count=0
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
10
1
10⁻¹

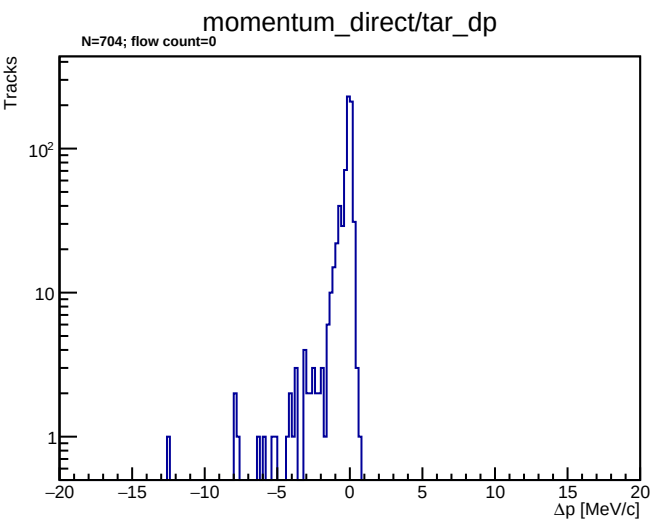

momentum_direct/ent_dx
N=853; flow count=4
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

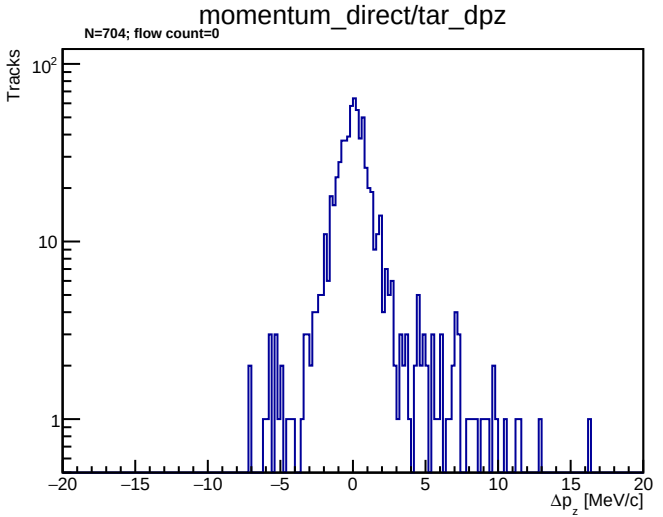

momentum_direct/ent_dy
N=853; flow count=7
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

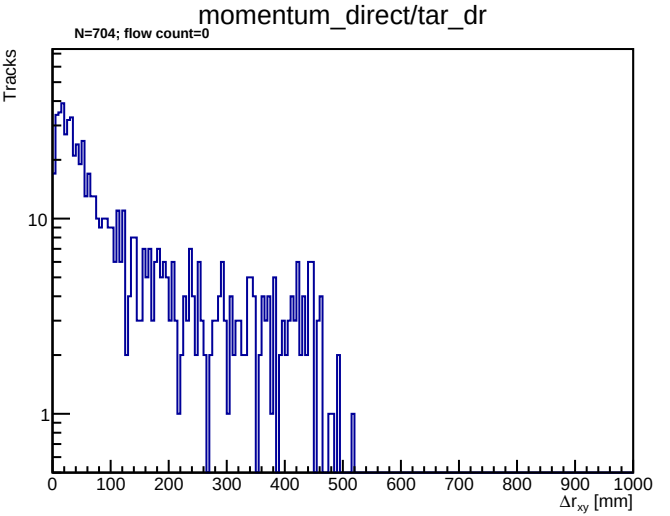

momentum_direct/ent_pitch
N=853; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

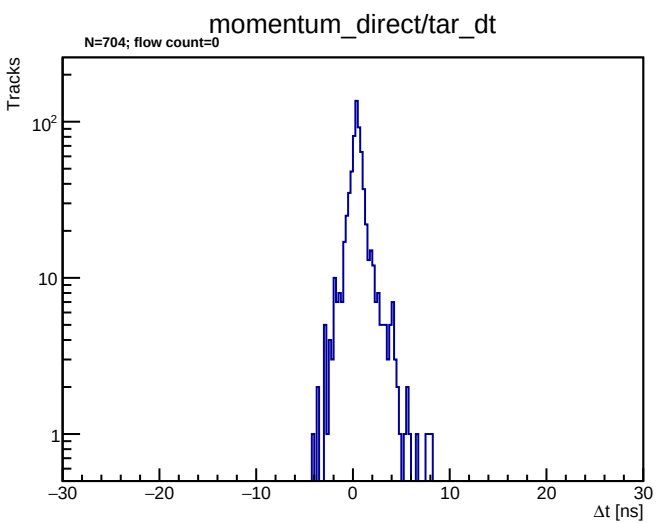

momentum_direct/ent_pitch_vs_tar_phase
N=704; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^-1
10^-2

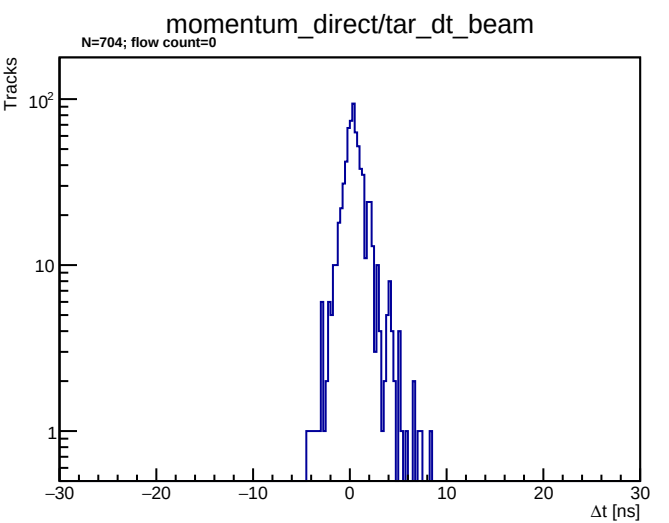

momentum_direct/tar_center
N=704; flow count=1
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

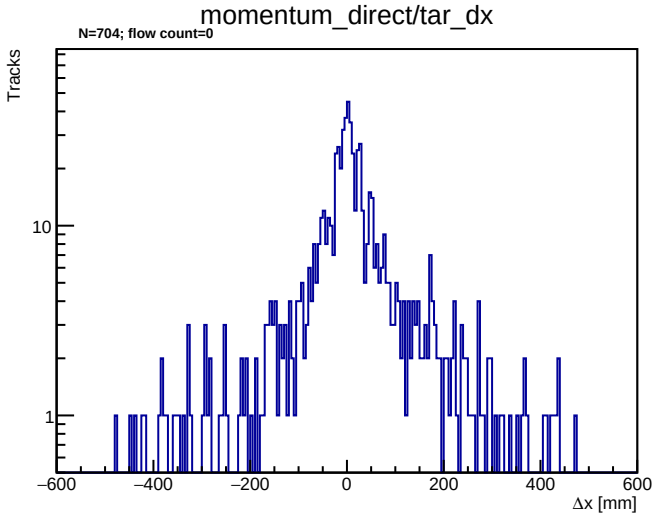

momentum_direct/tar_dp
N=704; flow count=0
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

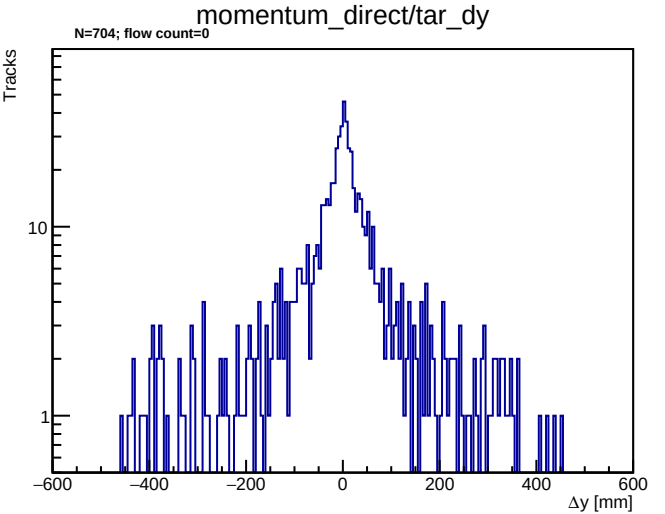

momentum_direct/tar_dpz
N=704; flow count=0
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

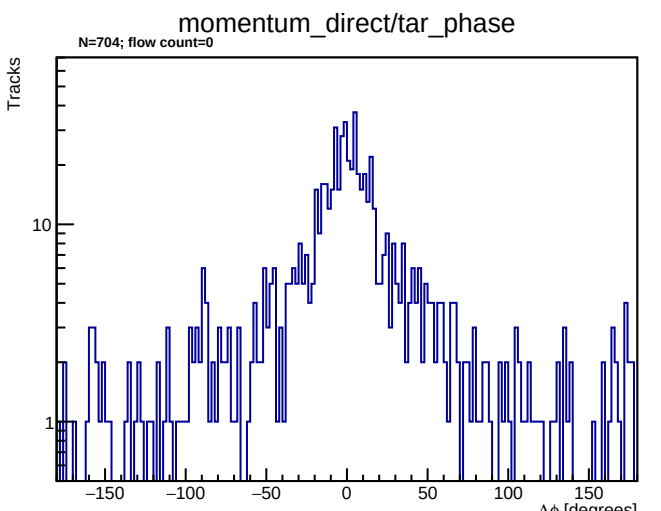

momentum_direct/tar_dr
N=704; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

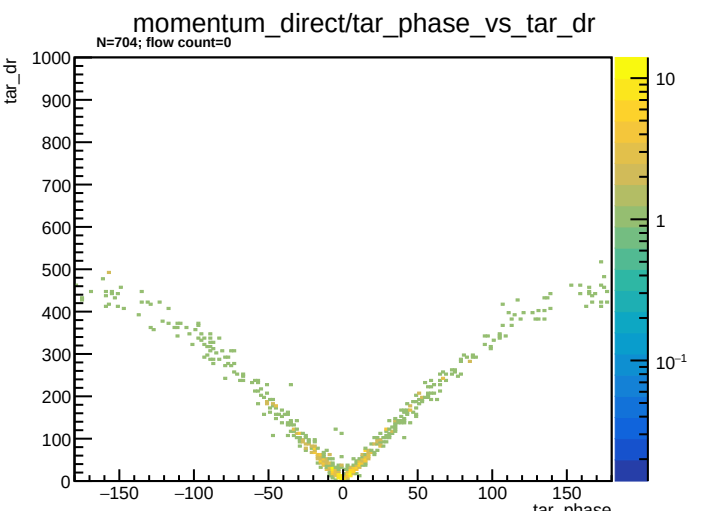

momentum_direct/tar_dt
N=704; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

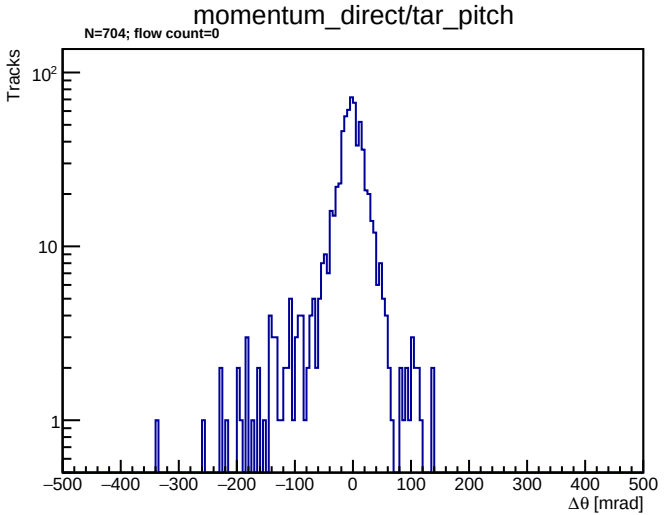

momentum_direct/tar_dt_beam
N=704; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

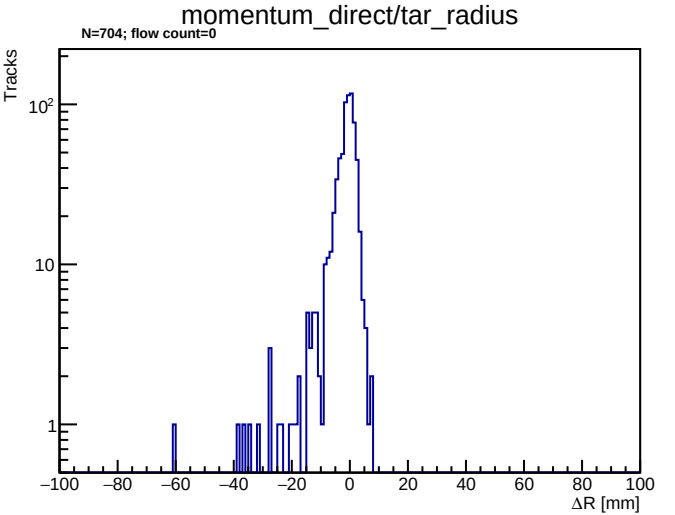

momentum_direct/tar_dx
N=704; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

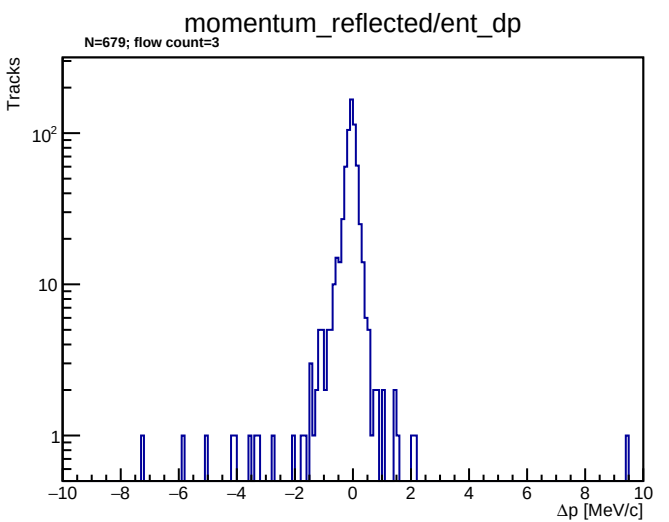

momentum_direct/tar_dy
N=704; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

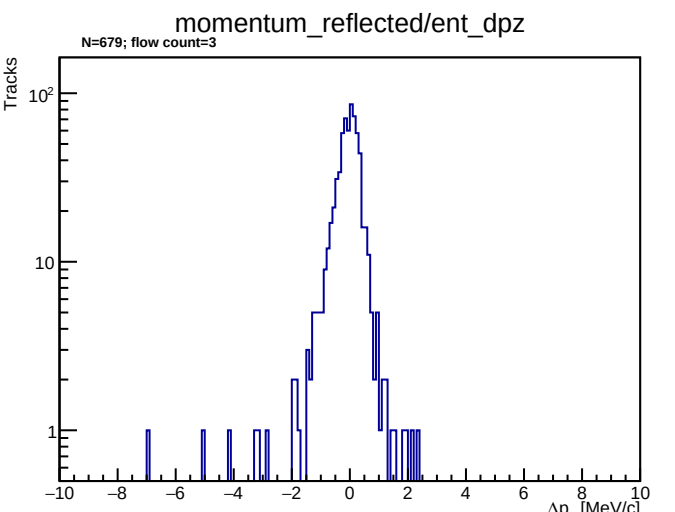

momentum_direct/tar_phase
N=704; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

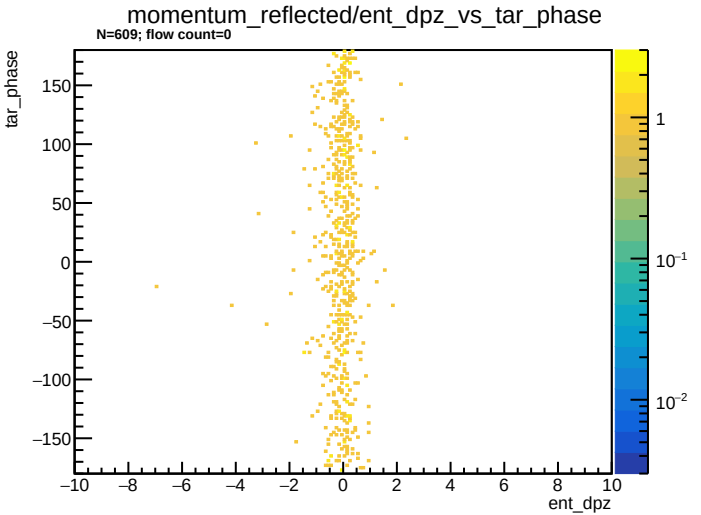

momentum_direct/tar_phase_vs_tar_dr
N=704; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
10
1
10⁻¹

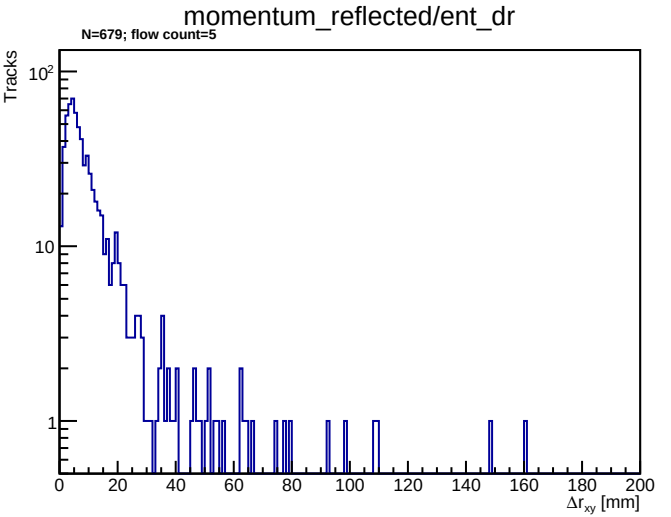

momentum_direct/tar_pitch
N=704; flow count=0
Tracks
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

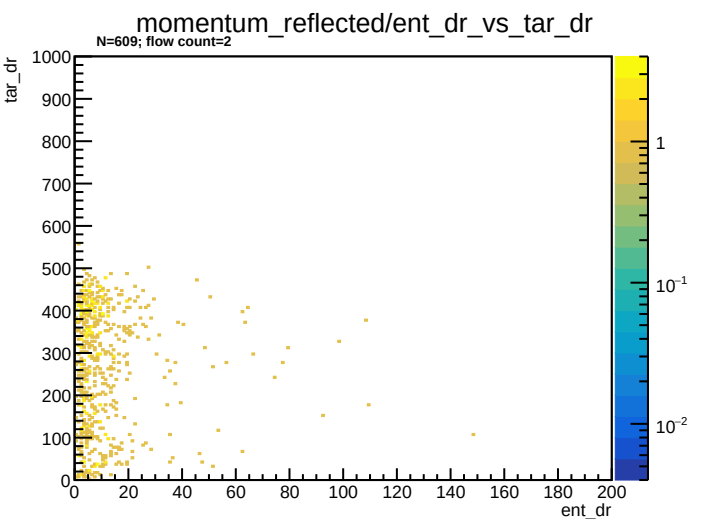

momentum_direct/tar_radius
N=704; flow count=0
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

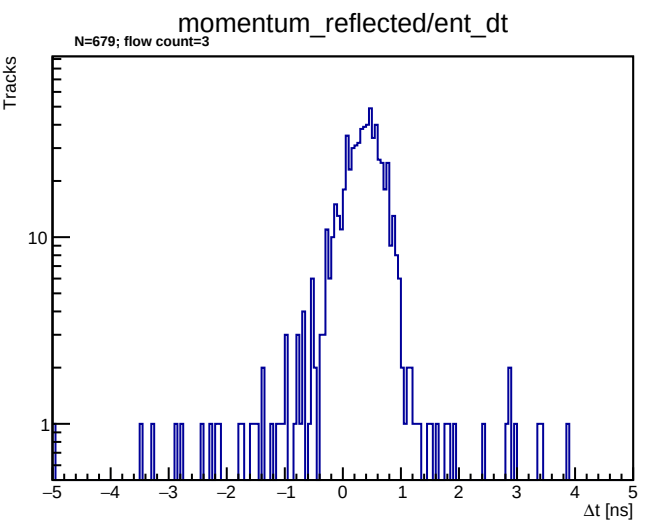

momentum_reflected/ent_dp
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

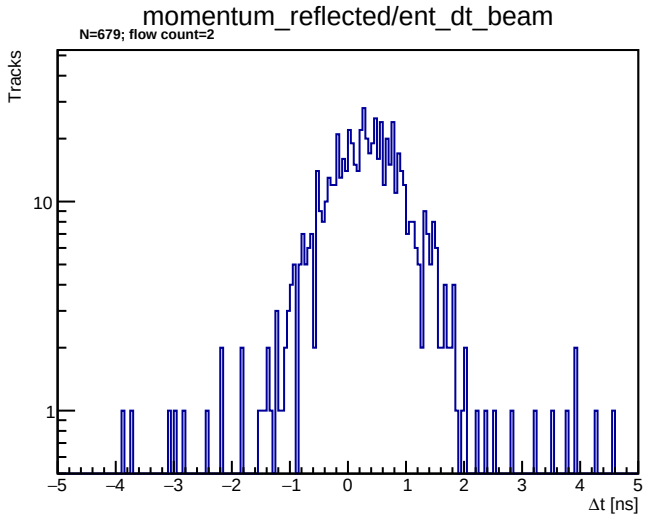

momentum_reflected/ent_dpz
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

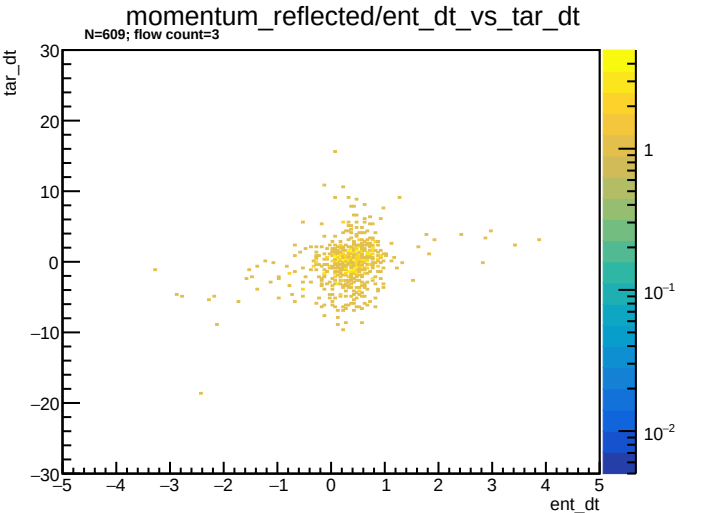

momentum_reflected/ent_dpz_vs_tar_phase
N=609; flow count=0
tar_phase
ent_dpz

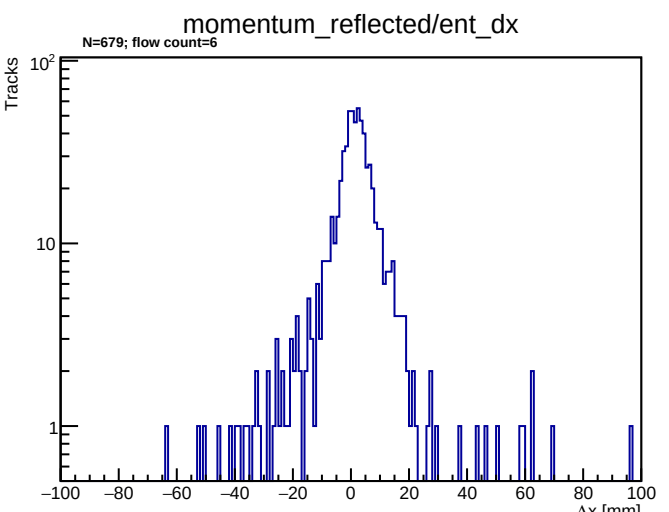

momentum_reflected/ent_dr
N=679; flow count=5
Tracks
Δr_xy [mm]

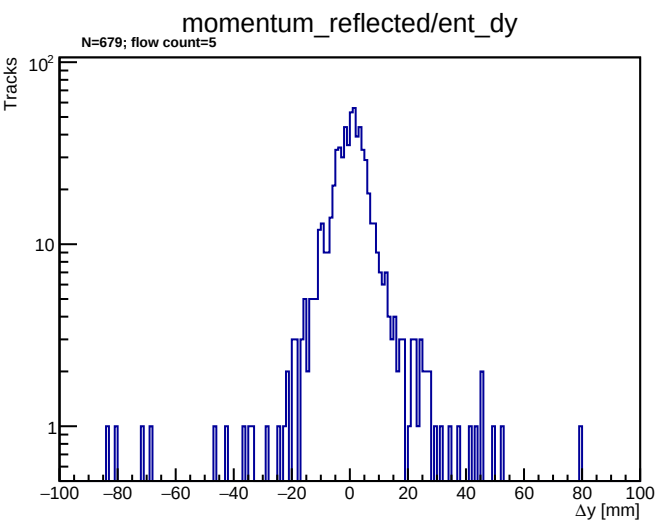

momentum_reflected/ent_dr_vs_tar_dr
N=609; flow count=2
tar_dr
ent_dr

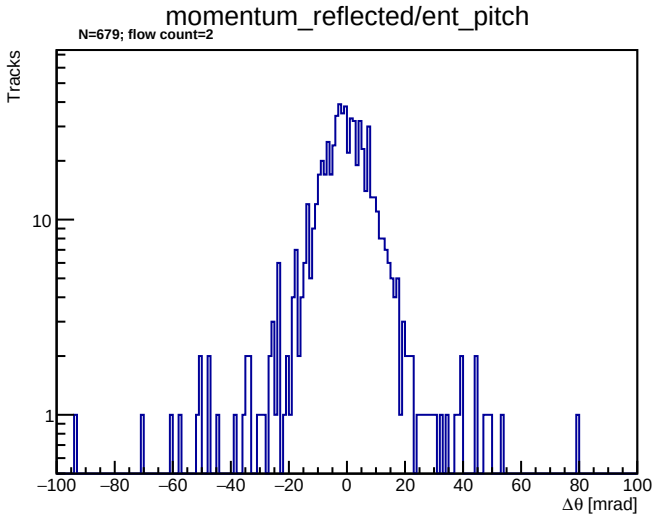

momentum_reflected/ent_dt
N=679; flow count=3
Tracks
Δt [ns]

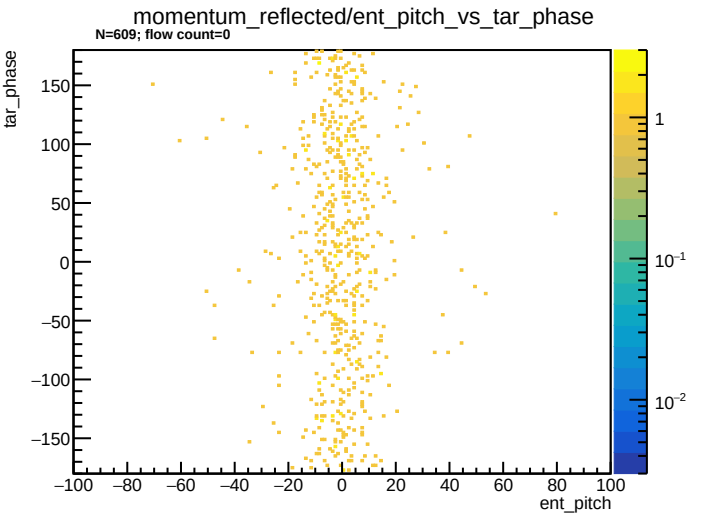

momentum_reflected/ent_dt_beam
N=679; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

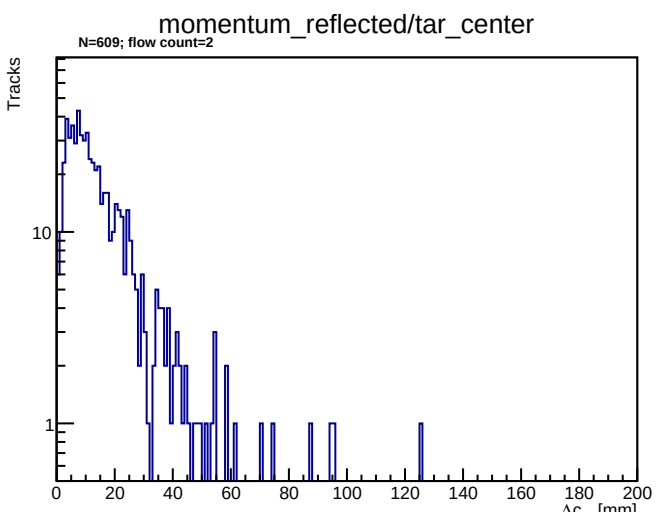

momentum_reflected/ent_dt_vs_tar_dt
N=609; flow count=3
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^{-1}
10^{-2}

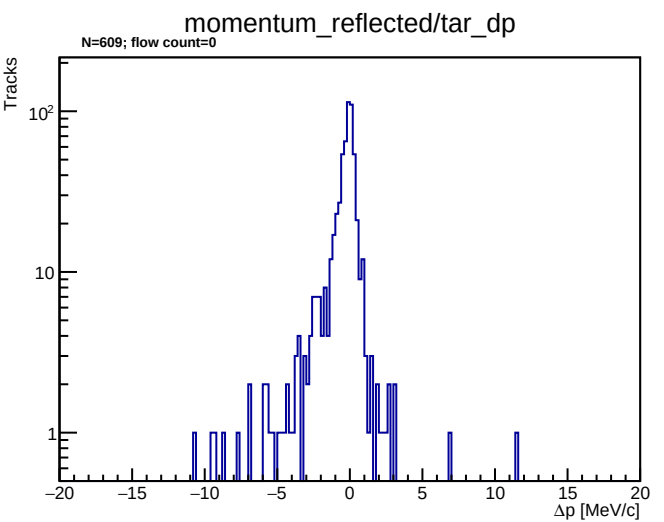

momentum_reflected/ent_dx
N=679; flow count=6
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

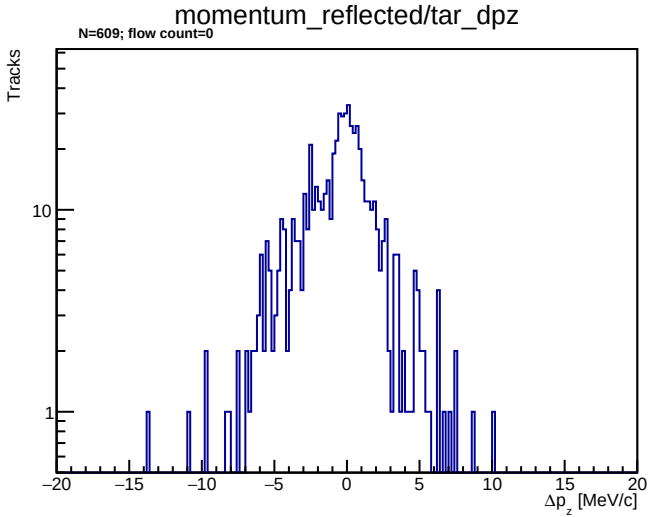

momentum_reflected/ent_dy
N=679; flow count=5
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

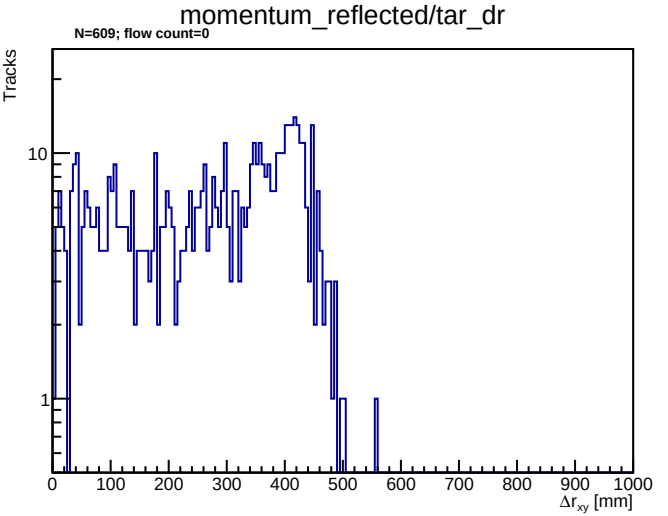

momentum_reflected/ent_pitch
N=679; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

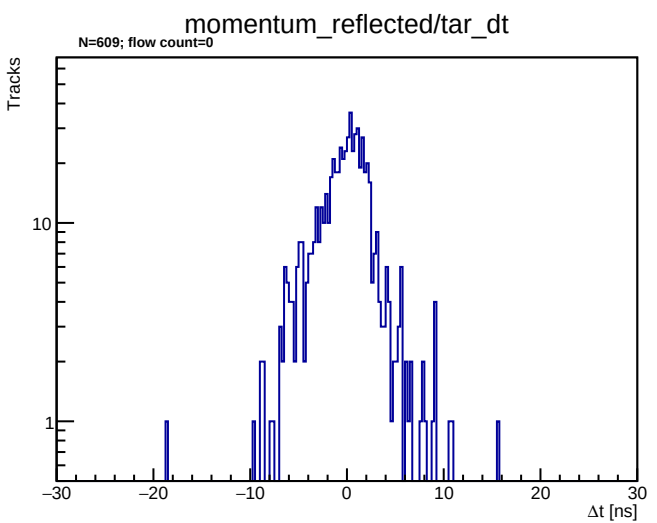

momentum_reflected/ent_pitch_vs_tar_phase
N=609; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²

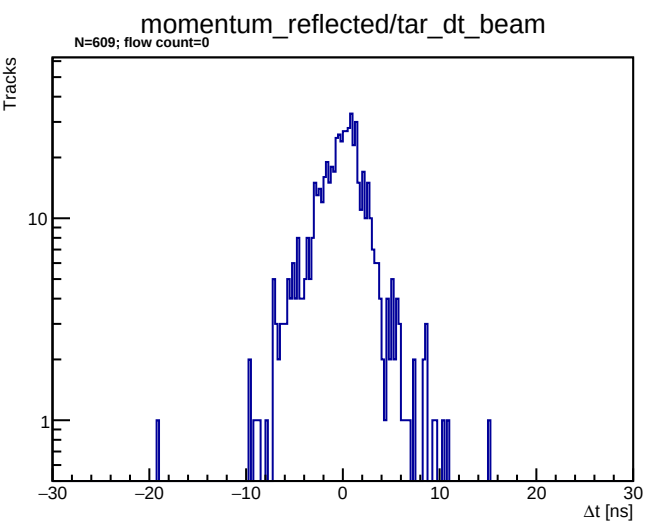

momentum_reflected/tar_center
N=609; flow count=2
Tracks
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

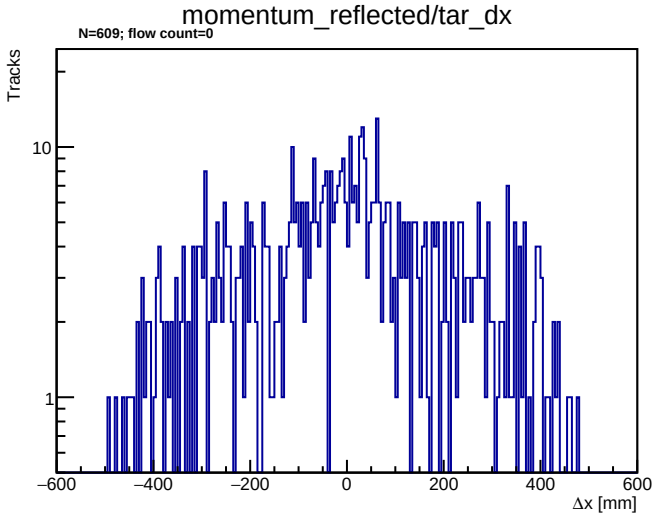

momentum_reflected/tar_dp
N=609; flow count=0
Tracks
10²
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

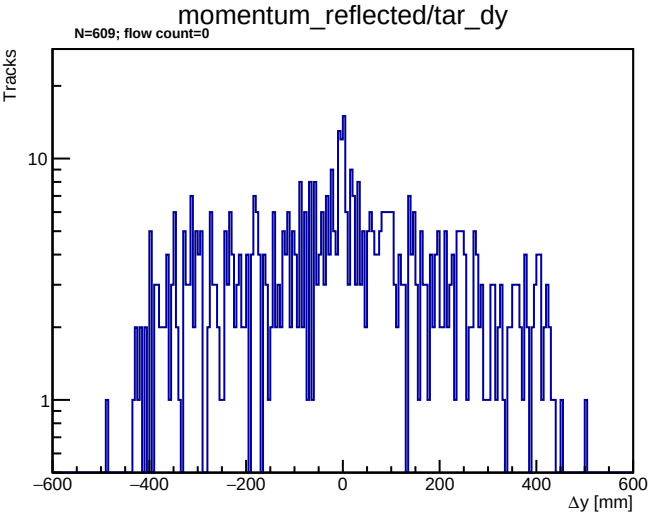

momentum_reflected/tar_dpz
N=609; flow count=0
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

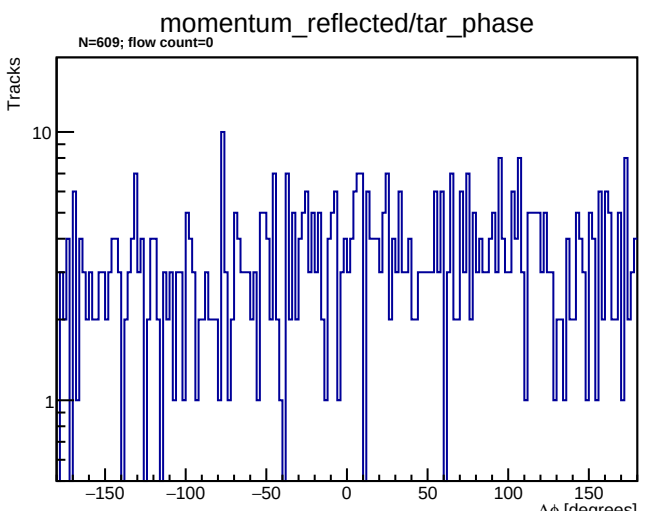

momentum_reflected/tar_dr
N=609; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

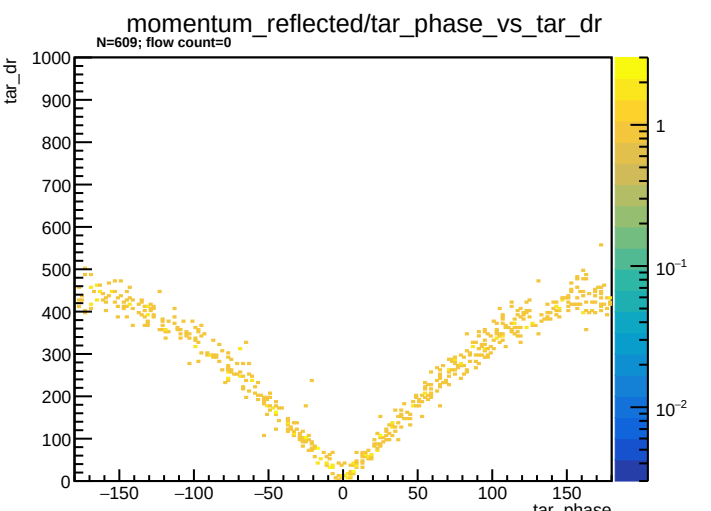

momentum_reflected/tar_dt
N=609; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

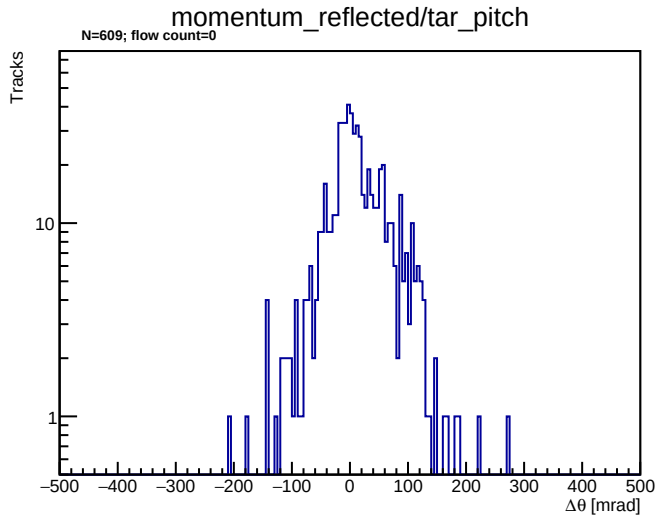

momentum_reflected/tar_dt_beam
N=609; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

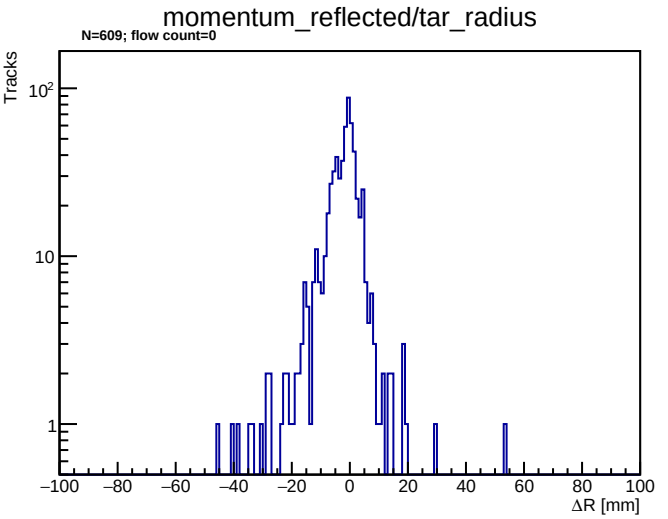

momentum_reflected/tar_dx
N=609; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

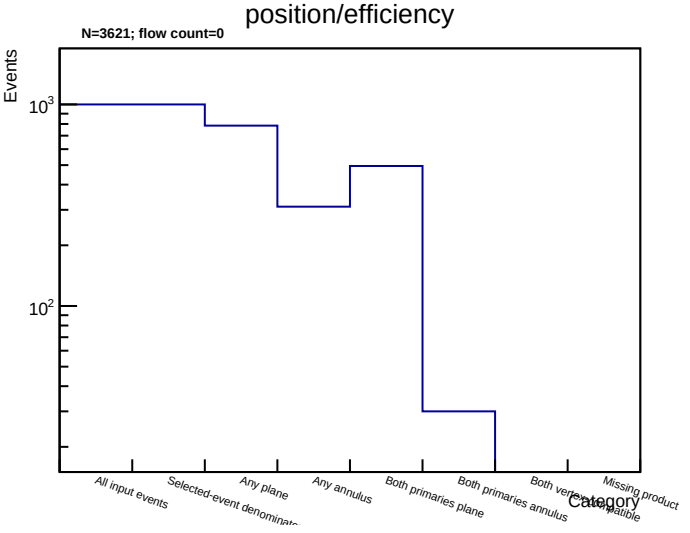

momentum_reflected/tar_dy
N=609; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

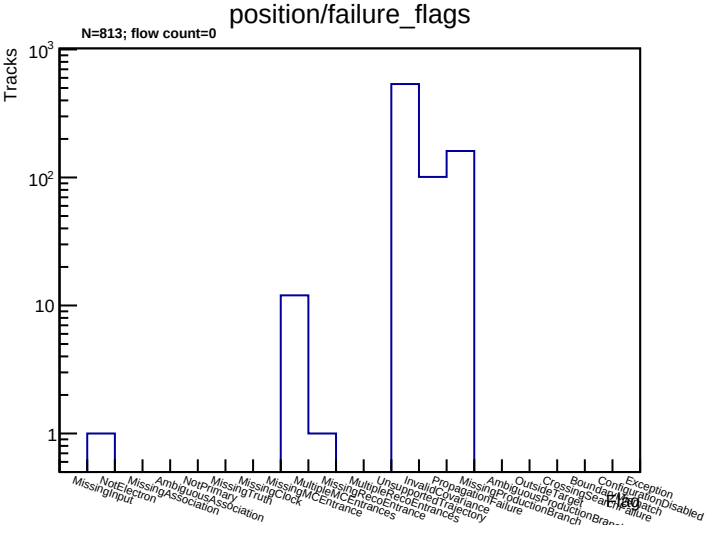

momentum_reflected/tar_phase
N=609; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

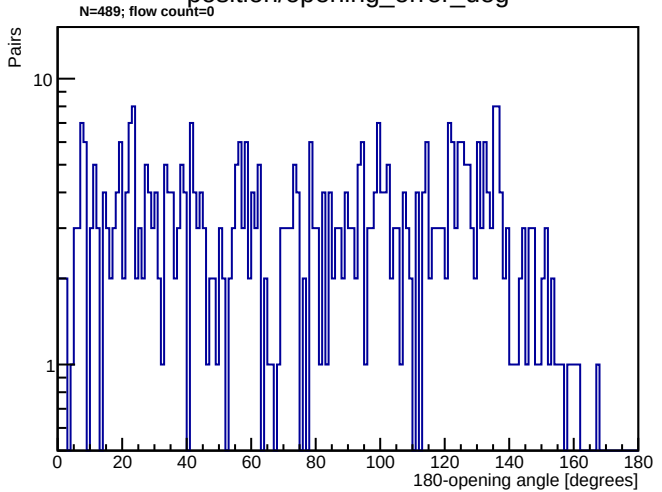

momentum_reflected/tar_phase_vs_tar_dr
N=609; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

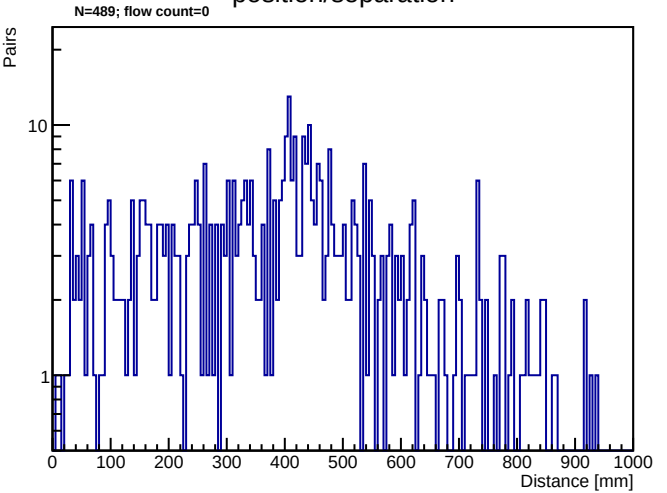

momentum_reflected/tar_pitch
N=609; flow count=0
Tracks
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

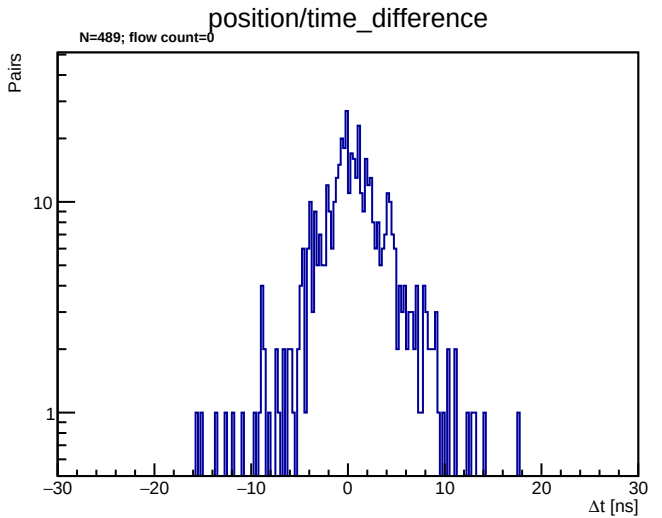

momentum_reflected/tar_radius
N=609; flow count=0
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

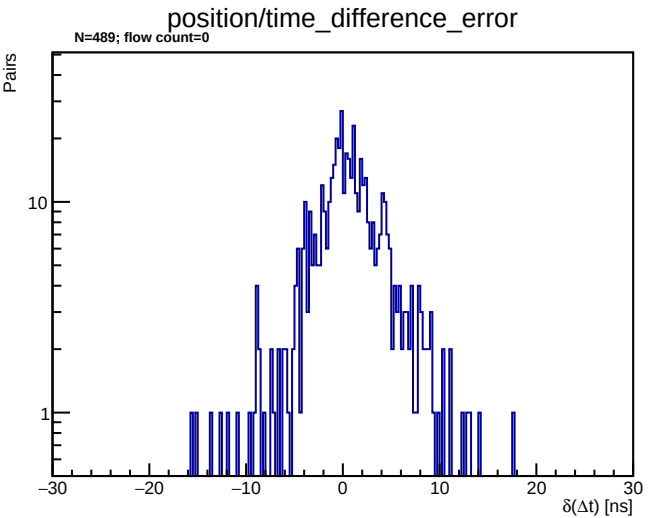

position/efficiency
N=3621; flow count=0
Events
10³
10²
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product
Category

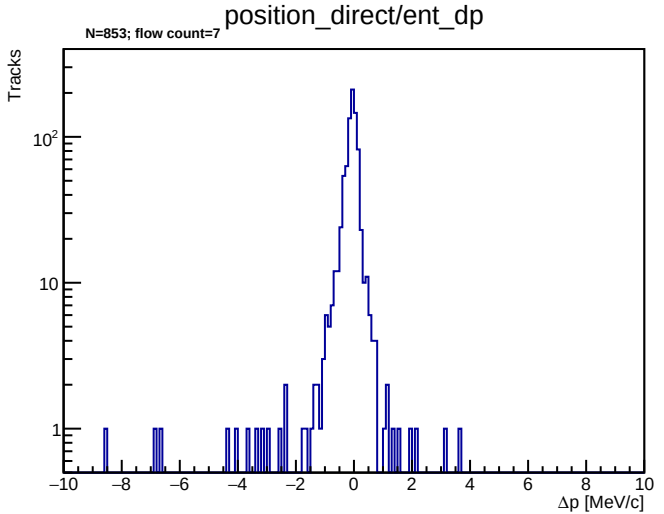

position/failure_flags
N=813; flow count=0
Tracks
10³
10²
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleReferenceEntrance
MissingReferenceEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProductionBranch
AmbiguousProductionBranch
OutsideTarget
CrossingSeamSearch
BoundaryConfiguration
ConfigurationDisabled
Exception
Error

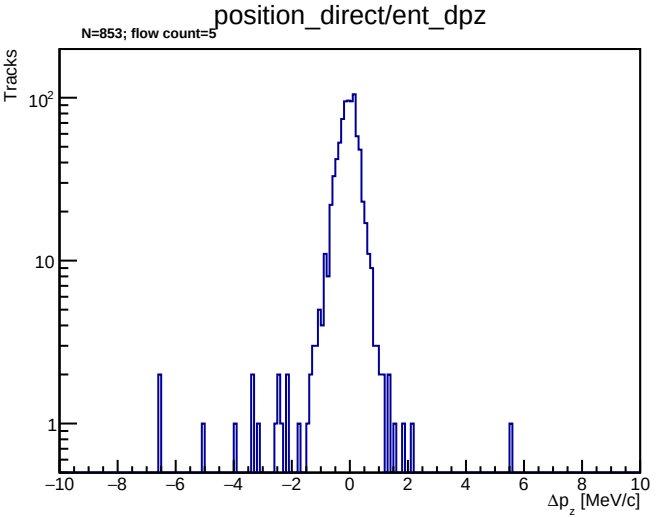

position/opening_error_deg
N=489; flow count=0
Pairs
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

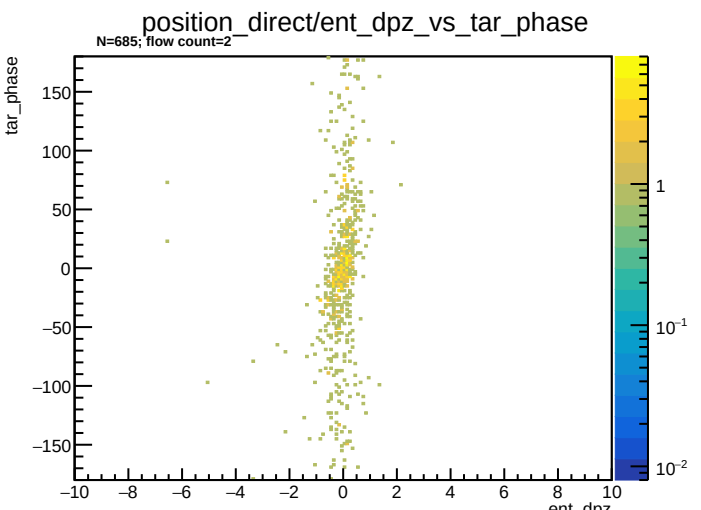

position/separation
N=489; flow count=0
Pairs
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

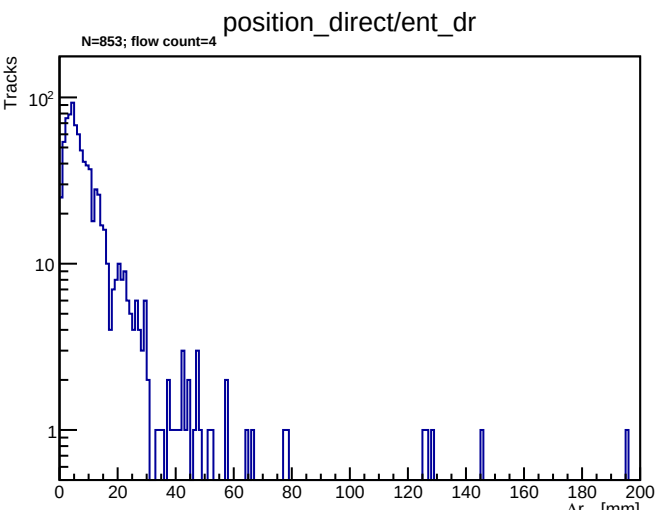

position/time_difference
N=489; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

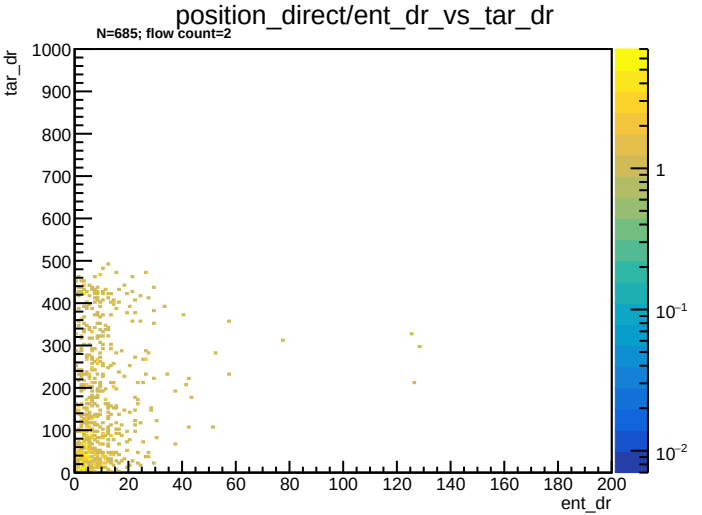

position/time_difference_error
N=489; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

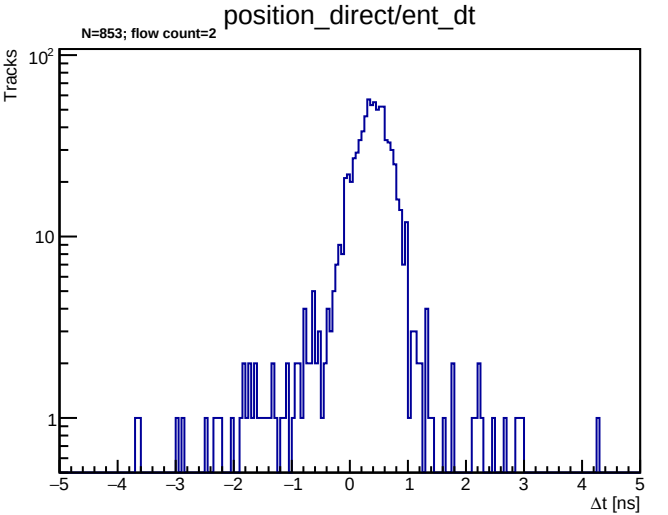

position_direct/ent_dp
N=853; flow count=7
Tracks
Δp [MeV/c]

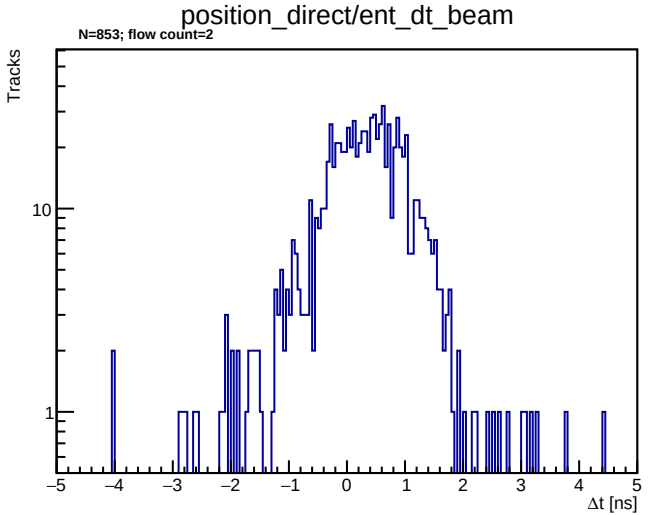

position_direct/ent_dpz
N=853; flow count=5
Tracks
Δp_z [MeV/c]

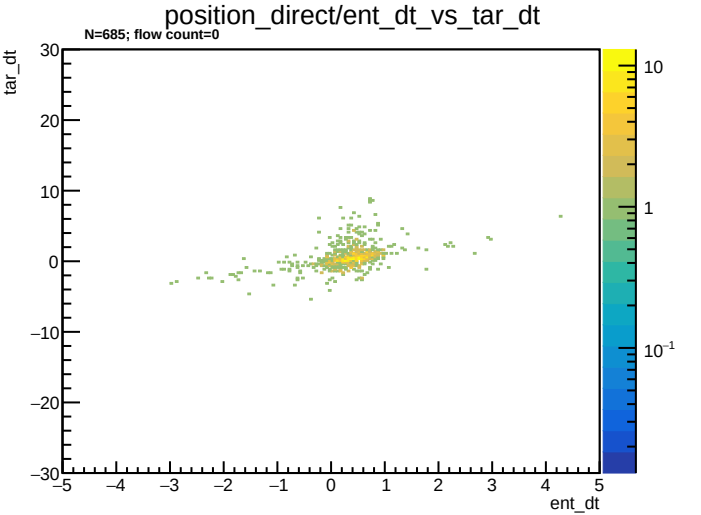

position_direct/ent_dpz_vs_tar_phase
N=685; flow count=2
tar_phase
ent_dpz

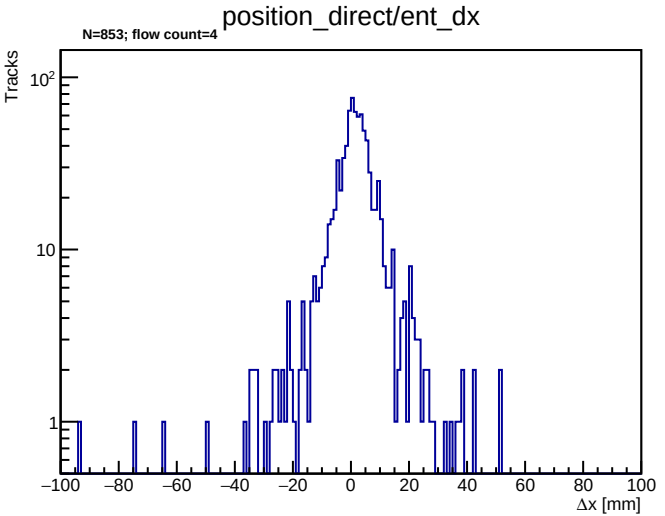

position_direct/ent_dr
N=853; flow count=4
Tracks
Δr_xy [mm]

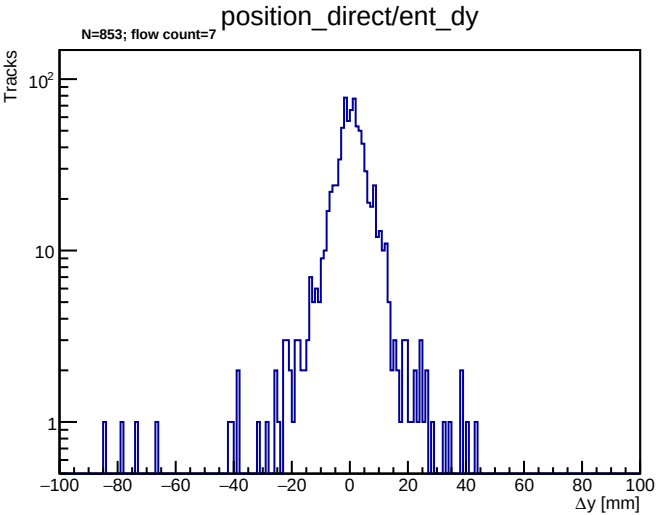

position_direct/ent_dr_vs_tar_dr
N=685; flow count=2
tar_dr
ent_dr

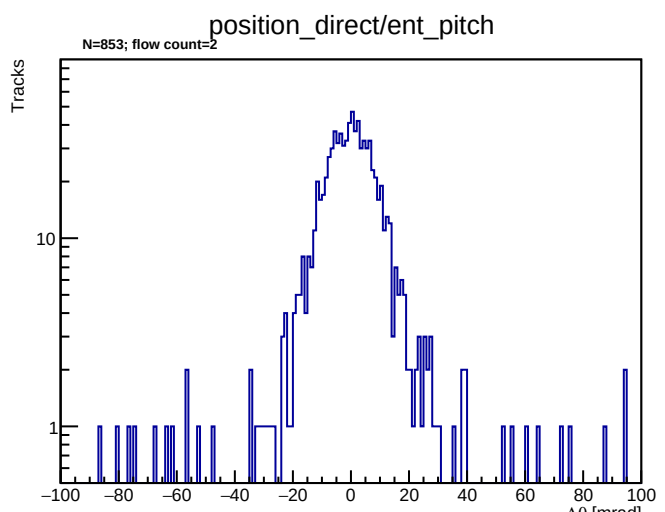

position_direct/ent_dt
N=853; flow count=2
Tracks
Δt [ns]

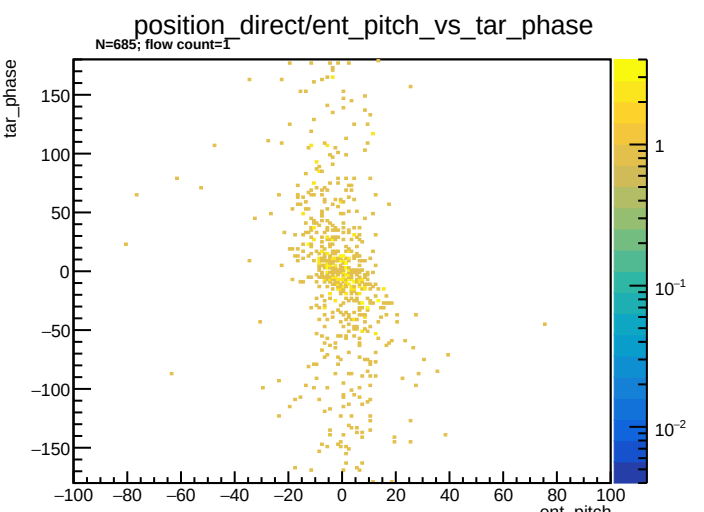

position_direct/ent_dt_beam
N=853; flow count=2
Tracks
Δt [ns]

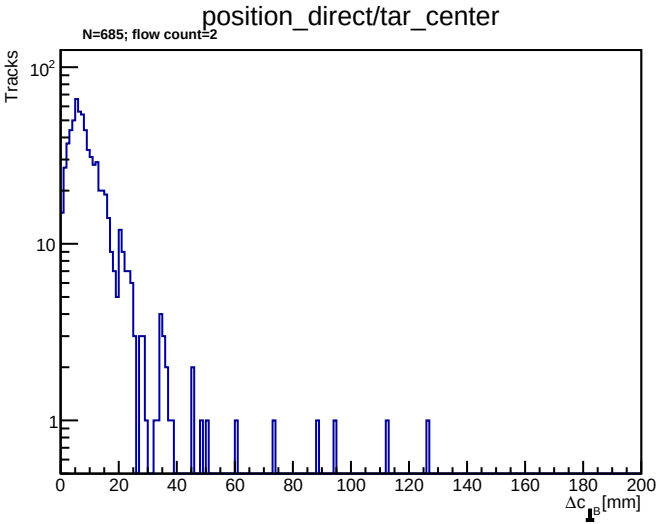

position_direct/ent_dt_vs_tar_dt
N=685; flow count=0
tar_dt
ent_dt

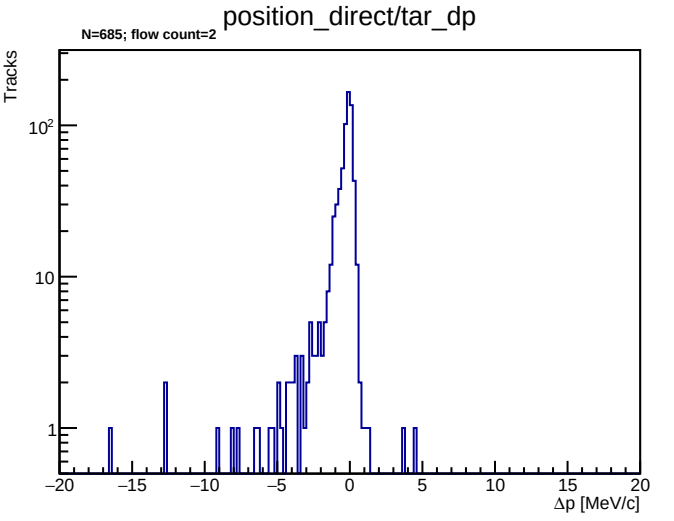

position_direct/ent_dx
N=853; flow count=4
Tracks
Δx [mm]

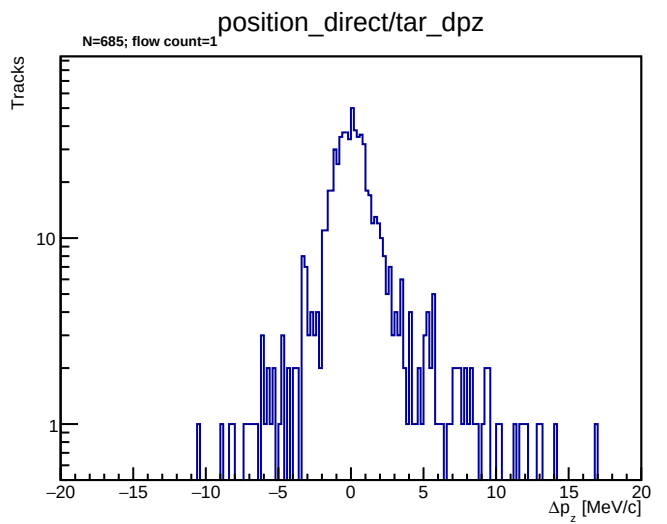

position_direct/ent_dy
N=853; flow count=7
Tracks
Δy [mm]

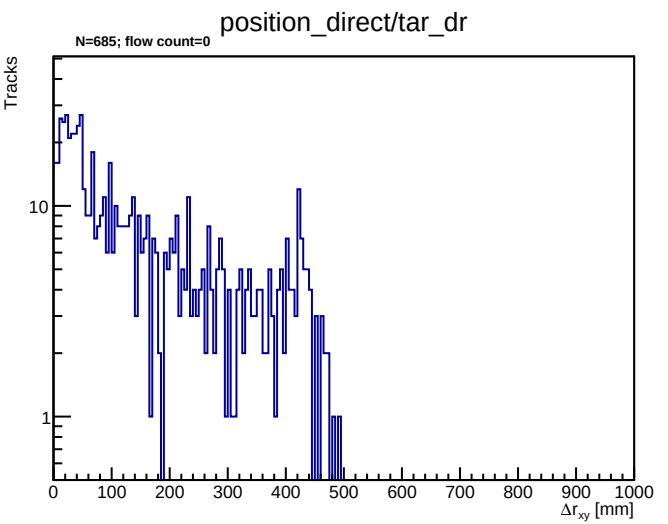

position_direct/ent_pitch
N=853; flow count=2
Tracks
Δθ [mrad]

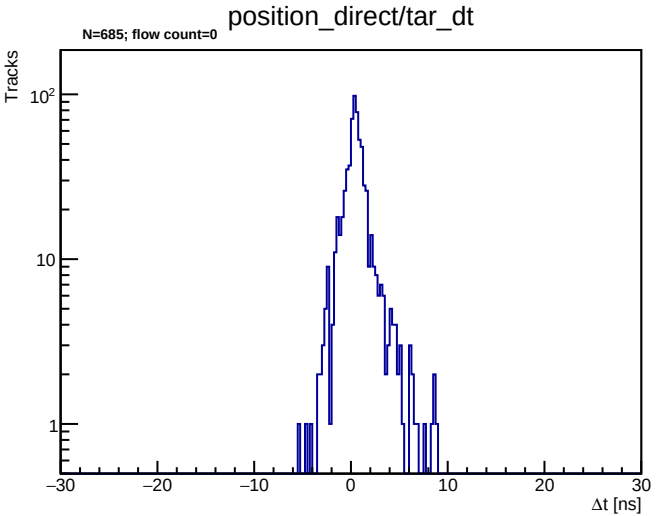

position_direct/ent_pitch_vs_tar_phase
N=685; flow count=1
tar_phase
ent_pitch

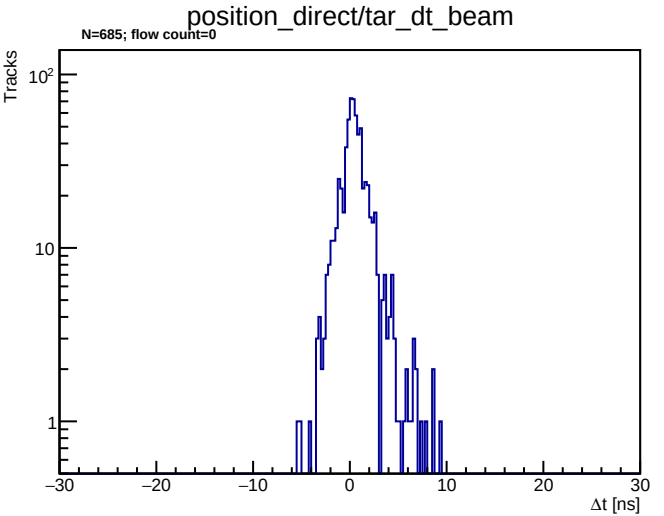

position_direct/tar_center
N=685; flow count=2
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

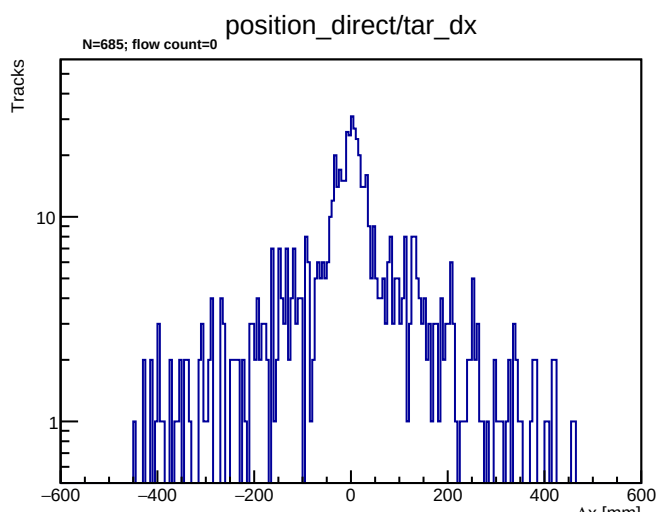

position_direct/tar_dp
N=685; flow count=2
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

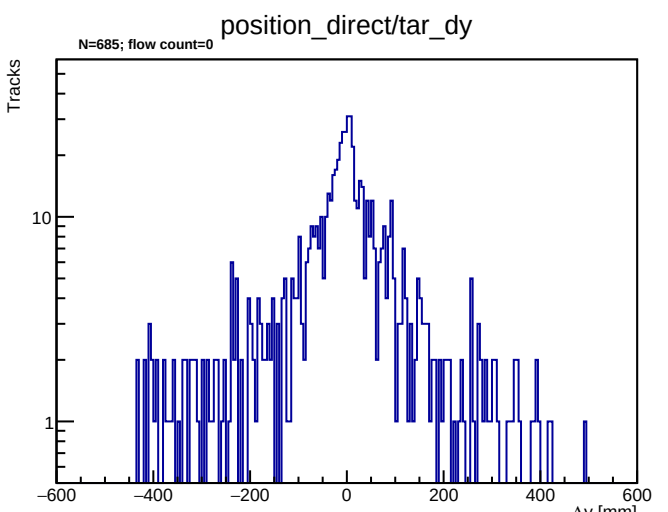

position_direct/tar_dpz
N=685; flow count=1
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

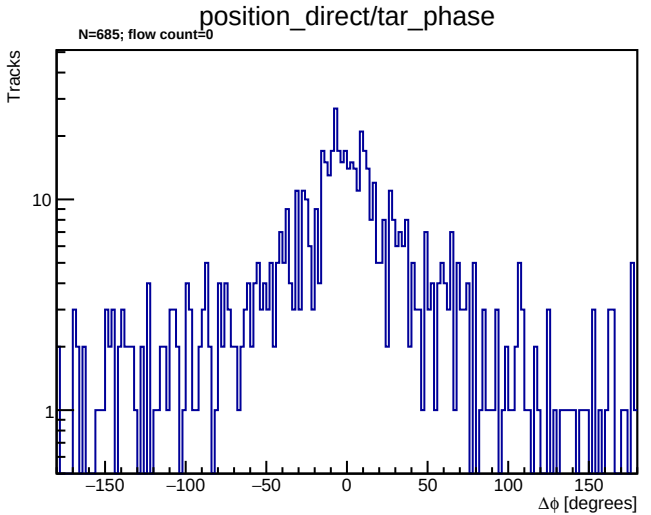

position_direct/tar_dr
N=685; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

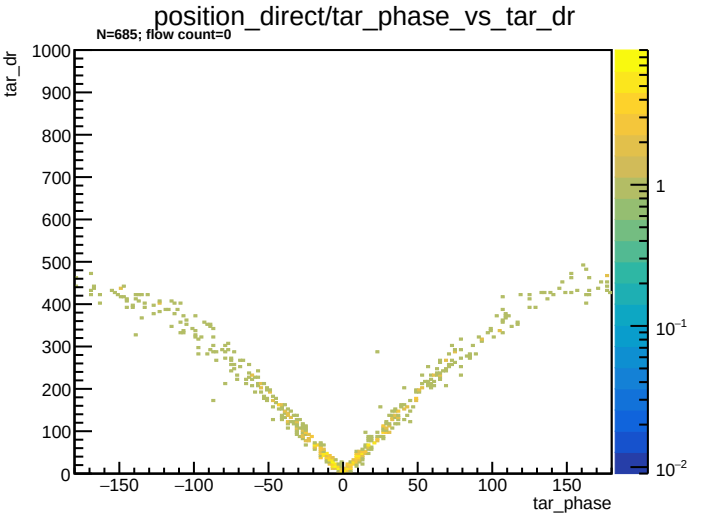

position_direct/tar_dt
N=685; flow count=0
Tracks
10²
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

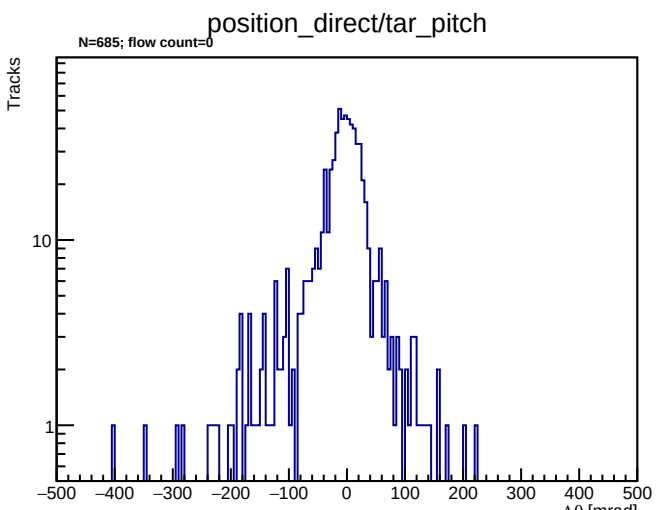

position_direct/tar_dt_beam
N=685; flow count=0
Tracks
10²
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

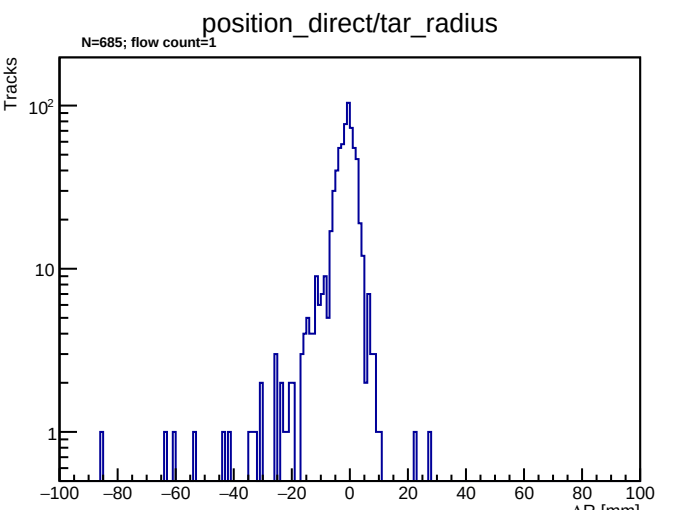

position_direct/tar_dx
N=685; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

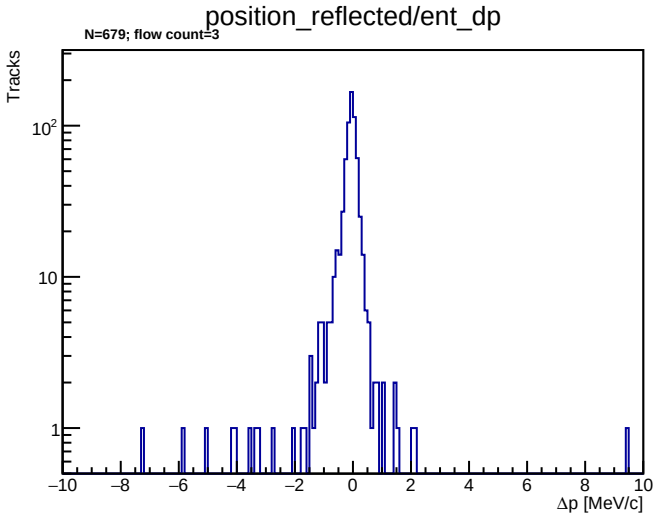

position_direct/tar_dy
N=685; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

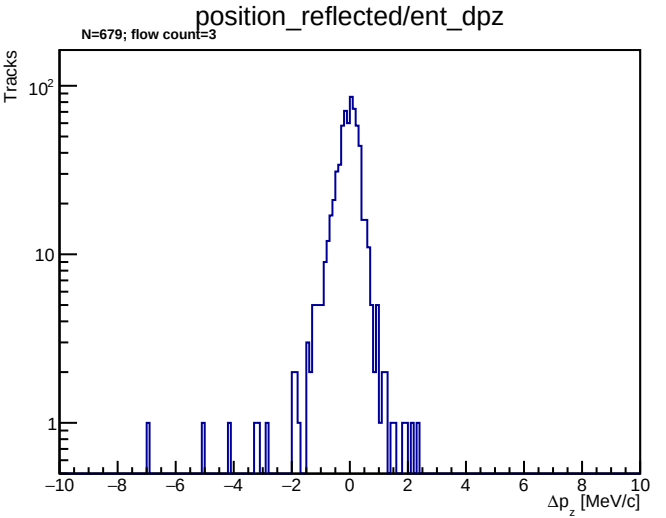

position_direct/tar_phase
N=685; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

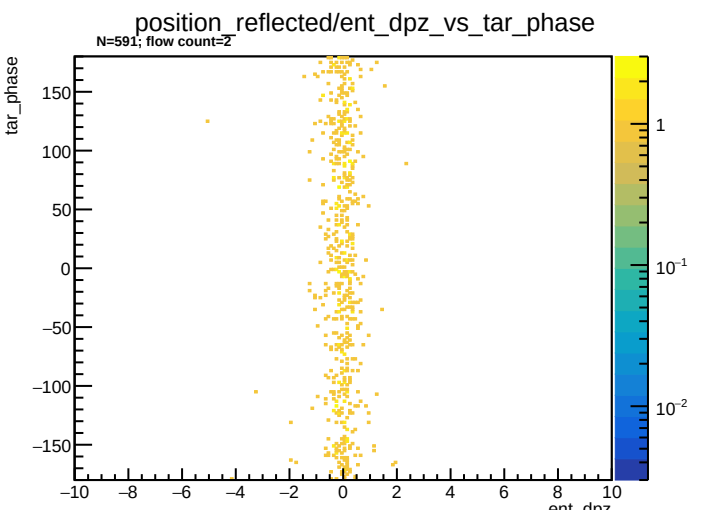

position_direct/tar_phase_vs_tar_dr
N=685; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10^−1
10^−2

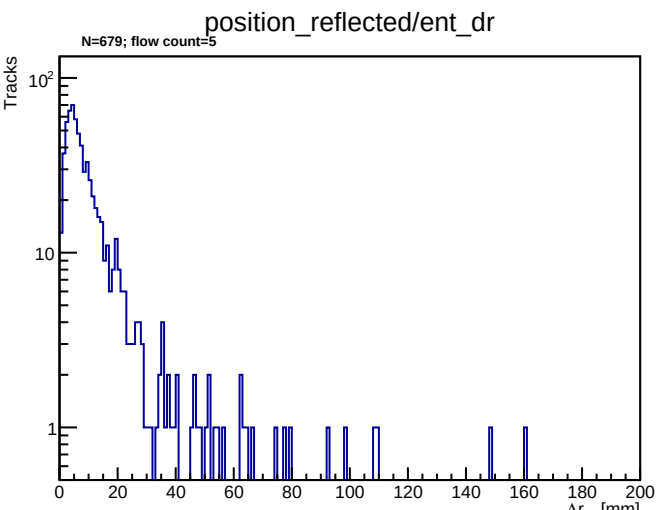

position_direct/tar_pitch
N=685; flow count=0
Tracks
10
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

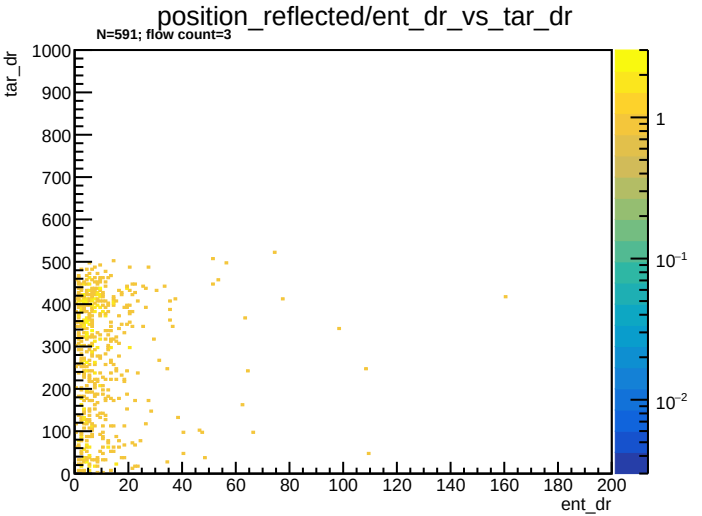

position_direct/tar_radius
N=685; flow count=1
Tracks
10^2
10
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

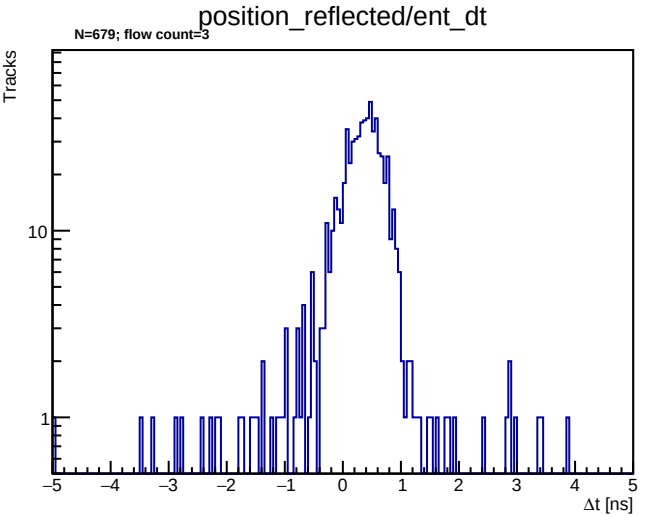

position_reflected/ent_dp
N=679; flow count=3
Tracks
Δp [MeV/c]

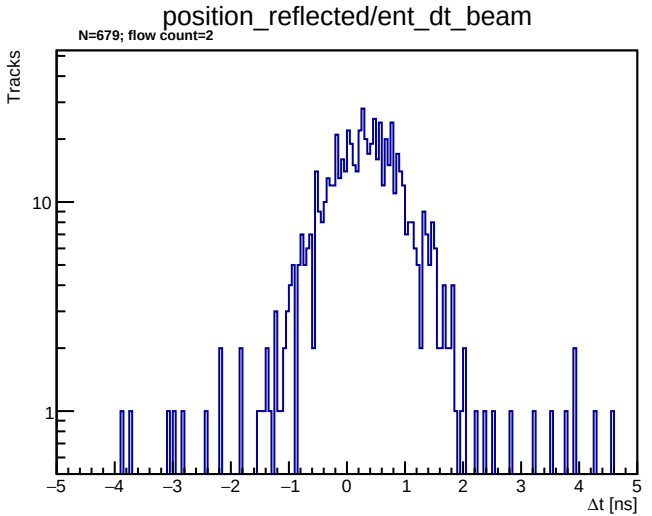

position_reflected/ent_dpz
N=679; flow count=3
Tracks
Δp_z [MeV/c]

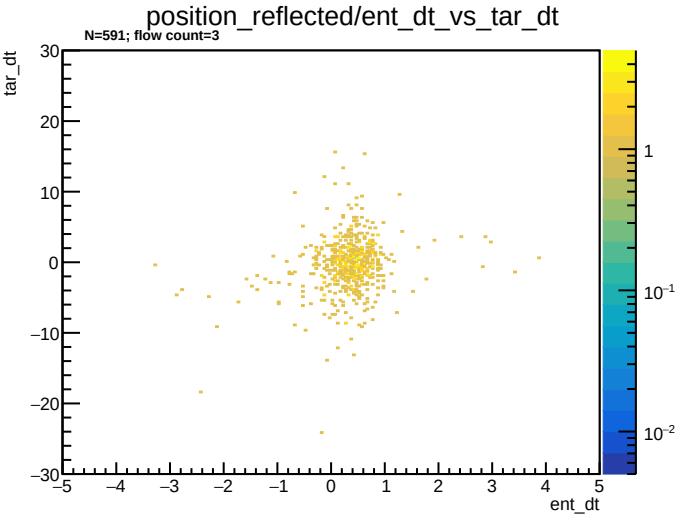

position_reflected/ent_dpz_vs_tar_phase
N=591; flow count=2
tar_phase
ent_dpz

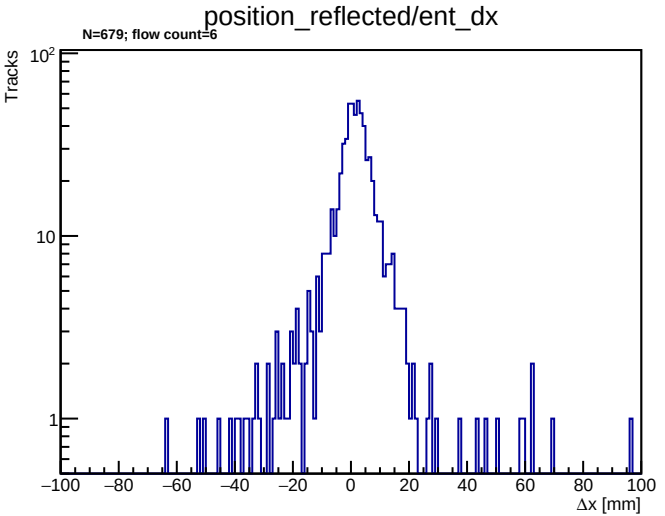

position_reflected/ent_dr
N=679; flow count=5
Tracks
Δr_xy [mm]

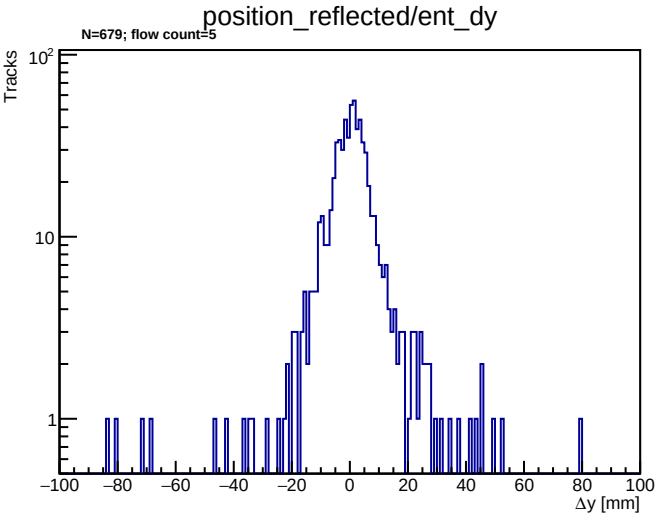

position_reflected/ent_dr_vs_tar_dr
N=591; flow count=3
tar_dr
ent_dr

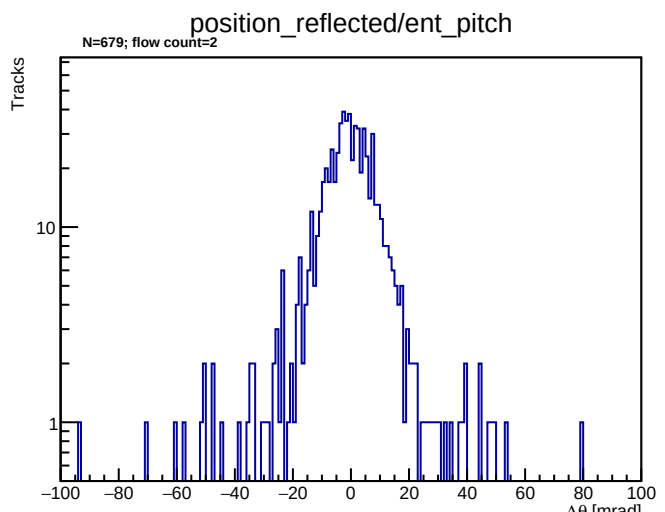

position_reflected/ent_dt
N=679; flow count=3
Tracks
Δt [ns]

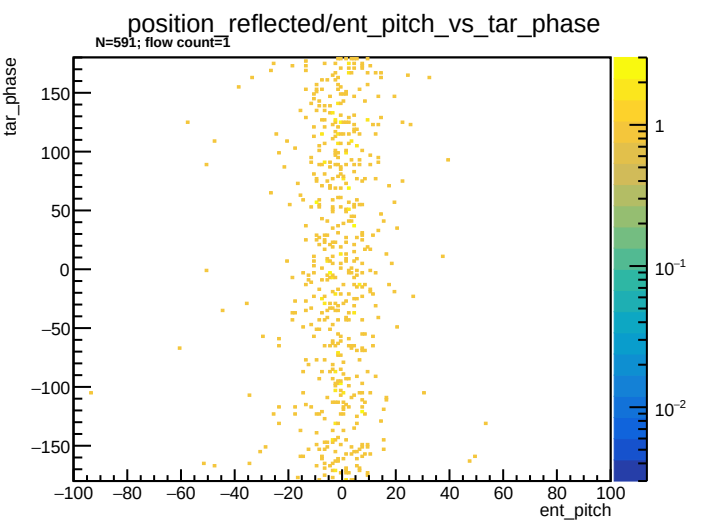

position_reflected/ent_dt_beam
N=679; flow count=2
Tracks
Δt [ns]

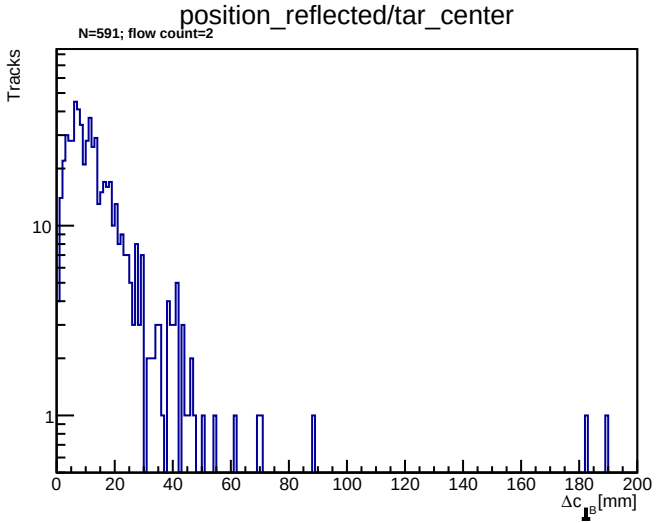

position_reflected/ent_dt_vs_tar_dt
N=591; flow count=3
tar_dt
ent_dt

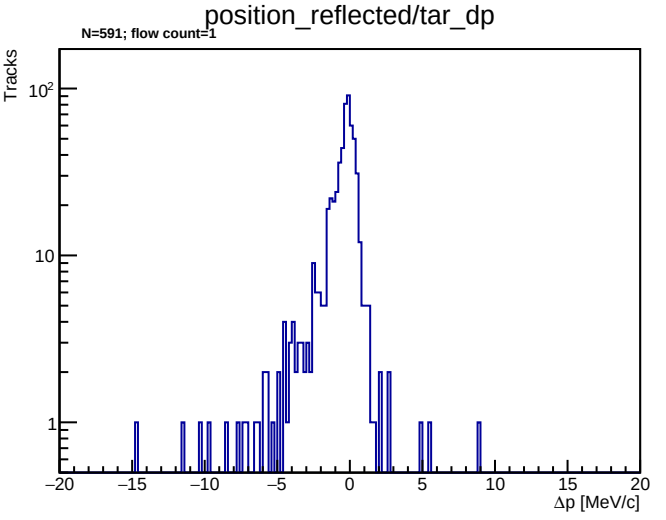

position_reflected/ent_dx
N=679; flow count=6
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

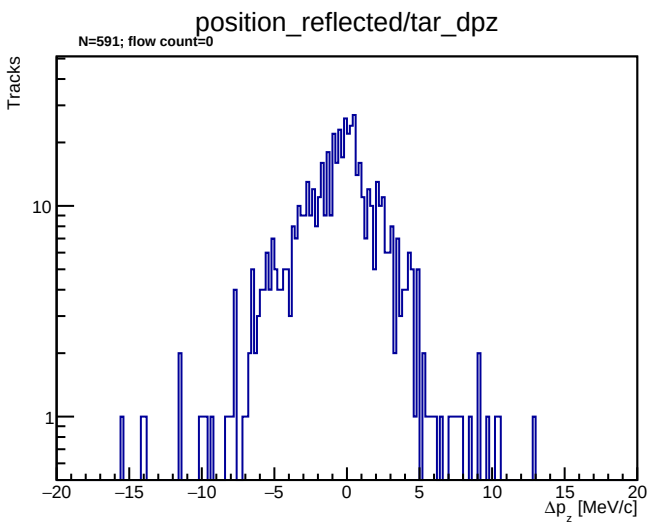

position_reflected/ent_dy
N=679; flow count=5
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

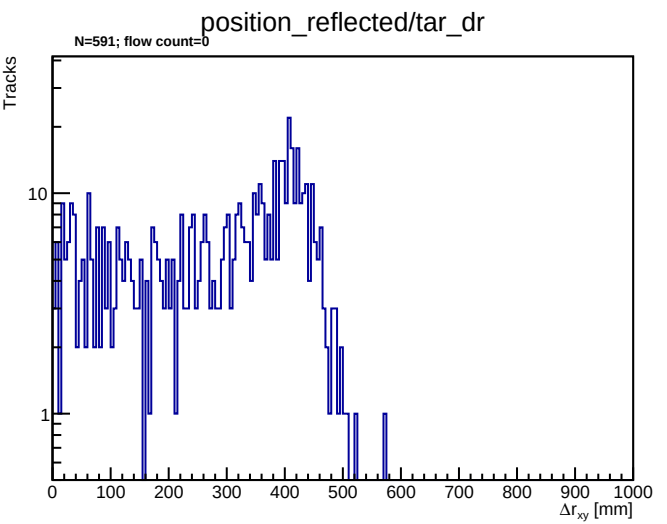

position_reflected/ent_pitch
N=679; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

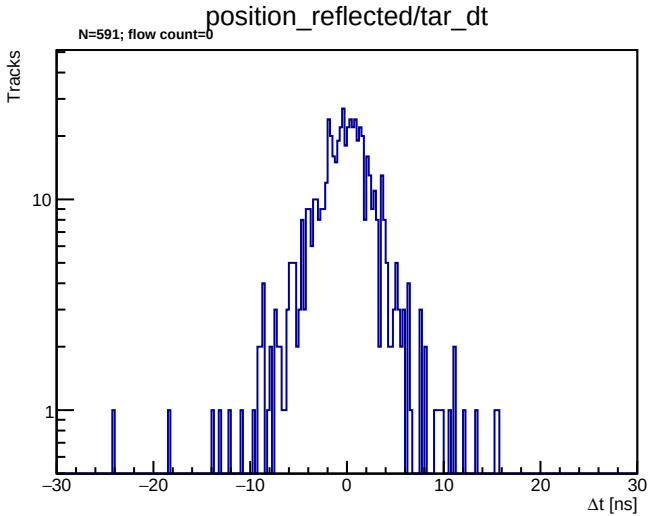

position_reflected/ent_pitch_vs_tar_phase
N=591; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^{-1}
10^{-2}

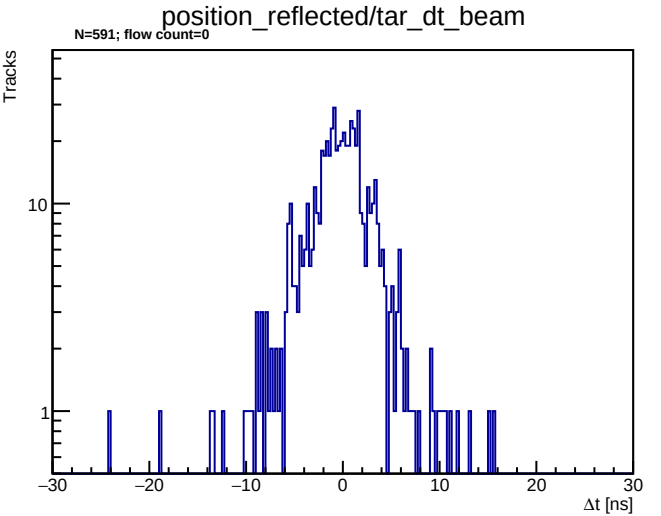

position_reflected/tar_center
N=591; flow count=2
Tracks
10
1
0
20
40
60
80
100
120
140
160
180
200
Δc_B [mm]

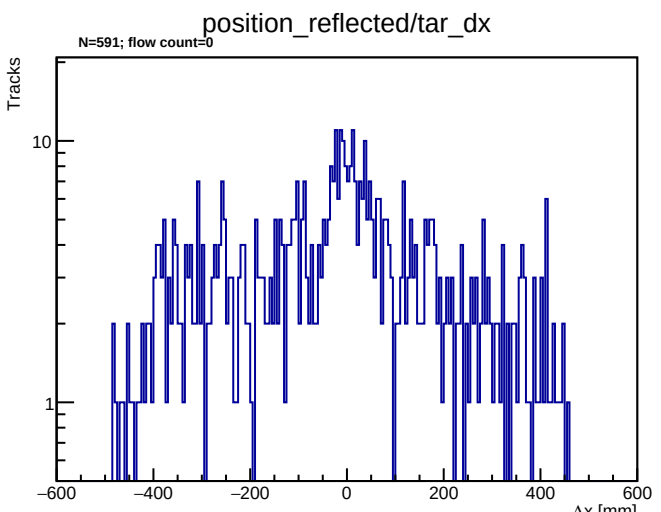

position_reflected/tar_dp
N=591; flow count=1
Tracks
10^2
10
1
−20
−15
−10
−5
0
5
10
15
20
Δp [MeV/c]

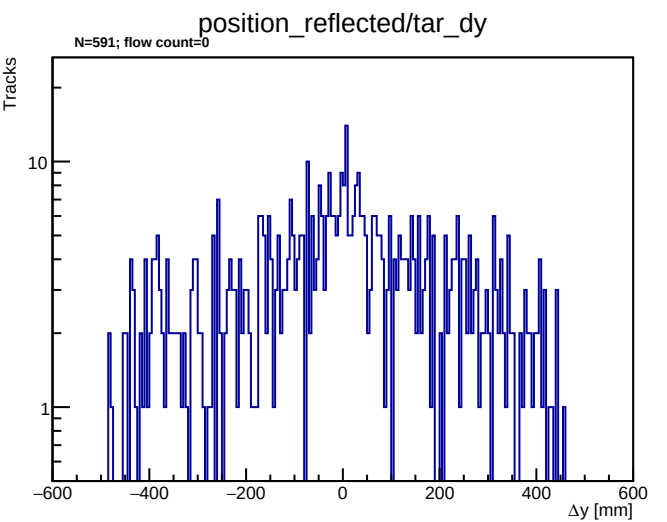

position_reflected/tar_dpz
N=591; flow count=0
Tracks
10
1
−20
−15
−10
−5
0
5
10
15
20
Δp_z [MeV/c]

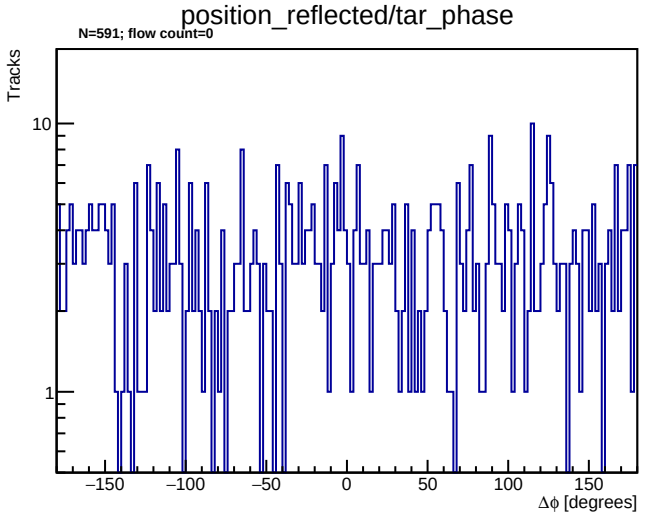

position_reflected/tar_dr
N=591; flow count=0
Tracks
10
1
0
100
200
300
400
500
600
700
800
900
1000
Δr_xy [mm]

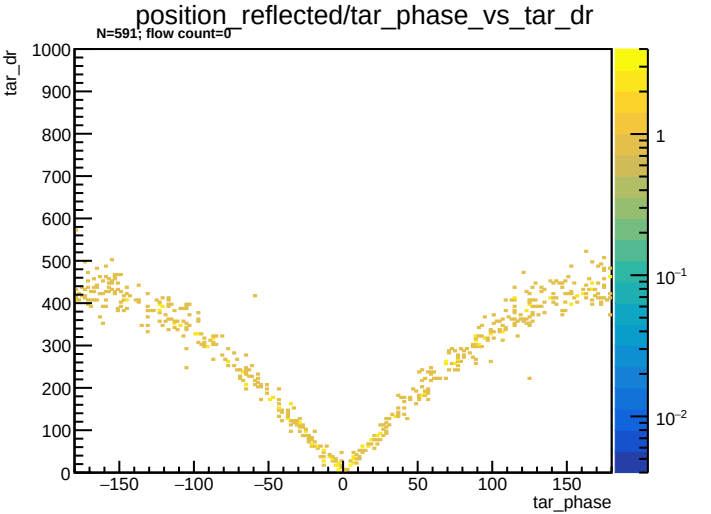

position_reflected/tar_dt
N=591; flow count=0
Tracks
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

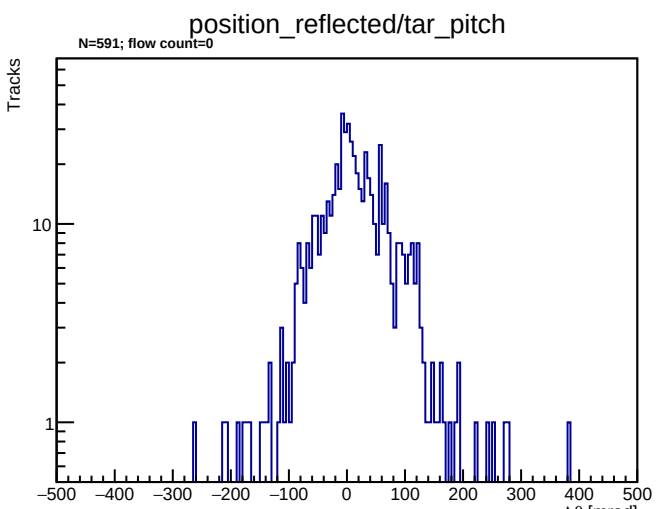

position_reflected/tar_dt_beam
N=591; flow count=0
Tracks
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

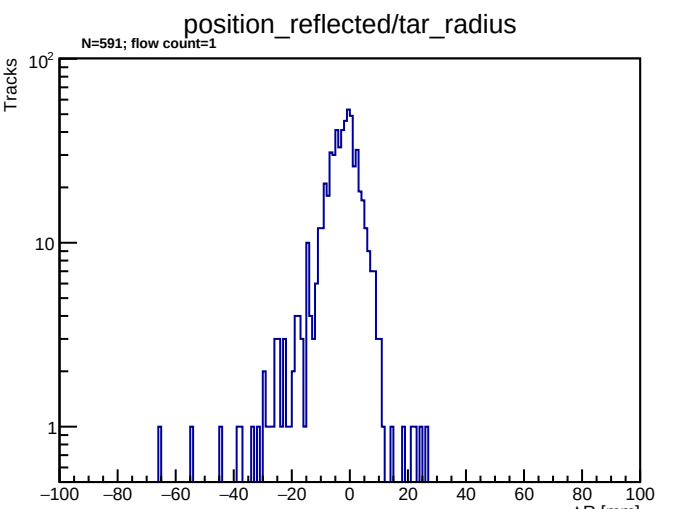

position_reflected/tar_dx
N=591; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

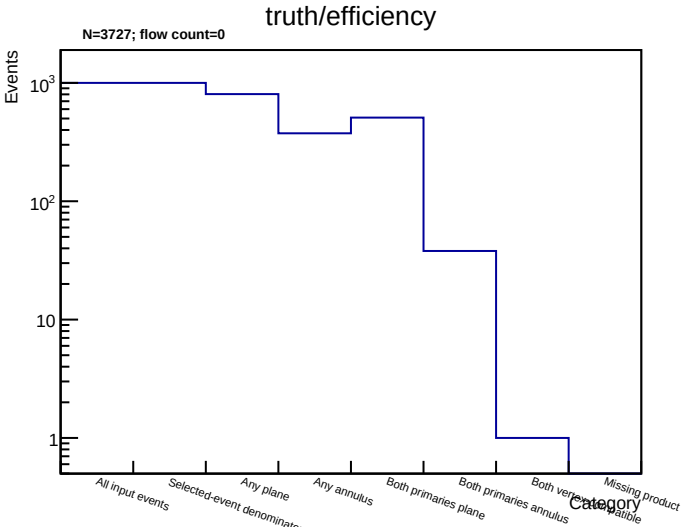

position_reflected/tar_dy
N=591; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

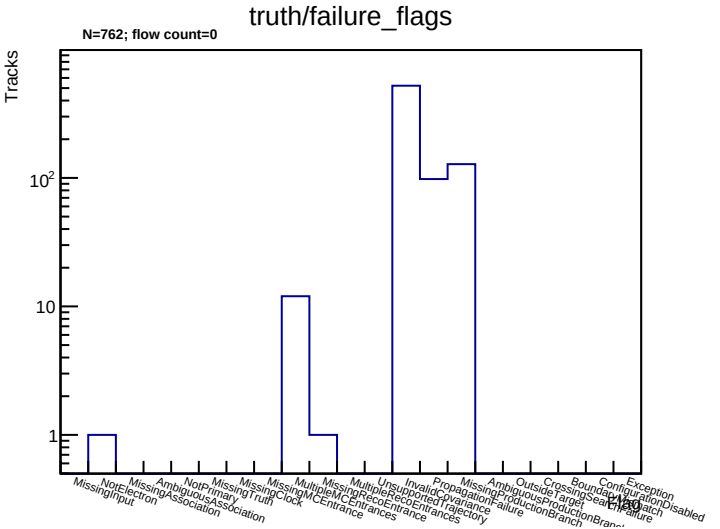

position_reflected/tar_phase
N=591; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

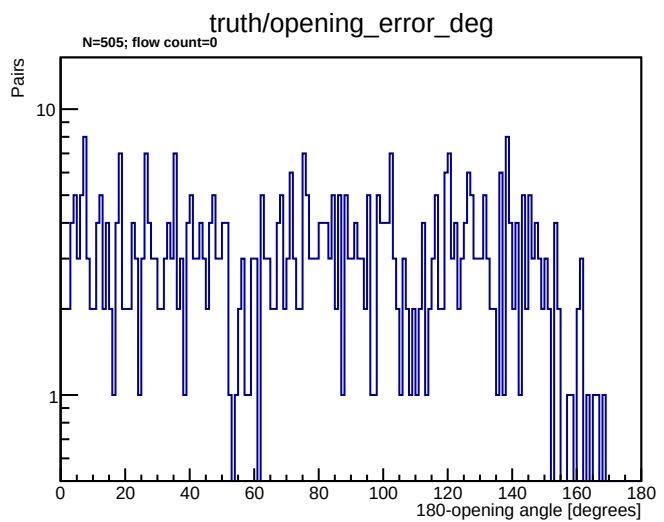

position_reflected/tar_phase_vs_tar_dr
N=591; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10^-1
10^-2

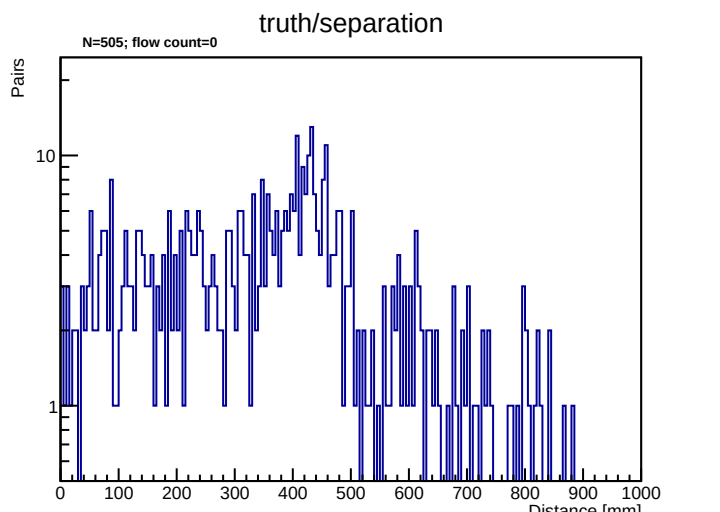

position_reflected/tar_pitch
N=591; flow count=0
Tracks
10
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

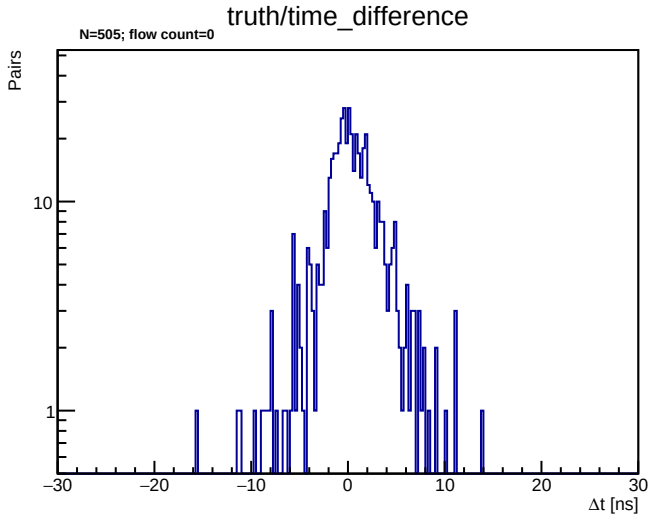

position_reflected/tar_radius
N=591; flow count=1
Tracks
10^2
10
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

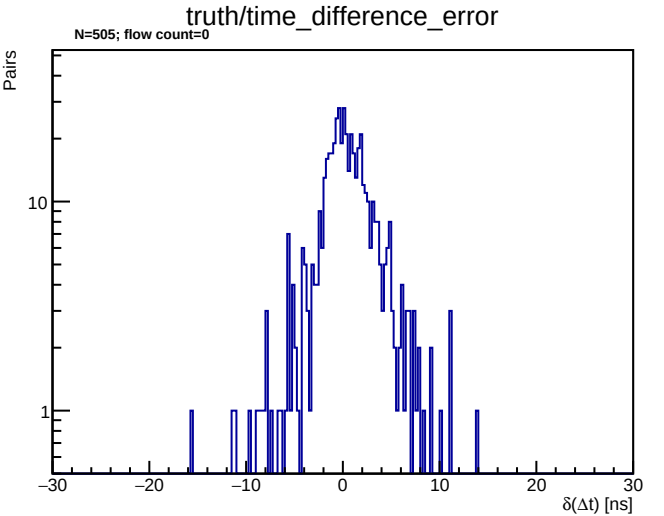

truth/efficiency
N=3727; flow count=0
Events
Category
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product

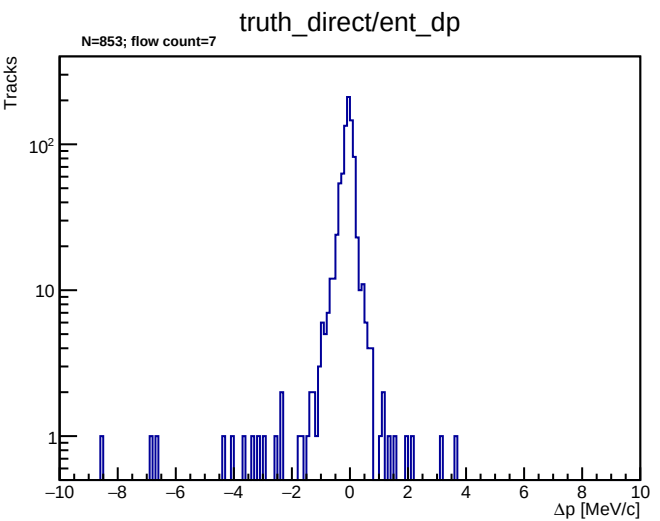

truth/failure_flags
N=762; flow count=0
Tracks
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MissingMCEntrance
MultipleMCEntrance
MissingRecoEntrance
MultipleRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeed
BoundaryMatch
Configuration
Exception
Disabled
Failed

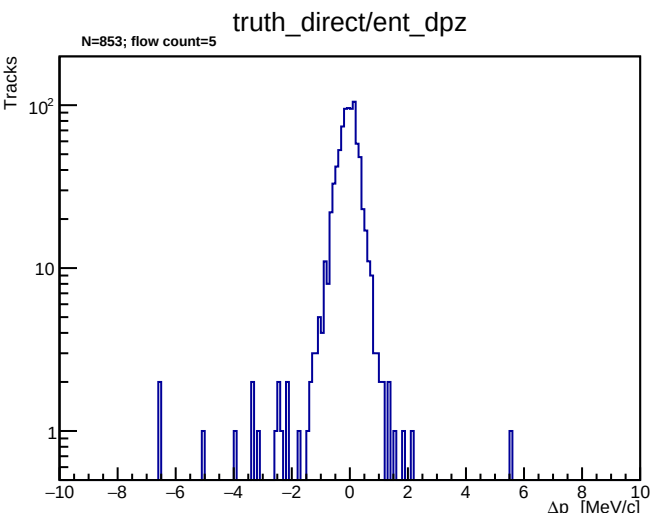

truth/opening_error_deg
N=505; flow count=0
Pairs
180-opening angle [degrees]

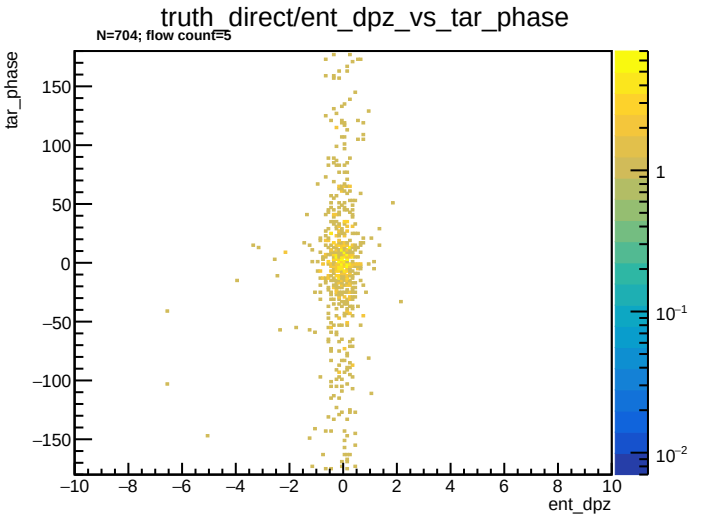

truth/separation
N=505; flow count=0
Pairs
Distance [mm]

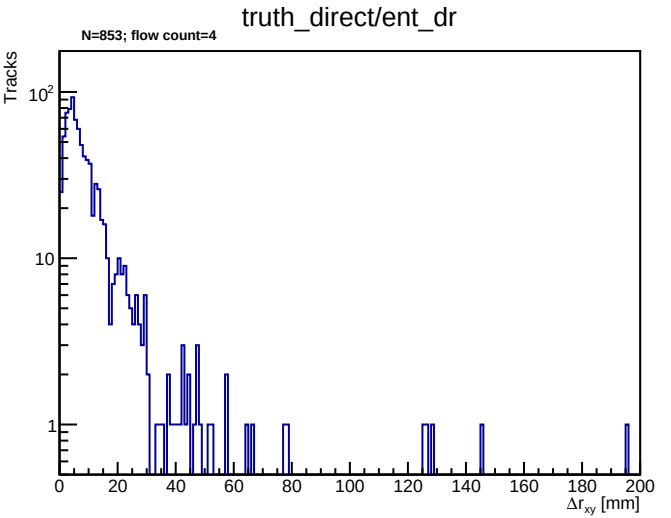

truth/time_difference
N=505; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

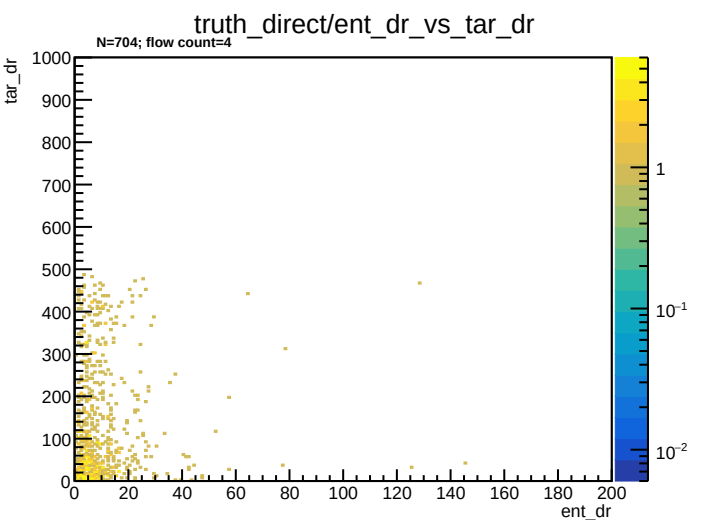

truth/time_difference_error
N=505; flow count=0
Pairs
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

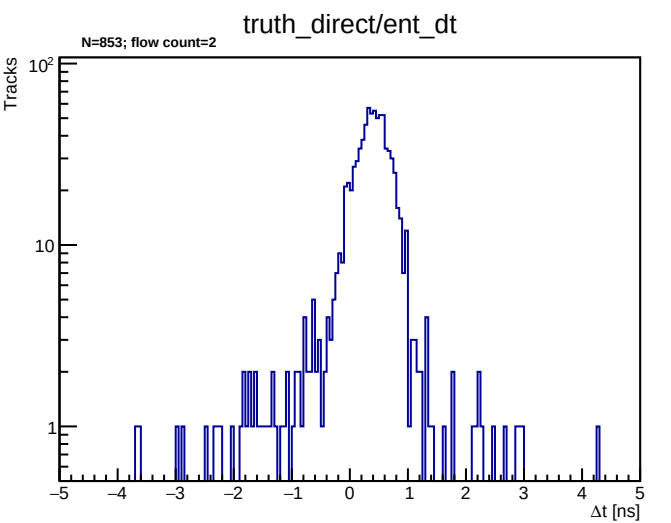

truth_direct/ent_dp
N=853; flow count=7
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp [MeV/c]

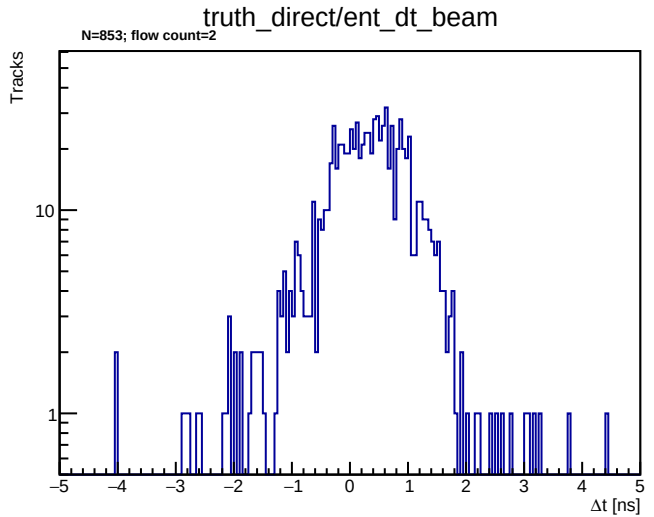

truth_direct/ent_dpz
N=853; flow count=5
Tracks
10²
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp_z [MeV/c]

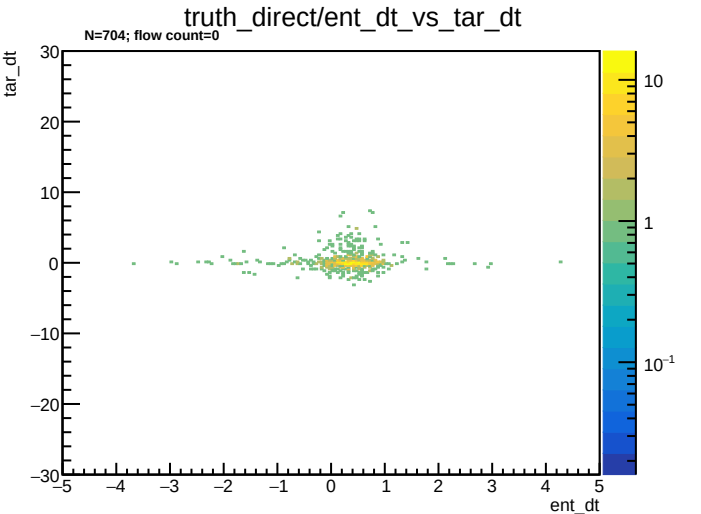

truth_direct/ent_dpz_vs_tar_phase
N=704; flow count=5
tar_phase
ent_dpz

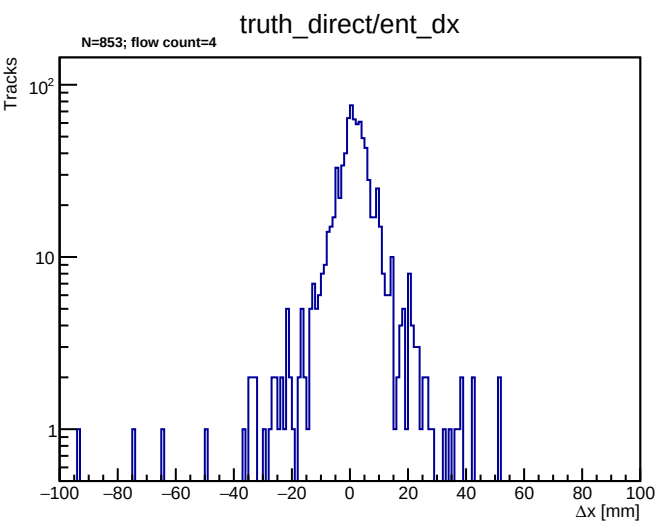

truth_direct/ent_dr
N=853; flow count=4
Tracks
Δr_xy [mm]

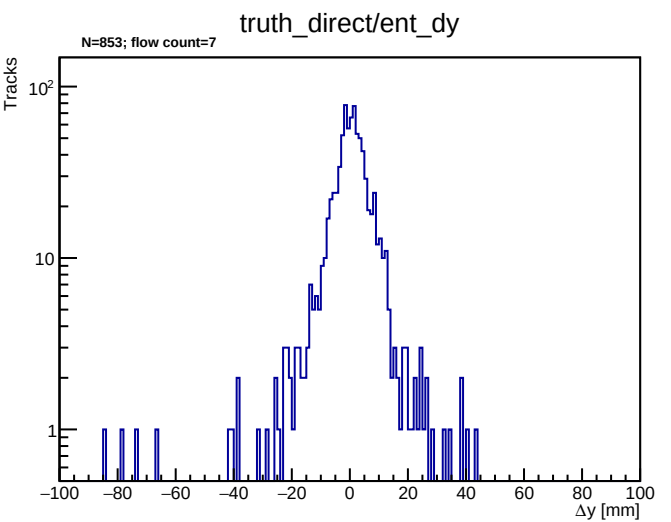

truth_direct/ent_dr_vs_tar_dr
N=704; flow count=4
tar_dr
ent_dr

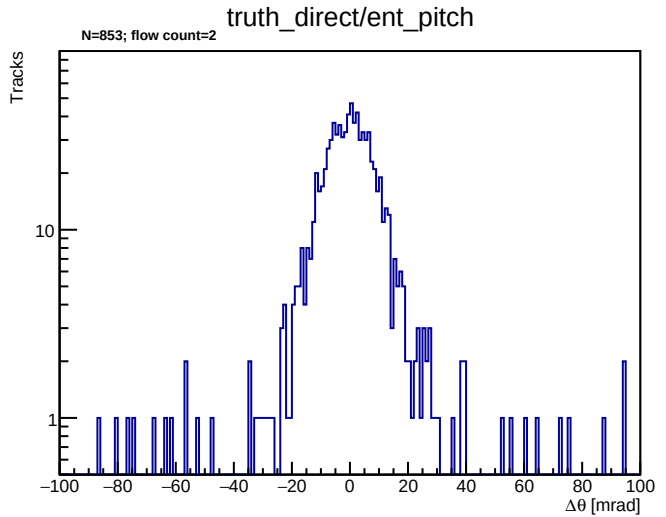

truth_direct/ent_dt
N=853; flow count=2
Tracks
Δt [ns]

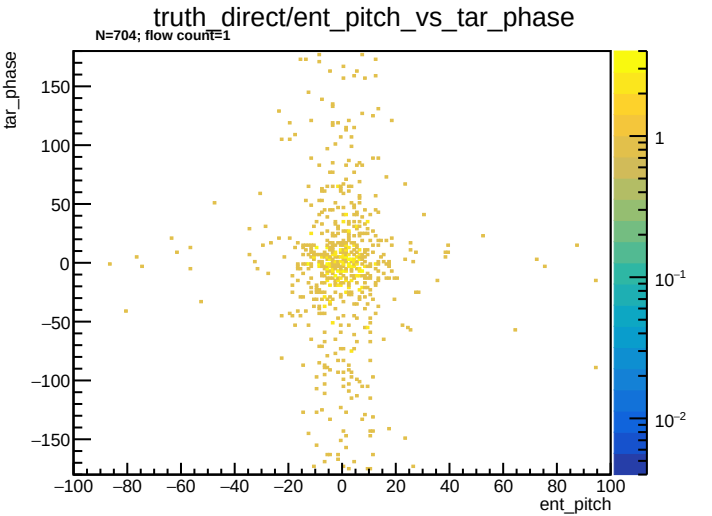

truth_direct/ent_dt_beam
N=853; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

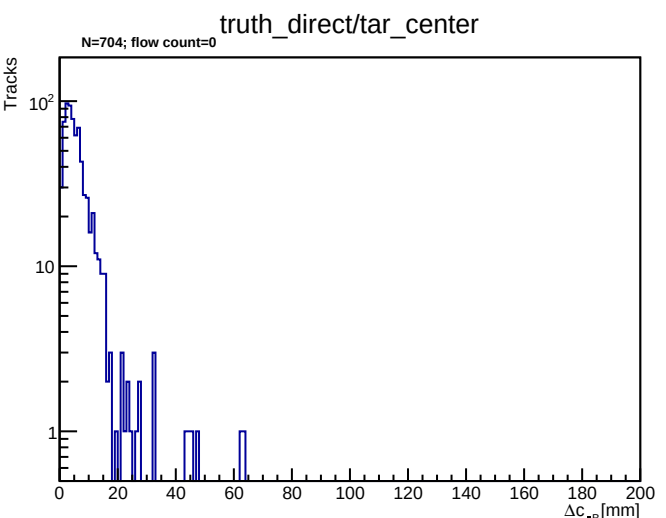

truth_direct/ent_dt_vs_tar_dt
N=704; flow count=0
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
10
1
10⁻¹

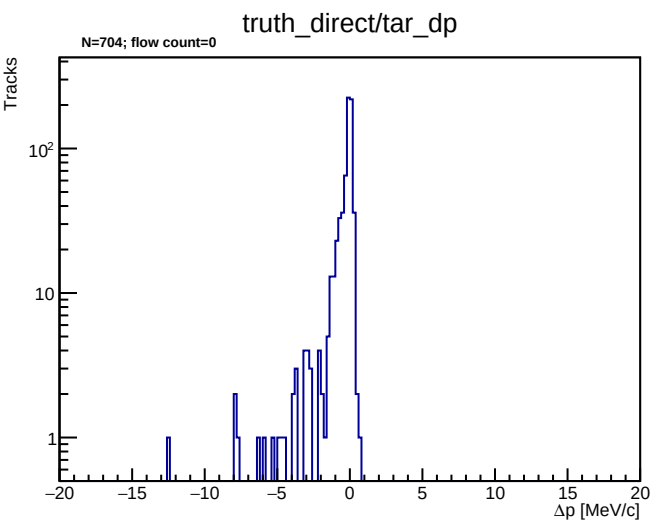

truth_direct/ent_dx
N=853; flow count=4
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

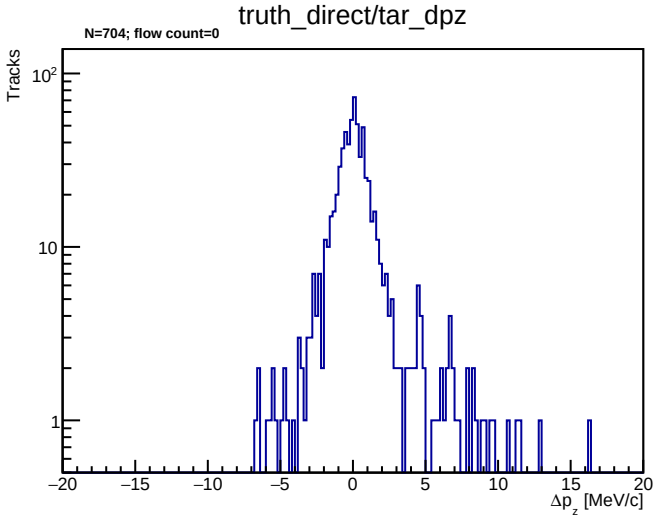

truth_direct/ent_dy
N=853; flow count=7
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

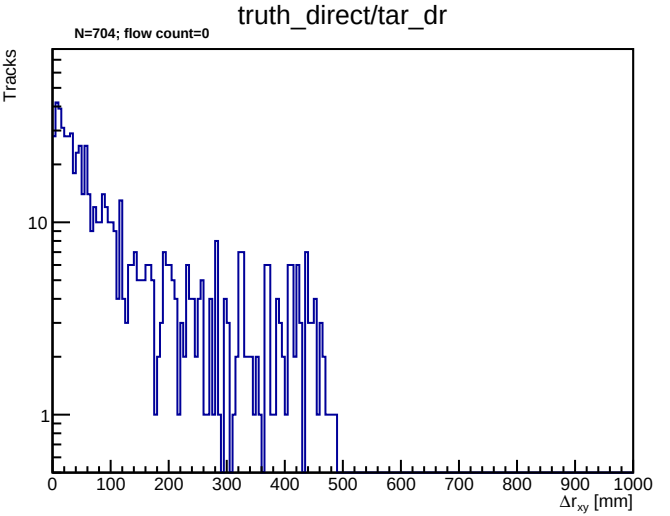

truth_direct/ent_pitch
N=853; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

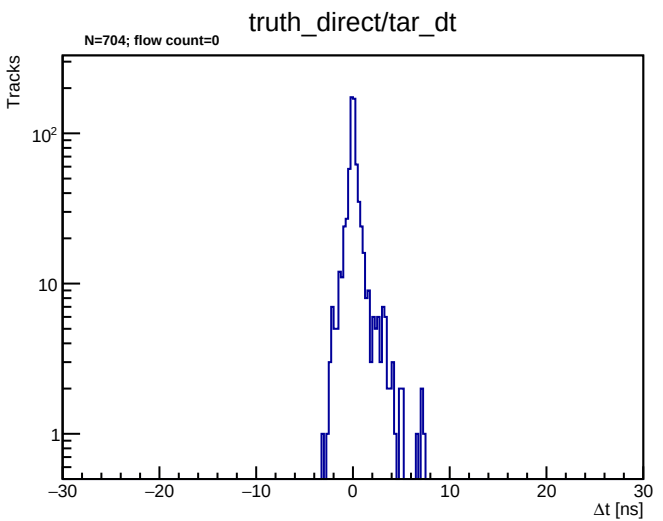

truth_direct/ent_pitch_vs_tar_phase
N=704; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^−1
10^−2

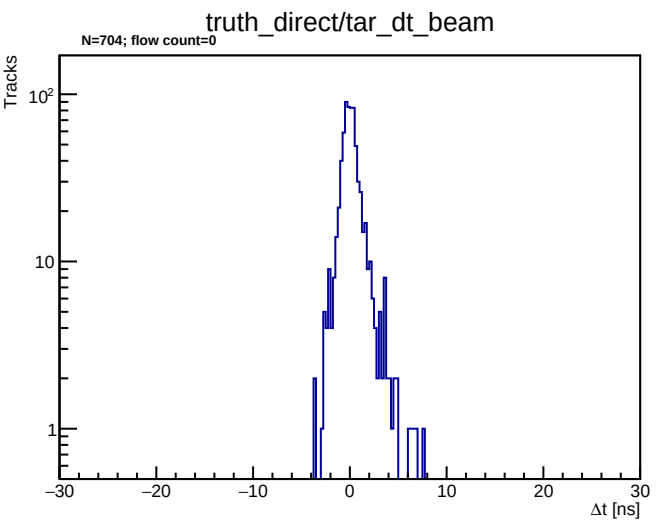

truth_direct/tar_center
N=704; flow count=0
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_tB [mm]

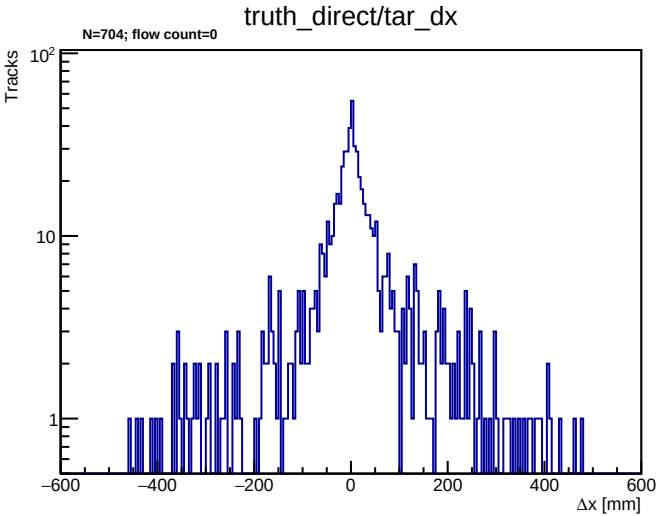

truth_direct/tar_dp
N=704; flow count=0
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

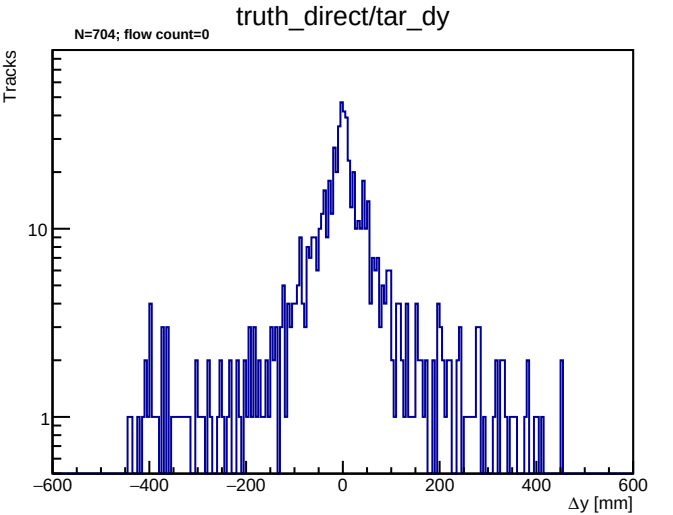

truth_direct/tar_dpz
N=704; flow count=0
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

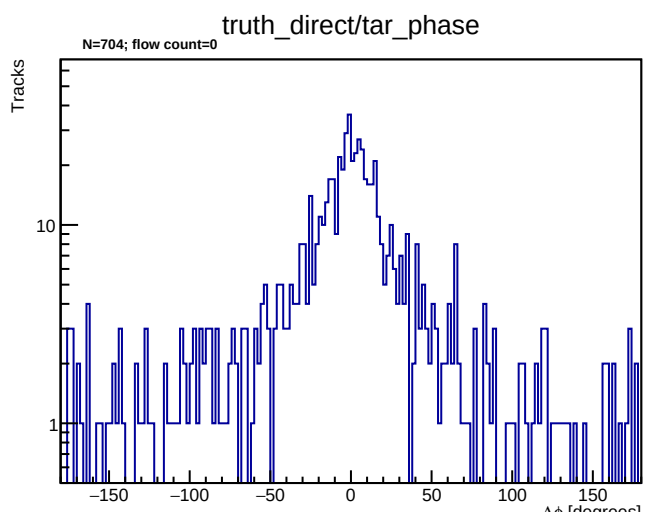

truth_direct/tar_dr
N=704; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

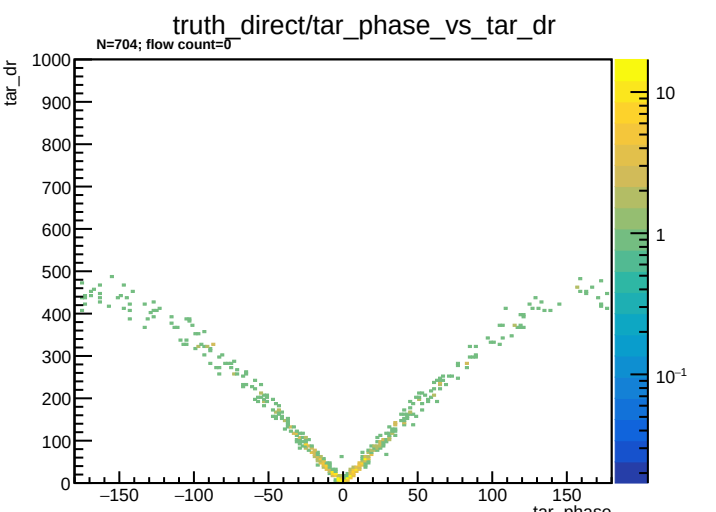

truth_direct/tar_dt
N=704; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

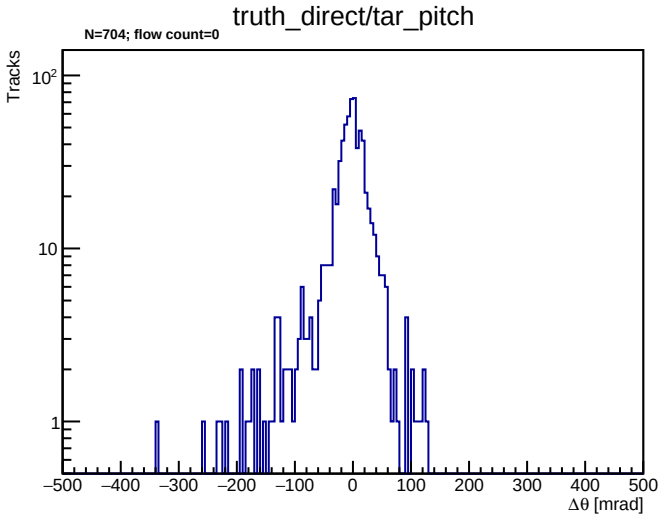

truth_direct/tar_dt_beam
N=704; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

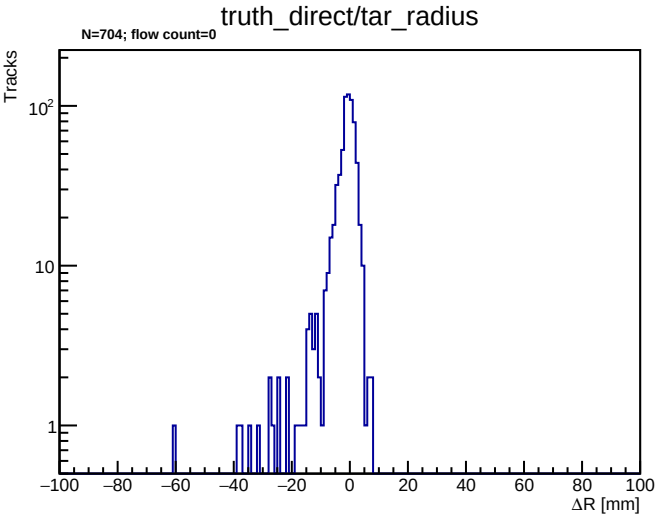

truth_direct/tar_dx
N=704; flow count=0
Tracks
Δx [mm]

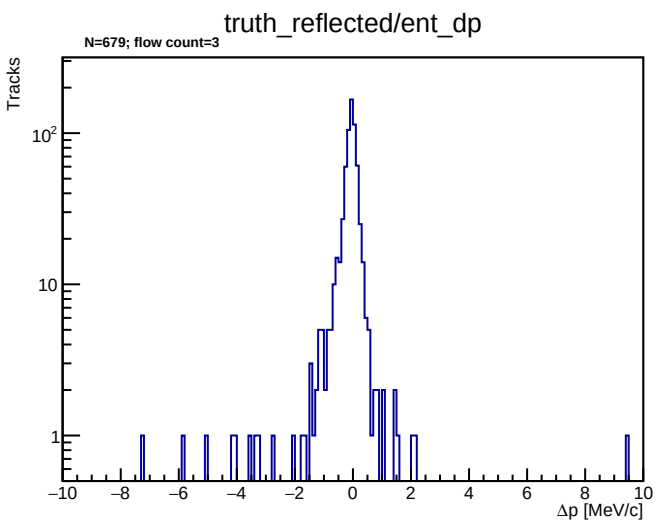

truth_direct/tar_dy
N=704; flow count=0
Tracks
Δy [mm]

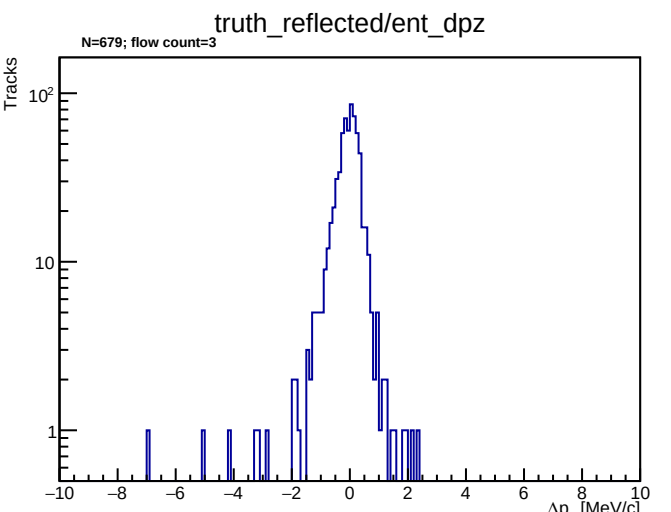

truth_direct/tar_phase
N=704; flow count=0
Tracks
Δϕ [degrees]

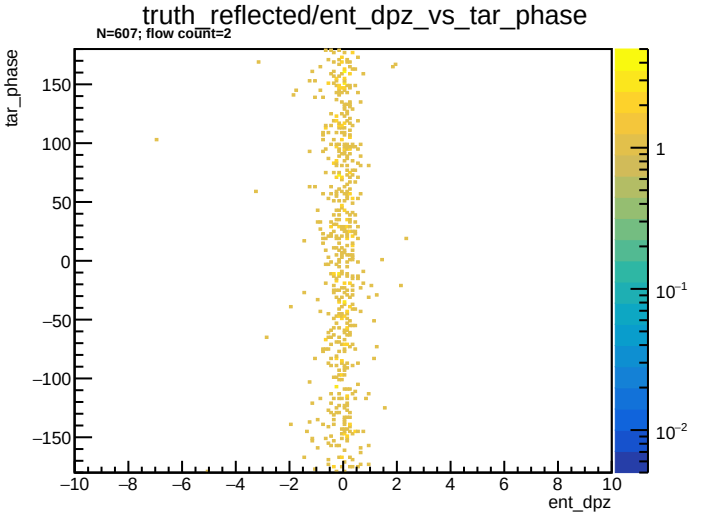

truth_direct/tar_phase_vs_tar_dr
N=704; flow count=0
tar_dr
tar_phase

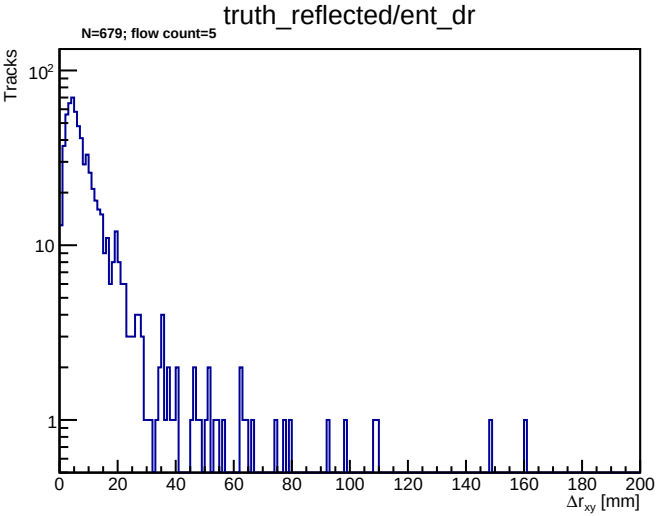

truth_direct/tar_pitch
N=704; flow count=0
Tracks
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

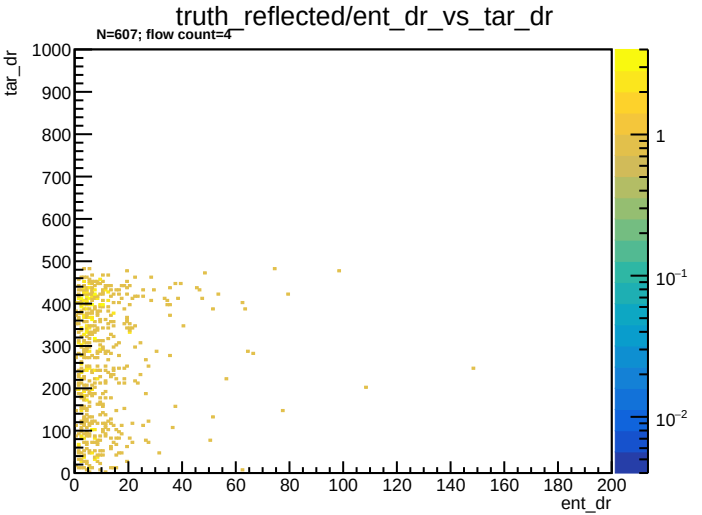

truth_direct/tar_radius
N=704; flow count=0
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

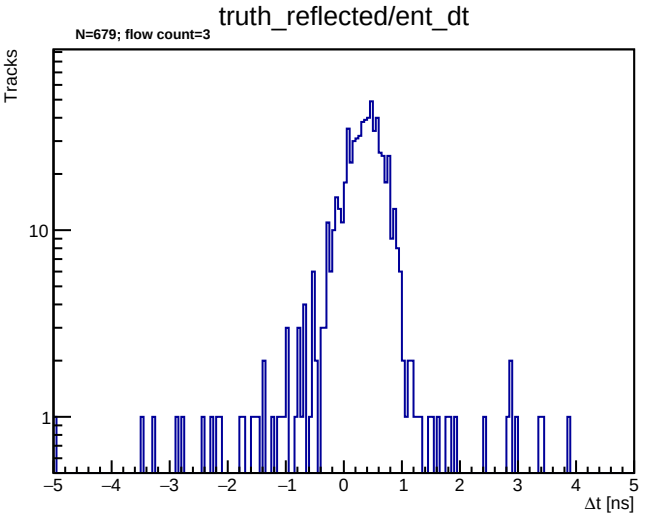

truth_reflected/ent_dp
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

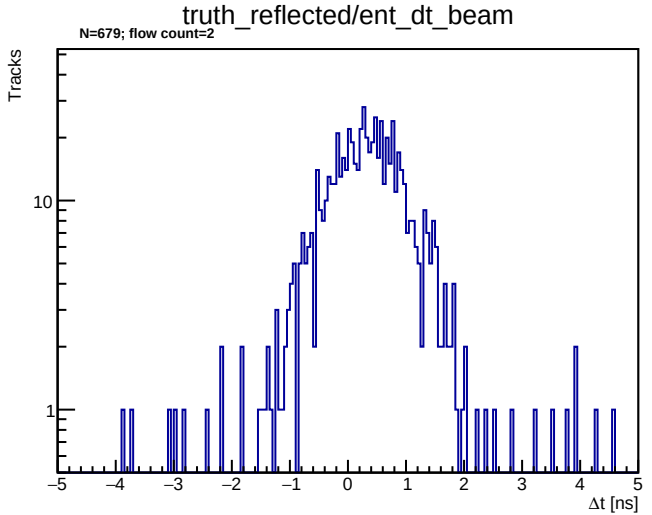

truth_reflected/ent_dpz
N=679; flow count=3
Tracks
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

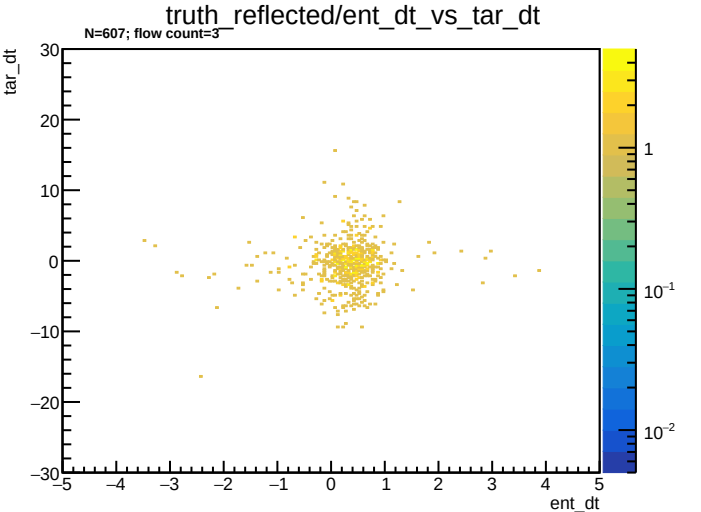

truth_reflected/ent_dpz_vs_tar_phase
N=607; flow count=2
tar_phase
ent_dpz

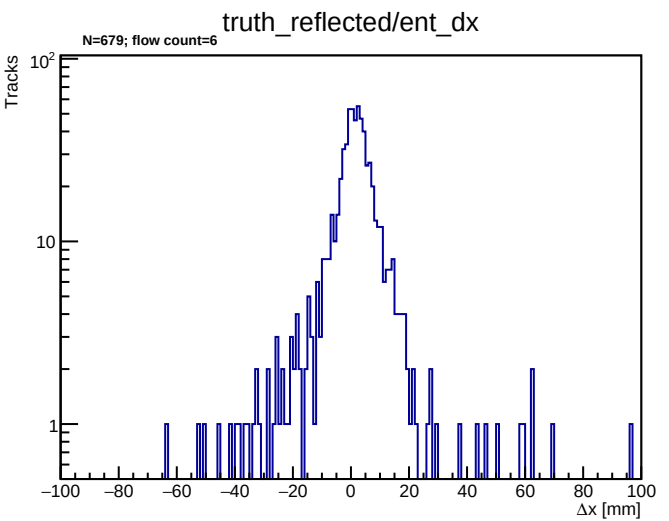

truth_reflected/ent_dr
N=679; flow count=5
Tracks
Δr_xy [mm]

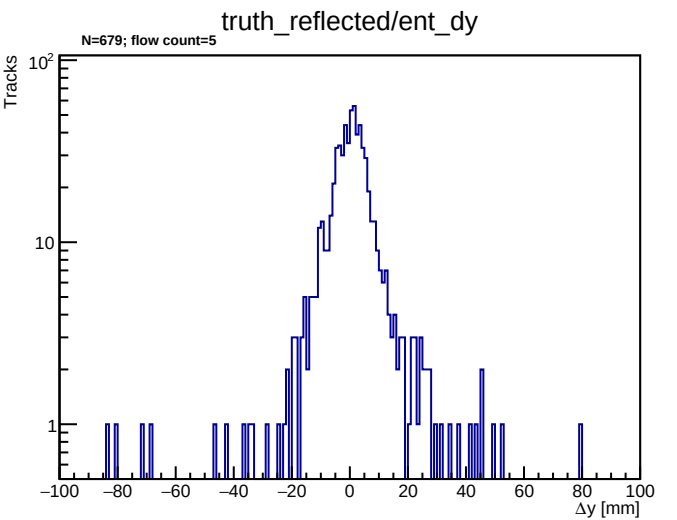

truth_reflected/ent_dr_vs_tar_dr
N=607; flow count=4
tar_dr
ent_dr

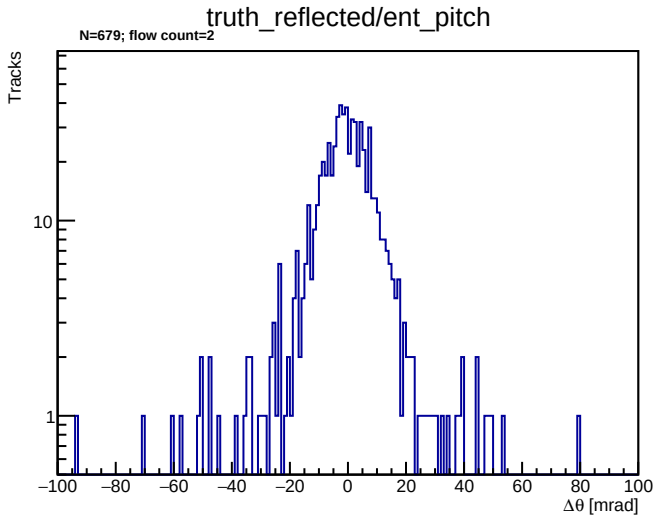

truth_reflected/ent_dt
N=679; flow count=3
Tracks
Δt [ns]

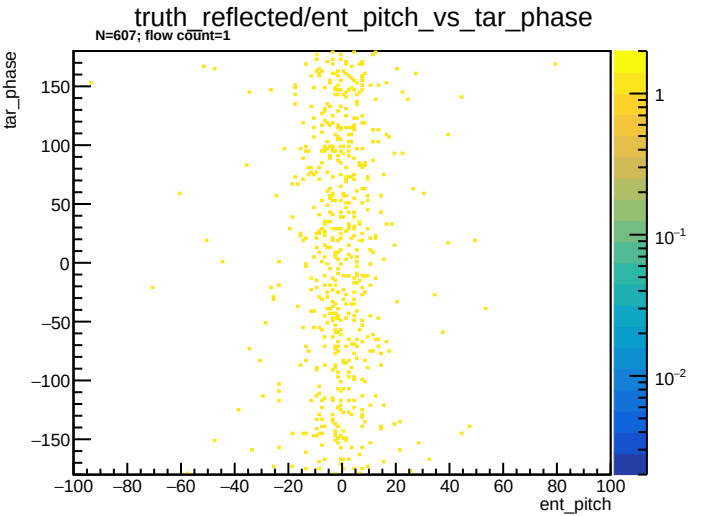

truth_reflected/ent_dt_beam
N=679; flow count=2
Tracks
10
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

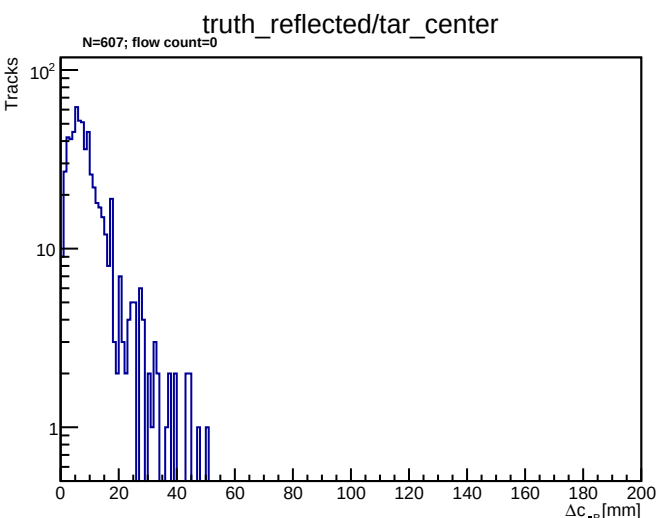

truth_reflected/ent_dt_vs_tar_dt
N=607; flow count=3
tar_dt
30
20
10
0
−10
−20
−30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^{-1}
10^{-2}

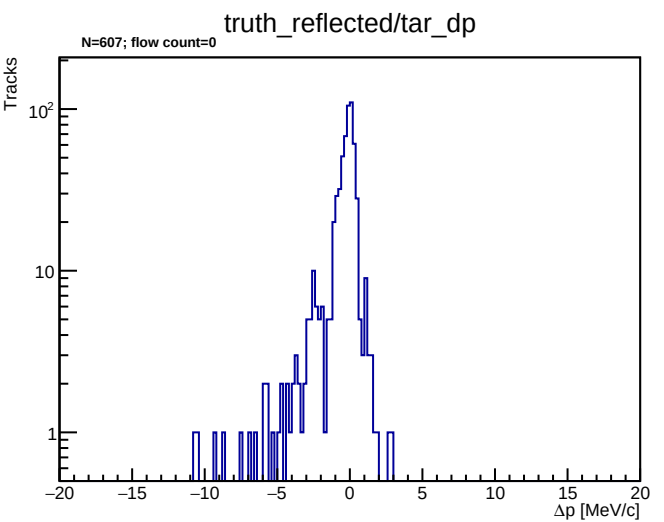

truth_reflected/ent_dx
N=679; flow count=6
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

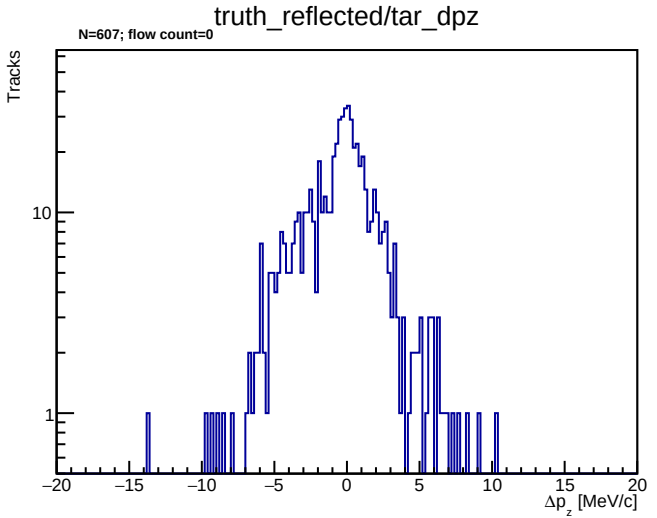

truth_reflected/ent_dy
N=679; flow count=5
Tracks
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

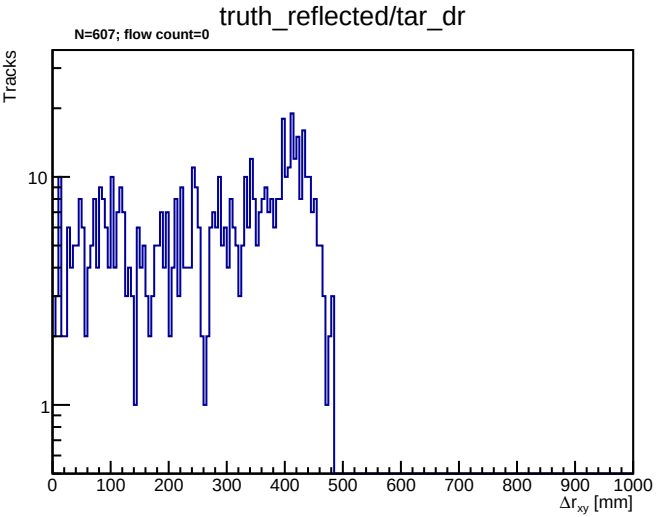

truth_reflected/ent_pitch
N=679; flow count=2
Tracks
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

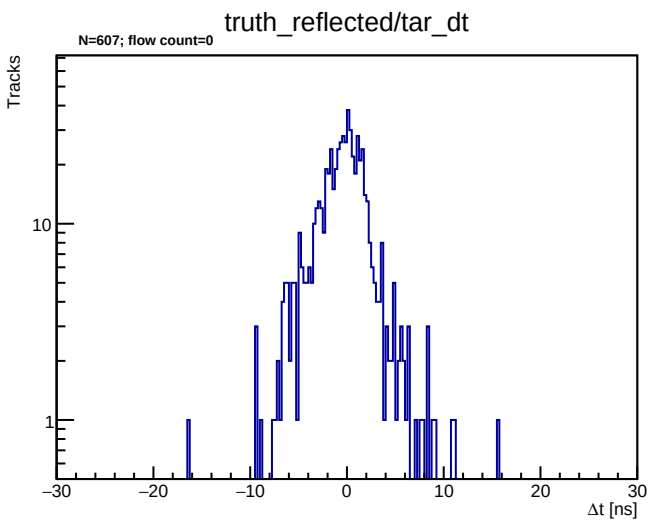

truth_reflected/ent_pitch_vs_tar_phase
N=607; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^−1
10^−2

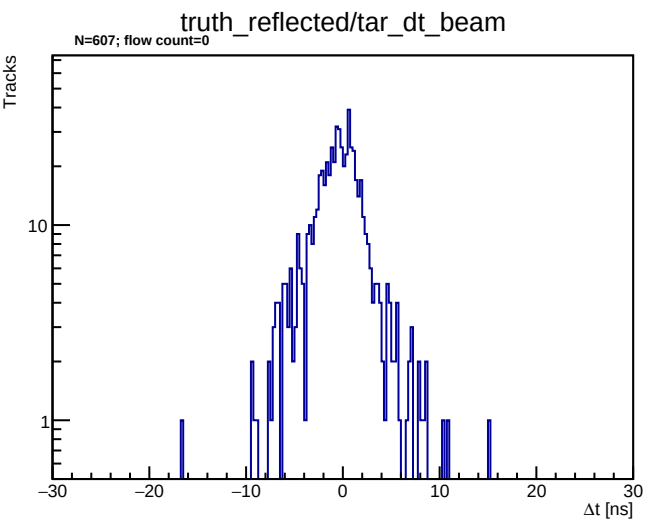

truth_reflected/tar_center
N=607; flow count=0
Tracks
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

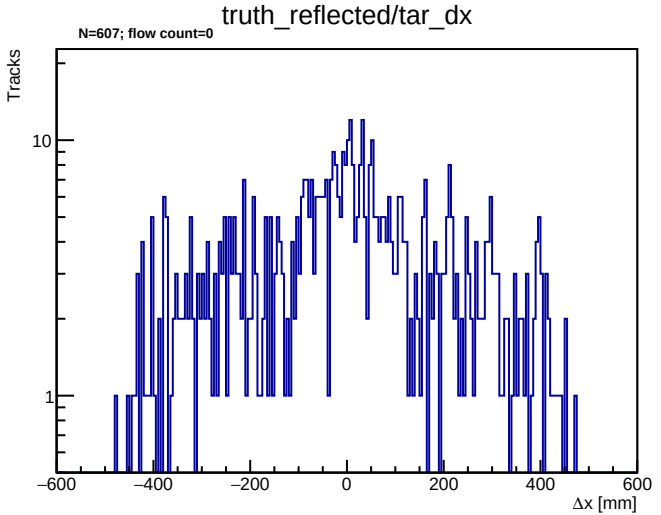

truth_reflected/tar_dp
N=607; flow count=0
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

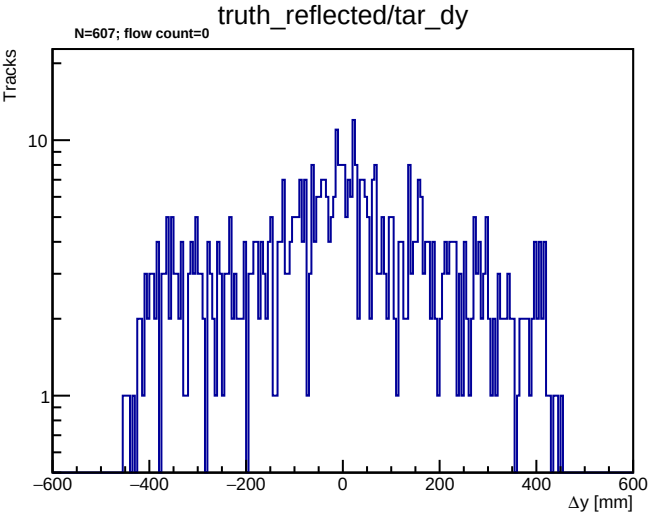

truth_reflected/tar_dpz
N=607; flow count=0
Tracks
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

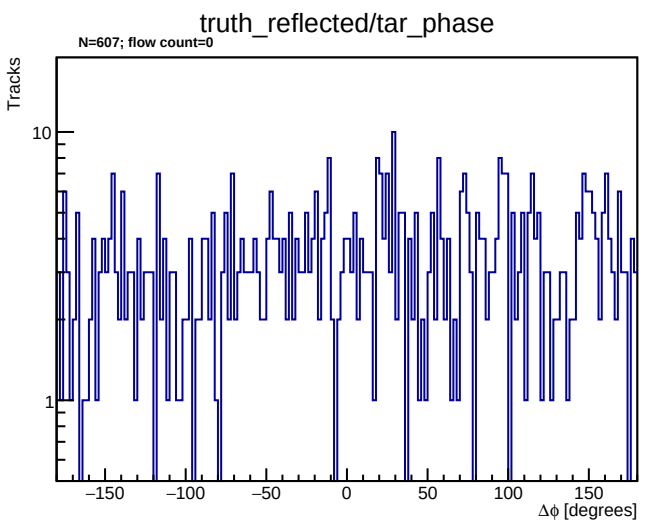

truth_reflected/tar_dr
N=607; flow count=0
Tracks
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

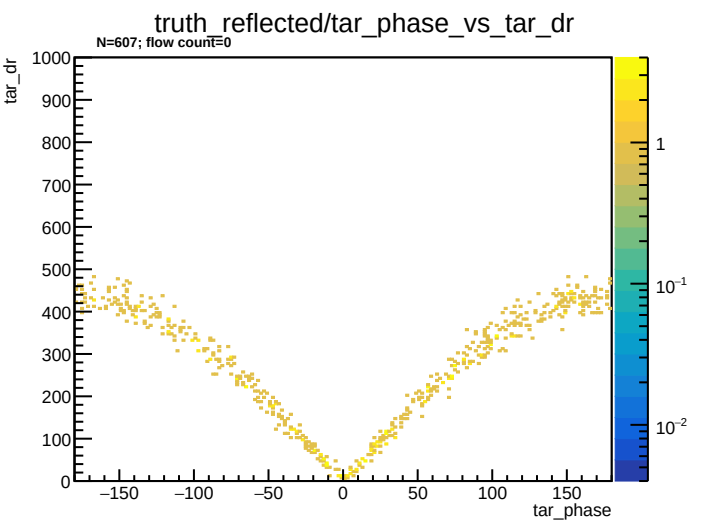

truth_reflected/tar_dt
N=607; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

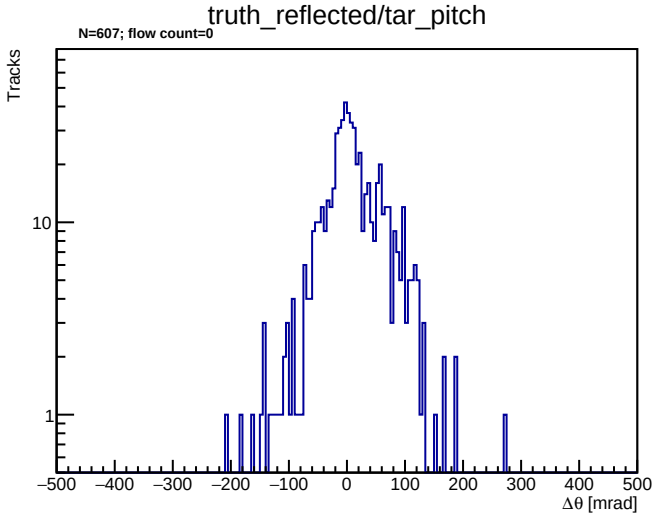

truth_reflected/tar_dt_beam
N=607; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

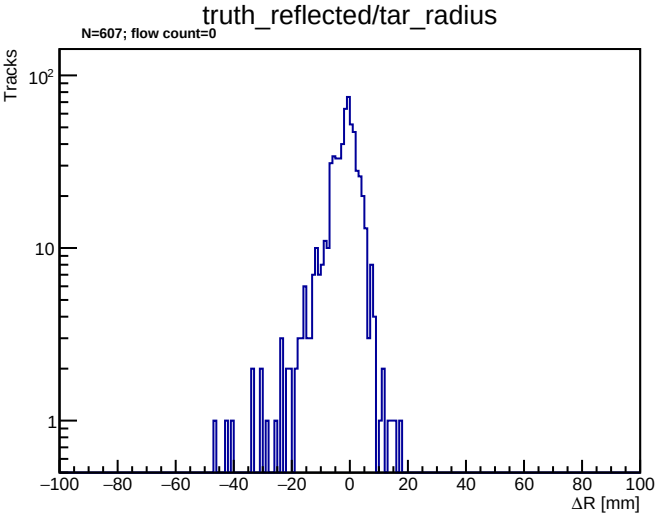

truth_reflected/tar_dx
N=607; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δx [mm]


truth_reflected/tar_dy
N=607; flow count=0
Tracks
10
1
−600
−400
−200
0
200
400
600
Δy [mm]


truth_reflected/tar_phase
N=607; flow count=0
Tracks
10
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]


truth_reflected/tar_phase_vs_tar_dr
N=607; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²


truth_reflected/tar_pitch
N=607; flow count=0
Tracks
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]


truth_reflected/tar_radius
N=607; flow count=0
Tracks
10²
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]